Theory of learned KKL observers for input-driven systems: where the input enters, how much memory the observer needs, and what no finite-memory observer can do

draft v3, generated from theory/v3/{m1,m2,m3,lemma_information,m4}

Reading guide. [v3-G1: rewritten for the final method] The theory is about the observer as finally defined: a fixed linear contracting latent filter, a strided window of $K = 20$ past latents (z-window), a spectral fading-memory (SFM) input context, and a low-rank hypernetwork decoder trained on mixed histories with the C5 regulariser (Definition 1). There is no input-dependent latent gain $B(c)$ and no latent injection in the default. The benchmark covers autonomous and input-driven state estimation; the autonomous case is Proposition 12.

Every statement carries a status tag: [proven] under the stated assumptions, [proven, linear plants], [conjecture], or [empirical]; each is followed by an intuition, a proof, what it predicts and which experiment tests it (or that none does). Proofs are written for the stated assumptions only; no claim is stronger than its proof. Statements that are new or changed in v3 are marked [v3-ID: note]. Statements for the main text:

1. Lemma 31 (*main theoretical contribution*): an information limit for every observer with a finite memory, a necessary condition. Its computed numbers at a window W use the climatology as a Gaussian stand-in for the prior at the window start, so they are an approximation (Remark 32); window starvation is not proven. The effective window of the default is at least $6/\alpha$ (Remark 33), and every example uses that corrected window.
2. Lemma 15 and Theorem 16: the z-window makes the latent an exact finite-memory map, so the truncation tail and the initial transient cancel, and the $K = 1$ lower bound does not carry over.
3. Proposition 8 (where the input enters) and Proposition 27 (when a learned observer beats a filter: model incompleteness).
4. Proposition 12: with no input the observer is a fixed filter plus a *stationary* decoder of the z-window, so the autonomous KKL guarantees apply in the form stated there.

The Lyapunov dependence of the learned observer's error is an empirical observation (Remark 35). Proposition 23 is a decomposition with measured constants, not a certificate. All numerical constants quoted are produced by `theory/checks/*.py`; the v3 checks are in `theory/checks/v3_checks.py`.

1 Where the input enters a learned KKL observer

Setting and conventions. Plant $\dot{x} = f(x, u)$, $y = h(x)$, $x \in \mathbb{R}^{n_x}$, $y \in \mathbb{R}^{n_y}$; no input-affine structure. Inputs are applied through a zero-order hold at the sample rate with values in a compact $U \ni 0$, $\bar{u} := \max_U |u|$; u_t is the window $u|_{[t-L,t]}$ shifted to $[-L, 0]$, a point of $W_L = U^{N_L}$ at sample

instants when $L/dt = N_L \in \mathbb{N}$ (the decoder runs only there), with the sup norm. $X(x,t;s;u)$ is the value at s of the solution through x at t. The latent filter is $\dot z = Az + By$. *Latent initial condition (one convention for Sections 1–3).* The *label* and training latents start from $z(0) = 0$, so $z(t) = T^0(x(t),t;u) := \int_0^t e^{A(t-s)}Bh(X(x(t),t;s;u))\,ds$ exactly; this finite-start map is the only time-varying map used. The *deployed* observer (decoder-side and latent-injection alike, `evaluate.py` `z0_map`) starts from $\hat z(0) = \hat T_0(\hat x_0)$, which adds the transient $e^{At}\hat T_0(\hat x_0)$ to the latent; it is carried as the Δz_0 term of Proposition 23 and never absorbed into a map. The decoder target is the windowed map

$$T_L(x,w) := \int_{-L}^{0} e^{-A\tau}B\,h\big(X(x,0;\tau;w)\big)\,d\tau, \tag{1}$$

which by Lemma 14(a) is within $\frac{1}{2}\varepsilon(L)$ of $T^0(x(t),t;u)$ at $w = u_t$. $\mathcal{P}_L$ is the set of pairs $(x(t),u_t)$, $t \ge L$, generated by admissible trajectories.

Definition 1 (the deployed default observer)**.** [v3-M1-1: new: the final method, A3 + mixed histories + C5] Latent filter (fixed, never trained): $\hat z_{k+1} = A_d\hat z_k + W_0\tilde y_k + W_1\tilde y_{k+1}$, $\hat z_0 = \hat T_0(\hat x_0)$ (Lemma 42(a)). Latent window: $Z_k := (\hat z_k, \hat z_{k-s}, \dots, \hat z_{k-(K-1)s})$, entries with negative index set to 0, $K = 20$, stride $s = \max\{1, \mathrm{round}(6/(\alpha\,dt\,K))\}$, span $S := (K-1)s\,dt$; on the twelve benchmark plants $\alpha S \in [5.50, 5.90]$ (check V8). Input context (SFM): $\xi_{k+1} = A_d^\xi\xi_k + B_d u_k/u_{\mathrm{ref}}$, $\xi_0 = 0$, $c_k = \psi([\xi_k, u_k/u_{\mathrm{ref}}])$ with ψ a bias-free tanh MLP. Decoder:

$$\hat x_k = D(Z_k,c_k) := T^+\big(\hat z_k;\,[c_k, g(Z_k)]\big),$$

g a bias-free tanh MLP of the normalised stack, T^+ the base MLP (activation tanh in the stored default models, check V7) whose layer l is modulated by $\Delta W_l(t) = U_l(t)V_l(t)/\sqrt{r}$, $t = \mathrm{Trunk}([c,g])$ (bias-free SiLU), U_l affine, V_l linear, $r = 4$, with the spectral cap $M \mapsto M\min\{1, \gamma/\|M\|_2\}$ on every head weight. The base decoder is retrained in phase 2. Training uses the mixed histories $D_0 \cup D_1$ and the loss with the C5 terms (jump penalty and off-manifold slope penalties); these change what the decoder is fitted to, not the structure above. *Not in the default:* an input-dependent latent gain $B(c)$ and a latent injection Φ. *Special case* $K = 1$ (no z-window, frozen base): $D(Z,c) = T^+(\hat z_k; c)$; this is the observer of the C1 and C2 comparison nodes (recipe A0), and every statement below that is written for $D(z,c)$ refers to it.

Two exact consequences used below. (i) The stack enters only through D, so every result that bounds the latent deviation (Proposition 23(i)) is unchanged; what changes is the Jacobian with respect to Z (Proposition 23). (ii) The spectral cap gives, for every context, $\|\Delta W_l(t)\|_2 \le \|U_l(t)\|_F\|V_l(t)\|_F/\sqrt{r} \le (\gamma\|t\| + \|b_{U_l}\|)\gamma\|t\|/\sqrt{r}$ (b_{U_l} the head bias), checked on the stored default models (V7c: ratio ≤ 0.67). It bounds the head part of the context Lipschitz constant on bounded sets; the trunk and base layers still enter as norm products, so it does not make Proposition 23(iii) informative a priori. [v3-M1-2: new]

Assumption 1 (standing; shared by all sections)**.** Regularity: $f(\cdot,u) \in C^1$ uniformly in $u \in U$, $h \in C^2$.

(A1) *Forward invariance.* A compact $\mathcal{X}$ contains every solution from the initial set $\mathcal{X}_0 \subset \mathcal{X}$ under every admissible input, for all $t \ge 0$.

(A2) *Bounded output.* $\|h(x)\| \le \bar y$ on $\mathcal{X}$.

(A3) *Hurwitz filter.* $-\alpha < 0$ is the spectral abscissa of A and $\kappa := \sup_{t\ge0} e^{\alpha t}\|e^{At}\| < \infty$ ($\kappa = 1$ for the normal Butterworth A of the code).

(A4) *Uniform Lipschitz injectivity of the windowed map.* There are L_0 and $\ell < \infty$ with $|x_1 - x_2| \le \ell\,|T_L(x_1, w) - T_L(x_2, w)|$ for all $(x_1, w), (x_2, w) \in \overline{\mathcal{P}_L}$ and all $L \ge L_0$ (i.e. $\ell := \sup_{L \ge L_0} \ell(L) < \infty$; Theorem 16(a) varies L at fixed ℓ).

Chaotic plants. (A1)–(A3) hold for the dissipative chaotic benchmarks (Lorenz, Rössler), (A4) is assumed for them as for every other plant, but its constant is not under control when α is below the *backward* expansion rate λ_b of the flow (for Lorenz $\lambda_b \approx 14.6$ against $\alpha = 2$): $\partial_x T^0$ contains $\int_0^t e^{A(t-s)} B\, \partial h\, \partial_x X(x, t; s; u)\, ds$ and $\|\partial_x X\|$ grows like $e^{\lambda_b (t-s)}$ along the backward flow, so the forward Lipschitz constant of $T^0(\cdot, t)$ grows with t when $\alpha < \lambda_b$; this is the eigenvalue condition of Andrieu and Praly [2] (the spectrum of A must dominate the Lipschitz constant of f for a C^1 stationary map; a Hölder map needs less), and it is a plausible mechanism (not tested: no node sweeps α or measures λ_b) behind the phase-1 PDE residual $\varepsilon_{\mathrm{pde,rms}} = 74$ on Lorenz against ≤ 0.3 elsewhere. Which results survive on such plants is stated at each result: everything that uses only the true past and (A1)–(A2) does (Lemma 14(a), Theorem 16(a), Proposition 23); the counterfactual-past statements (Lemma 14(b)) and the linear lower bounds (Theorem 16(b), Proposition 18(b)) do not apply to them.

Remark 2 (existence and injectivity: cited, and what is assumed)**.** T^0 is a C^1 solution of $\partial_x T\, f(x, u(t)) + \partial_t T = AT + Bh(x)$ whenever backward solutions exist on $[0, t]$ [3, Lemma 1], and an observer $\hat{x} = T^*(\hat{z}, t)$ converges if there are $\bar{t}$ and a concave class-$\mathcal{K}_\infty$ ρ_T with $|x_1 - x_2| \le \rho_T(|T(x_1, t) - T(x_2, t)|)$ for all $t \ge \bar{t}$ [3, Theorem 1]. For a fixed input that is backward distinguishable in finite time, $T^0(\cdot, t)$ is injective for almost every complex diagonal A with $n_y(n_x + 1)$ eigenvalues [3, Theorem 3], but the exceptional set depends on the input [3, Remark 5]; the Lipschitz, uniform-in-time version [3, Theorem 2] needs strong differential observability, inputs of class C^{m+1} (their time derivatives enter) and a gain k above a threshold, which excludes held, switching and PRBS inputs and a fixed (A, B). For autonomous systems, $2n_x + 1$ generic real dimensions suffice [5, Theorem 3.4], $2n_x + 2$ for time-varying ones [5, Remark 3.6]. (A4) is therefore *assumed*: it is the quantitative, uniform-in-input version of what these results give for each regular input. The code's $n_z = 5$ for 2-D plants is the autonomous count; the time-varying count $2n_x + 2$ is not met, so (A4) at $n_z = 5$ is an assumption whose constant must be reported, not a consequence of the cited results; the phase-1 proxy for ℓ (`l_dec_q95`, Remark 29) is 24 (vdp), 44 (bioreactor), 299 (duffing), 10 (lorenz), 14 (quad_tank), 816 (battery) for the C1-where phase 1 (used in `check_apriori_vacuity`); the C2-window-a1 phase 1 of the results layer gives $23, 44, 295, 10, 14, 806$ (Table 1).

Remark 3 (Koopman reading; which model classes the variants are)**.** For the autonomous plant and a diagonalised filter $A = \mathrm{diag}(\lambda_i)$, $B = (b_i)$, the KKL PDE $\partial_x T_i\, f = \lambda_i T_i + b_i h$ reads $(L_f - \lambda_i) T_i = b_i h$ with $L_f = f \cdot \nabla$ the Koopman generator on observables, so

$$T_i = b_i\,(L_f - \lambda_i)^{-1} h = b_i \int_0^\infty e^{\lambda_i \tau}\, h\big(X(x, 0; -\tau; 0)\big)\, d\tau,$$

the resolvent of the generator at the filter pole, i.e. the Laplace transform of the backward Koopman semigroup applied to the output observable; if $h = \sum_j c_j \varphi_j$ with $L_f \varphi_j = \mu_j \varphi_j$ then $T_i = \sum_j b_i c_j \varphi_j / (\mu_j - \lambda_i)$. For a linear plant this is the row identity $p_i = b_i C (F - \lambda_i I)^{-1}$ of the Sylvester solution (check [7]); for a nonlinear plant it gives the spectral reading of the "A fast enough" condition above ($\mathrm{Re}\,\lambda_i$ must clear the backward spectrum) and of the zero-input floor ε_0 of Remark 7(i) as a spectral truncation. Under an input the same object is the resolvent for the skew-product (state $\times$ input-shift) system, and the windowed map T_L is its finite-horizon version with the window as a truncated coordinate on the shift space; for a linear plant the context $\dot{q} = Aq - PGu$ of Proposition 6(a) has the structure of a linear predictor with inputs [20, 13]. In these terms, *latent injection*

with a stationary T_0 is a lifted Koopman model with inputs (its input-affine case $\Omega(z)u$, and the NLOX baseline $\omega(z)u$, are bilinear lifted observers with a fixed output-injection gain B), whose stability requires the input-dependent generator $A + \partial_z\Phi$ to stay contracting (Proposition 8(a)); *decoder-side conditioning* keeps a fixed generator and parametrises the left inverse of the lifting by the input context. Proposition 10 is then the statement that the Koopman spectrum moves with u (eigenvalue $u - a$; rotation rate $\omega_0 + u$) while a fixed A does not. We state this as positioning only; no result below depends on it.

1.1 The decoder target and sufficient contexts

Definition 4 (sufficient context)**.** A context map $E : W_L \to \mathbb{R}^{d_c}$ is *sufficient* if the latent and the context determine the state: for $(x_i, w_i) \in \overline{\mathcal{P}_L}$, $E(w_1) = E(w_2) \Rightarrow |x_1 - x_2| \le \ell\,|T_L(x_1, w_1) - T_L(x_2, w_2)|$. It is ϵ_E-sufficient if the same holds with an additive ϵ_E on the right.

The window itself ($E = \mathrm{id}$) is sufficient by (A4). A lower-dimensional E cannot be injective on W_L (invariance of domain), but it need not be: it only has to resolve what the latent leaves open.

Lemma 5 (decoder target)**.** *Let E be continuous and ϵ_E-sufficient, and $S_L := \{(T_L(x, w), E(w)) : (x, w) \in \overline{\mathcal{P}_L}\}$. If $\epsilon_E = 0$ there is a continuous $G^* : S_L \to \mathcal{X}$ with $G^*(T_L(x, w), E(w)) = x$, ℓ-Lipschitz in z for each c; extend $G^*(\cdot, c)$ to $\mathbb{R}^{n_z}$ with the same constant (Kirszbraun). Then along every admissible trajectory, for $t \ge L$, $|G^*(z(t), E(u_t)) - x(t)| \le \frac{\ell}{2}\varepsilon(L)$. For $\epsilon_E > 0$, any decoder D satisfies $\sup |D(T_L(x, w), E(w)) - x| \ge \frac{1}{2}\delta_E$, where δ_E is the largest $|x_1 - x_2|$ over pairs with equal latent and equal context.*

Intuition. Over a fixed window the plant has no clock, so the state is a fixed function of (latent, context) as soon as the context resolves the latent's ambiguity.

Proof. $\Psi(x, w) = (T_L(x, w), E(w))$ is continuous on the compact $\overline{\mathcal{P}_L}$ (under the hold, the backward solution is a composition of N_L C^1 flows with constant parameters). By sufficiency, $\Psi(x_1, w_1) = \Psi(x_2, w_2)$ implies $x_1 = x_2$, so π_x factors as $G^* \circ \Psi$; G^* is continuous because Ψ is a closed map (continuous on a compact set into a Hausdorff space), and ℓ-Lipschitz in z by sufficiency. Then $|G^*(z(t), c) - x(t)| = |G^*(z(t), c) - G^*(T_L(x(t), u_t), c)| \le \ell\,|z(t) - T_L(x(t), u_t)|$ and Lemma 14(a). The lower bound: D returns one value for two states δ_E apart. □

Proposition 6 (sufficient contexts exist, and must carry history)**.** *(a)* Linear plants. *For $\dot{x} = Fx + Gu$, $y = Cx$, the windowed map is affine in x with a w-independent slope, $T_L(x, w) = P_L x + Q_L w$, $P_L = P - e^{AL}Pe^{-FL}$, $PF = AP + BC$. Hence the n_z-dimensional linear context $c = Q_L w$ is exactly sufficient whenever P_L is injective ($\ell = \|P_L^+\|$), and as $L \to \infty$ (under the washout $e^{AL}Pe^{-FL} \to 0$, (H1) of Theorem 16) it is the output q of the stable filter $\dot{q} = Aq - PGu$: a spectral filter bank with identity readout. (b)* Nonlinear plants. *The window ($d_c = n_u N_L$) is sufficient, and by (a) of Lemma 14 any context that determines u_t for some L is $\ell\varepsilon(L)$-sufficient for the infinite past: $|x_1 - x_2| \le \ell\,|T_1^0 - T_2^0| + \ell\,\varepsilon(L)$ (each T_i^0 is within $\frac{1}{2}\varepsilon(L)$ of T_L). (c)* History is necessary. *The current input alone is not sufficient in general (Proposition 18).*

Proof. (a) $X(x, 0; \tau; w) = e^{F\tau}x +$ (linear in w); substitute in (1) and use $\int_{-L}^{0} e^{-A\tau}BCe^{F\tau}d\tau = P - e^{AL}Pe^{-FL}$ (integrate $\frac{d}{d\tau}(e^{-A\tau}Pe^{F\tau}) = e^{-A\tau}BCe^{F\tau}$). If $Q_L w_1 = Q_L w_2$ then $T_L(x_1, w_1) - T_L(x_2, w_2) = P_L(x_1 - x_2)$. The limit: $z - Px$ obeys $\dot{(z - Px)} = A(z - Px) - PGu$. (b) (A4) and Lemma 14(a). □

For nonlinear plants and contexts of fixed dimension below $n_u N_L$, finite stable filter banks with a static readout uniformly approximate every time-invariant fading-memory operator on slew-limited bounded inputs [4, Theorem 2]; applying it to the observer operator $(y,u) \mapsto x$ (a fading-memory operator under (A1)–(A4), on slew-limited signals, and for the filters the theorem chooses rather than ours) motivates the spectral context but is not a guarantee for the trained encoder.

Remark 7 (expressivity at a fixed base; what C1 compares)**.** (i) *Shared floor* [v3-M1-3: scope: the C1 arms, not the default]. Every conditioner with exact zero-input collapse, $D(\cdot,0) = T_0^+$, errs at $c = 0$ by $\varepsilon_0 := \sup_z |G^*(z,0) - T_0^+(z)|$: the zero-input target is the inverse of the zero-input windowed map, not the learned stationary base. This describes the C1 comparison arms ($K = 1$, frozen base). It does not describe the default: there $D(\cdot,[0,g(Z)]) \neq T_0^+$ and the base is retrained (Proposition 12(a)). (ii) *Additive (concat).* $T_0^+(z) + R(z,c) - R(z,0)$ approximates on S_L, as the width of R grows, every continuous target that equals T_0^+ on the zero slice: SiLU satisfies $\mathrm{SiLU}(s) - \mathrm{SiLU}(-s) = s$, so a two-hidden-layer SiLU network contains every one-hidden-layer one, and the universal approximation theorem for non-polynomial activations [14] applies (with a Tietze extension of the target). (iii) *Base-capped (FiLM, LoRA, hypernetwork).* For each fixed c the decoder is a network $N_{\theta(c)}$ of the frozen base's architecture, so for every conditioner budget

$$\inf_{D} \sup_{S_L} |D - G^*| \;\geq\; \sup_{c} \inf_{\theta\in\Theta} \sup_{z} |N_\theta(z) - G^*(z,c)|,$$

with Θ the reachable parameter set. At the matched budget Θ is much smaller than the base's weight space: the hypernetwork's linear bias-free heads place $\theta(c)$ in an affine subspace of dimension equal to its trunk width; LoRA updates have the configured rank and leave biases fixed; FiLM never modulates the readout layer, so $D(\cdot,c) = W_L\phi_c(z) + b_L$ with (W_L, b_L) frozen. In the ReLU setting, a LoRA rank of order width times a depth ratio suffices to represent a smaller target network exactly [29]; that result does not cover tanh bases or small fixed ranks. (iv) *Latent injection.* Φ is a universal class, but as an observer it represents only targets of the stationary route: a stationary T_0 with an exact correction. [v3-M1-4: default] For the default the base weights are retrained in phase 2, so Θ also contains the base's own weights; the display stays valid with that larger Θ. The theory therefore does not rank the variants by expressivity; at a matched budget on one base, C1 measures inductive bias and optimisation within these classes.

1.2 Where the input enters: a stability–extrapolation trade-off

Both observers run the filter on $y+\nu$, $\|\nu\|_\infty \leq \delta_y$, with a context $\hat{c} = E(\hat{u})$ of the input $\hat{u}$ they are given; $c = E(u)$ and $\delta_c := \sup_t |\hat{c} - c| \leq \ell_E \|\hat{u} - u\|_\infty$ (Lemma 19).

$$\begin{aligned} &\text{(latent)} && \dot{\hat{z}} = A\hat{z} + B(y+\nu) + \Phi(\hat{z},\hat{c}), && \hat{x} = T_0^+(\hat{z}), \quad \hat{z}(0) = T_0(\hat{x}_0), \\ &\text{(decoder side)} && \dot{\hat{z}} = A\hat{z} + B(y+\nu), && \hat{x} = D(\hat{z},\hat{c}), \quad \hat{z}(0) = \hat{T}_0(\hat{x}_0). \end{aligned}$$

For the latent variant, $T_0 \in C^1$ is a stationary map with left inverse T_0^+ (ℓ_{dec}-Lipschitz, $\varepsilon_{\mathrm{dec}} = \sup_{\mathcal{X}} |T_0^+ T_0(x) - x|$), the residual along trajectories is $r = \partial_x T_0(x) f(x,u) - AT_0(x) - Bh(x) - \Phi(T_0(x),c)$ with $|r| \leq \varepsilon_{\mathrm{res}}$, and $\ell_{\Phi,c}$ is the Lipschitz constant of $\Phi(z,\cdot)$ *uniformly over* $z \in T_0(\mathcal{X}) + B(r)$ (the set used in (a)). $\mu_2(M) = \lambda_{\max}(\frac{1}{2}(M + M^\top))$ is the logarithmic norm.

Proposition 8 (where the input enters)**.** *Under Assumption 1:*

(a) Latent injection: stability depends on the input *[proven]. Let* $\bar{d} := \varepsilon_{\mathrm{res}} + \|B\|\delta_y + \ell_{\Phi,c}\,\delta_c$ *and let* $r \geq |e(0)| + \bar{d}/\mu$. *If* $\mu_2\big(A + \partial_z\Phi(z,\hat{c})\big) \leq -\mu < 0$ *for all* z *in the set* $\mathcal{Z}_r := T_0(\mathcal{X}) + B(r)$ *and*

all contexts $\hat{c}$ the observer meets, then $e = \hat{z} - T_0(x)$ stays in $B(r)$ for all t (so the hypothesis is checked on a set fixed in advance, not on the solution) and

$$|e(t)| \le e^{-\mu t}|e(0)| + \tfrac{1}{\mu}\big(\varepsilon_{\mathrm{res}} + \|B\|\delta_y + \ell_{\Phi,c}\,\delta_c\big), \qquad |\hat{x} - x| \le \ell_{\mathrm{dec}}|e| + \varepsilon_{\mathrm{dec}}.$$

(b) Decoder side: stability is free, extrapolation is not *[proven]. The latent is an input-free LTI filter, so $\hat{z} - z$ is driven only by ν and the initial transient, and $|\hat{x} - x|$ obeys the bound of Proposition 23 for every input, with the decoder's Jacobian norms $\ell_z(\hat{c}), \ell_c(\hat{c})$ taken at the contexts actually queried. Both conditioner families have c-uniform bounds on a bounded context set, but with very different constants: for an additive conditioner $\ell_z(c) \le \ell_z(T_0^+) + 2\sup\|\partial_z R\|$; for multiplicative ones (FiLM, LoRA, hypernetwork) the available bound is the product form $\ell_z(c) \le \prod_l \|W_l + \Delta W_l(c)\| \sup|\sigma'|^d$, finite because $\theta(c)$ is bounded (a continuous trunk on a bounded context set: the SFM context and g have* tanh *output layers [v3-M1-5: was: tanh trunk; the trunk is SiLU]), but 10^1–10^5 times larger at random initialisation (check [5]) before the tanh base saturates.*

Intuition. Injecting the input makes the observer a recurrent network whose stability depends on how strongly the injection reacts to the latent, but its decoder never sees an unfamiliar context; conditioning the decoder keeps the recurrent part a fixed stable filter, but the network is queried at every unseen context directly.

Proof. (a) $\dot{e} = Ae + [\Phi(\hat{z},\hat{c}) - \Phi(T_0x,\hat{c})] + d$, $d = B\nu + [\Phi(T_0x,\hat{c}) - \Phi(T_0x,c)] - r$, $|d| \le \bar{d}$. While $e \in B(r)$ the segment $[T_0x, \hat{z}]$ lies in the ball $T_0x + B(r) \subseteq \mathcal{Z}_r$ (convexity of the ball, not of $\mathcal{Z}_r$, is what is used); write the bracket as $\bar{J}e$, $\bar{J} = \int_0^1 \partial_z\Phi(T_0x + se, \hat{c})ds$, so $\mu_2(A + \bar{J}) \le -\mu$ by subadditivity and convexity of μ_2, and $D^+|e| \le -\mu|e| + \bar{d}$. The comparison function $\phi(t) = e^{-\mu t}|e(0)| + \bar{d}(1 - e^{-\mu t})/\mu \le r$ bounds $|e|$ as long as $e \in B(r)$, and $\phi < r$ whenever $|e| = r$ would be reached, so $B(r)$ is forward invariant and the bound holds for all t (contraction, 17; equivalently the perturbation lemmas of 12, Ch. 9). The state bound is the triangle inequality through $T_0^+(T_0(x))$. (b) The latent statement is linearity; the bound is Proposition 23; the Jacobian bounds are the chain rule (R's z-Jacobian does not involve c multiplicatively, whereas modulated layers multiply z-features by c-dependent weights), and $\theta(c)$ is bounded because the trunk is continuous and its input is bounded (tanh output layers of ψ and g). For the default, $\hat{z}$ is replaced by the stack Z in the Jacobians (Proposition 23). □

Corollary 9 (the certified margin shrinks with the context)**.** *$\mu_2(A + \partial_z\Phi) \le \mu_2(A) + \ell_\Phi$, and $\mu_2(A) = -\alpha$ for normal A, so $\mu \ge \alpha - \ell_\Phi$ (for general A, the Gronwall version gives $\alpha - \kappa\ell_\Phi$). For an injection with exact zero-input collapse, $\Phi(z,c) = M(z,c) - M(z,0)$, $M \in C^2$: $\ell_\Phi(c) \le \min\big(|c|\,L_{zc}, 2\sup\|\partial_z M\|\big)$ with $L_{zc} = \sup\|\partial_c\partial_z M\|$, and $|c| \le \ell_E\|u\|_\infty$ (sup norm) since $E(0) = 0$; for an input-affine injection $\Omega(z)u$, $\ell_\Phi \le \ell_\Omega\|u\|_\infty$. The margin is evaluated at the* observer's *context $\hat{c}$, so a corrupted input shrinks it. The margin saturates: if $2\kappa\sup\|\partial_z M\| < \alpha$ the latent observer is certified at every amplitude, and when the plant input saturates (as through a saturating actuator) the exact correction's z-Lipschitz constant stays bounded too. The condition is* sufficient only*: for the code's trained injection the certified margin is already negative at moderate gain (-5.47, log-norm $+8.1$ in check [3]) while two observer copies still contract, so a negative margin predicts nothing, and only a measured incremental rate does.*

Proposition 10 (the margin cannot be removed at a fixed spectrum)**.** *[proven] There are bounded, observable plants with an* exact *injection ($\varepsilon_{\mathrm{res}} = 0$) whose latent observer diverges while a decoder-side observer with a bounded sufficient context is within $\frac{\ell}{2}\varepsilon(L)$ of exact. For $\dot{x} = (u - a)x + b$,*

$y = x$, $a > \alpha > 0$, *scalar latent* $\dot{z} = -\alpha z + y$ *and constant* $u \in (\alpha, a)$*: the plant converges to* $b/(a-u)$*, the stationary map is* $T_0(x) = Px + q$*,* $P = 1/(\alpha - a)$*,* $q = -Pb/\alpha$*, the exact injection is* $\Phi^*(z,u) = u(z-q)$ *(*$\ell_\Phi = |u|$*), and* $\dot{e} = (u-\alpha)e$*. The windowed map is* $T_L(x,w) = P_L(u)\,x + Q_L(u)$ *with* $P_L = \int_{-L}^{0} e^{(\alpha+u-a)\tau} d\tau > 0$ *and* $Q_L = \int_{-L}^{0} e^{\alpha\tau}\, b\, \frac{e^{(u-a)\tau}-1}{u-a}\, d\tau$ *(both bounded functions of the window input), so the decoder* $\hat{x} = (\hat{z} - Q_L)/P_L$ *errs by at most* $\frac{\ell}{2}\varepsilon(L)$ *with* $\ell = 1/P_L$ *(Lemma 5), for every* $u \in (\alpha, a)$*.*

Proof. $T_0'(x) f(x,0) = P(b - ax)$ and $-\alpha T_0 + x = (1 - \alpha P)x - \alpha q = -aPx + Pb$; so T_0 solves the zero-input PDE and $\partial_x T_0[f(x,u) - f(x,0)] = Pux = u(T_0(x) - q)$. Then $\dot{e} = -\alpha\hat{z} + x + u(\hat{z} - q) - P((u-a)x + b) = (u-\alpha)e$. For the window: $X(x,0;\tau;u) = e^{(u-a)\tau}x + b(e^{(u-a)\tau} - 1)/(u-a)$ in (1). □

Here the small-gain threshold $u = \alpha$ is attained exactly. The *exact* time-varying map $z = m(t)x + n(t)$, $\dot{m} = 1 - (\alpha + u - a)m$, $\dot{n} = -\alpha n - bm$, is also a sufficient context, but for $u < a - \alpha$ its coefficient $-(\alpha + u - a)$ is positive and $m, n \to \infty$ (check: $m(30) = 2 \cdot 10^{19}$ at $u = 0.5$); it is outside the bounded class of Proposition 8(b), which is why the windowed context is the one used. The obstruction is intrinsic to the stationary route *at a fixed spectrum*: a designer removes it by choosing $\alpha > \bar{u}$ (here) or $\alpha > \bar{u}\ell_\Omega\kappa$ (Corollary 9), i.e. by a gain that grows with the input bound, which is also what Bernard and Andrieu [3, Theorem 4] require. A second example, the rotating oscillator $\dot{x} = (\omega_0 + u)Jx$, $y = x_1$ ($|x|$ is conserved for every input), with the drift map P and the exact correction $\Omega(z) = PJP^+z$, has error rate $\max \mathrm{Re}\,\lambda(A + uPJP^+)$: stable for all tested $u > -\omega_0$, but unstable ($+2.65$, $+5.16$ at $u = -2, -4$, $\omega_0 = 1$, $n_z = 5$, $\alpha = 1.5$) when the rotation reverses, although the plant is bounded and observable there; this is the input-independence required by the stationary route [3, Theorem 4], which also needs an input-affine plant and a gain growing with $\bar{u}$. The decoder-side route needs only Assumption 1 and a sufficient context.

What the bounds predict, and what the current nodes test. Under protocol v2 no tier changes the input amplitude (every family lives in the same box), so the quantity that varies between tiers is the *distance in context space*, $d(\hat{c}, \mathcal{C}_{\text{train}})$, not $|c|$. (P1) The latent variant's excess error on `shape_ood` and `sequence_ood` is the extrapolation error of Φ at the new contexts, filtered at the measured incremental rate; it degrades sharply only if that rate approaches zero. Decoder-side variants pay $\ell_z(\hat{c})\,|\Delta z|$ and $\ell_c(\hat{c})\,|\Delta c|$ at the new contexts (Corollary 11), with the additive conditioner's ℓ_z small and c-uniform. (P2) Input noise moves the latent variant's operating point to a corrupted context $\hat{c}$ (its margin is evaluated there), whereas decoder-side variants stay stable for every input and pay $\ell_c\ell_E\delta_u$. *Status: untested.* No current node measures μ_2 or the incremental rate on visited segments, $\ell_z(\hat{c})$, $\ell_c(\hat{c})$, or the context distance per tier; under C3-faults `input_noise` every variant, latent included, degrades by about the same factor ($\approx 10\times$ on vdp, read from the C3-faults node report; not a number of the generated results layer), which neither supports nor falsifies (P2). The quantities to record are eval-only on saved checkpoints: $\min_{c' \in \mathcal{C}_{\text{train}}} \|\hat{c} - c'\|$ per tier (C1-where, TEST-C1), the latent margin $\alpha - \kappa\ell_\Phi$ already computed by `metrics.py`, and the realised gains of Definition 28. Measured Lipschitz constants and log-norms are estimates on the visited set (lower bounds of suprema), not certificates.

Corollary 11 (out of distribution: Lipschitz extrapolation, and nothing more)**.** *[proven] Let* $\varepsilon_{\text{train}} := \sup |D(z', E(w')) - x'|$ *over the training support* S_{train} *of triples (label latent* $z' = T^0$*, window* w'*, state* x'*), and* assume *the target* $x = G^*(z,w)$ *of Lemma 5 has, on its domain* S_L*, Lipschitz constants* ℓ *in* z *(this is (A4)) and* ℓ_w *in* w *(sup norm; a new assumption, not implied by sufficiency), and assume its extension beyond* S_L *keeps these constants (a componentwise McShane*

extension achieves them only up to a factor $\sqrt{n_x}$, which then multiplies ℓ and ℓ_w in (a)). (a) For a query (z, w) and any $(z', w') \in S_{\text{train}}$, with D's Jacobian norms ℓ_z^D, ℓ_c^D on the segment,

$$|D(z, E(w)) - G^*(z, w)| \le \varepsilon_{\text{train}} + \tfrac{\ell}{2}\varepsilon(L) + (\ell_z^D + \ell)|z - z'| + (\ell_c^D\, \ell_E + \ell_w)\|w - w'\|_\infty,$$

the $\frac{\ell}{2}\varepsilon(L)$ because $G^*(z', w') = x'$ only at $z' = T_L$, whereas the training latent is T^0; and $|G^*(z(t), u_t) - x(t)| \le \frac{\ell}{2}\varepsilon(L)$ (Lemma 5). [v3-M1-6: stack] For the default, replace z by the stack Z, $|z - z'|$ by $\|Z - Z'\|$ and ℓ by the Lipschitz constant of the target in Z; the proof is the same triangle inequality. (b) The linear rate cannot be improved without structure. Among targets that are 2λ-Lipschitz and agree with a λ-Lipschitz one on S_{train}, $G_\pm = G_0 \pm \lambda\, d(\cdot, S_{\text{train}})\, e$ ($|e| = 1$) both fit the data, so every learner errs by at least $\lambda\, d(q, S_{\text{train}})$ at some query q. At the plant level, if φ is a diffeomorphism of $\mathcal{X}$ with $h \circ \varphi = h$, equal to the identity on an open neighbourhood of the states reached under the training inputs (so that $D\varphi = I$ there), then the plants f and $\varphi_* f$ produce identical training data, and under an input that drives the state where $\varphi \ne \mathrm{id}$ they produce identical (y, u) from x_0 and $\varphi(x_0)$, so any observer errs by at least $\frac{1}{2}|\varphi(x(t)) - x(t)|$ on one of them.

Proof. (a) Triangle inequality through $D(z', E(w'))$, $G^*(z', w')$ and $G^*(T_L(x', w'), w') = x'$, with $|E(w) - E(w')| \le \ell_E\|w - w'\|_\infty$ and $|z' - T_L(x', w')| \le \frac{1}{2}\varepsilon(L)$. (b) $d(\cdot, S)$ is 1-Lipschitz and vanishes on S; $|G_+ - G_-| = 2\lambda d$ and one learner output is at distance $\ge \lambda d$ from one of them. For the plants: φ maps solutions of f to solutions of $\varphi_* f$ and preserves y; on the training set it is the identity. □

What it predicts under protocol v2. Distances are measured in input (window) space so that they are comparable across encoders. In v2 at most 0.1% of evaluation states lie outside the training box on any dev system and tier (C1-where node report; not in the generated results layer), and the input amplitude box is shared by all tiers, so $|z - z'|$ and the state-level construction of (b) are inert: the relevant distance is $\|w - w'\|_\infty$ (equivalently $d(\hat{c}, \mathcal{C}_{\text{train}})$ in context space), which `shape_ood` and `sequence_ood` change through the waveform and the event timing at the same amplitude. The corollary therefore predicts a shape/sequence-OOD excess that is at most linear in that context distance and, by (b), can be no better without structure; it does not predict its size, and the context distance per tier is not yet reported (eval-only; C1-where, TEST-C1). A model-based filter with the correct model extrapolates by structure; a learned observer cannot, beyond this Lipschitz rate.

1.3 The autonomous case

Proposition 12 (no input: a fixed filter and a stationary decoder of the latent window)**.** *[v3-M1-7: new; replaces the zero-input paragraph, which claimed collapse to the autonomous KKL observer and holds only for the C1 arms] [(a), (b), (c) proven; (d) is what transfers, with its hypotheses] Let $u \equiv 0$ (an autonomous plant, or an input held at zero) and $\xi_0 = 0$.*

*(a) (*exact reduction*) $c_k = 0$ for every k, and the deployed observer is $\hat{x}_k = D_0(Z_k)$ with $D_0(Z) := T^+(z_0; [0, g(Z)])$ (z_0 the first block of Z): the fixed input-free latent filter followed by a time-invariant map of the latent window. D_0 is in general neither the phase-1 decoder T_0^+ nor the retrained base: $\Delta W_l(\mathrm{Trunk}([0, g(Z)]))$ vanishes where $g(Z) = 0$ and not in general elsewhere.*

*(b) (*latent window along the solution*) Suppose there is a continuous $T : \mathcal{X} \to \mathbb{R}^{n_z}$ with $\frac{d}{dt}T(x(t)) = AT(x(t)) + Bh(x(t))$ along every solution in $\mathcal{X}$ (the autonomous KKL map; conditions for its existence and injectivity: 2, Theorems 2.2, 2.4, 5, Theorem 3.4). For $t_k \ge S$ let $\mathcal{T}(x(t_k)) := (T(x(t_k)), T(x(t_k - s\,dt)), \ldots, T(x(t_k - S)))$; the entries are values of T along the true past,*

which for an autonomous C^1 flow is a function of $x(t_k)$. Then, with $e_0 := \hat{z}_0 - T(x_0)$, noise-free measurements and $r_K(t) := (\sum_{j<K} e^{-2\alpha(t-js\,dt)})^{1/2}$,

$$\|Z_k - \mathcal{T}(x(t_k))\| \le \kappa\, r_K(t_k)\, \|e_0\| + \sqrt{K}\, e_{\text{disc}}.$$

*(c) (*error bound*) Let $\varepsilon_0 := \sup |D_0(\mathcal{T}(x)) - x|$ over the states $x(t)$, $t \ge S$, of the admissible solutions, and let D_0 be ℓ_Z-Lipschitz on a set containing the segments $[\mathcal{T}(x(t_k)), Z_k]$. Then for $t_k \ge S$*

$$|\hat{x}_k - x(t_k)| \le \varepsilon_0 + \ell_Z\big(\kappa\, r_K(t_k)\, \|e_0\| + \sqrt{K}\, e_{\text{disc}}\big),$$

and $r_K(t) \le e^{-\alpha(t-S)}/\sqrt{1 - e^{-2\alpha s\,dt}}$. Sensor deviations add the terms of Proposition 23 with $\Delta c = 0$.

*(d) (*what transfers from autonomous KKL theory*) Where T exists and is injective with a continuous left inverse T^* (the cited conditions), $\mathcal{T}$ is injective (its first block is) and $T^* \circ \pi_1$ is a stationary left inverse of $\mathcal{T}$ in the class of D_0; so the autonomous existence results apply to the target of D_0, and (c) has the form of the robustness bound of learned KKL observers [19, Lemma 17 and Theorem 19 of arXiv v2; Lemma 6 and Theorem 4 of v3] (their* cond(V) *is our κ), with the stack in place of the latent. What does not transfer is any a priori value of ε_0 or ℓ_Z: the trained D_0 is not $T^* \circ \pi_1$, and ε_0 is a measured quantity, exactly as in the input-driven case.*

Intuition. With no input the SFM context is identically zero, so nothing in the observer depends on time except the latent path; what remains is a KKL observer whose decoder reads a window of its own latent.

Proof. (a) $\xi_k = \sum_{l\ge1}(A_d^{\xi})^{l-1} B_d u_{k-l}/u_{\text{ref}} = 0$; ψ is bias-free with $\tanh(0) = 0$ in every layer, so $c_k = \psi(0) = 0$. The weights are fixed at deployment, so D_0 does not depend on k. The trunk receives $[0, g(Z)]$; V_l is linear and bias-free, so $\Delta W_l = 0$ when Trunk$([0, g(Z)]) = 0$, which holds at $g(Z) = 0$ (bias-free trunk) and not in general otherwise. (b) With noise-free $y = h(x)$, $e(t) := z(t) - T(x(t))$ obeys $\dot{e} = Ae$ for the continuous latent, so $\|e(t)\| \le \kappa e^{-\alpha t}\|e_0\|$; for $t_k \ge S$ every stack entry is at a time $t_k - js\,dt \ge 0$, and the Euclidean norm of the stack is the root sum of squares. The deployed latent is the exact step of Lemma 42(a) with a piecewise-linear y; its difference to the continuous latent from the same initial condition does not depend on that condition (linearity) and is at most e_{disc} per entry (Lemma 42(b)). (c) $|\hat{x}_k - x(t_k)| \le |D_0(Z_k) - D_0(\mathcal{T}(x(t_k)))| + |D_0(\mathcal{T}(x(t_k))) - x(t_k)|$, then (b); the bound on r_K is the geometric series. (d) $T^*(\pi_1\mathcal{T}(x)) = T^*(T(x)) = x$. □

Status of the C1 zero-input claim. The statement "with zero input HyperKKL collapses exactly to the autonomous KKL observer" holds for the C1 arms ($K = 1$, frozen base, $E(0) = 0$; `tests/test_core.py`) and is *false* for the default: on the stored default models (battery and cart-pole, two seeds each) the context is exactly 0 at $u = 0$ while $g(Z) \ne 0$ and $\|\Delta W_l\|_2$ is 0.2 to 204 on random stack probes (check V7b; off-data probes, so these magnitudes are not on-data values). *Check* V6: linear autonomous plant, perturbed stationary decoder of the stack, bound (c) holds (sup ratio 0.04).

1.4 The non-autonomous residual (optional; not in the default)

Remark 13 (the correct physics residual for an input-driven decoder)**.** [v3-M1-8: new; replaces nothing, but no statement in v3 says that a physics term helps] For the autonomous phase-1 map the KKL equation is $\partial_x T_0\, f(x, 0) = AT_0 + Bh$. For an input-driven observer the object that is deployed is the decoder,

and the equation it must satisfy is the *inverse* non-autonomous equation: if $D(Z(t), c(t)) = x(t)$ along trajectories (an exact left inverse), then between samples (input held at u_k)

$$R := \sum_{j=0}^{K-1} \partial_{z^{(j)}} D\left(Az^{(j)} + By(t - j\,s\,dt)\right) + \partial_c D\,\dot{c} - f\big(D(Z,c), u_k\big) = 0, \qquad \dot{c} = \partial_\xi \psi\,(A^\xi \xi + B^\xi u_k/u_{\text{ref}}),$$

where $z^{(j)}$ is the j-th stack entry, each obeying the latent ODE, and the direct path u_k/u_{ref} has zero derivative on the step. [proven: differentiate the identity with the chain rule; f is the true right-hand side.] Both chain-rule terms are needed: for an exact linear decoder the residual is 0 to rounding with them and $O(10^{-1})$ if either the $\dot{c}$ term or the delayed-latent term is dropped (check V5, $K = 1$ and a two-entry FIR decoder). This is the form implemented in `experiments/tv_physics` (`tv.py`). The term is *optional* and is not part of the default. It needs a model of f; with a wrong model its zero is the model's, not the plant's. Evidence (recipe A4+D3, not the default, three seeds): no significant effect in 10/10 cells, also with the true f (`tv_physics/RESULTS.md`); the phase-1 autonomous PDE term was neutral or harmful there. We make no claim that either physics term improves the learned observer.

Numerical checks. [v3-M1-9: v3 checks added] `theory/checks/v3_checks.py`: V5 (residual of Remark 13), V6 and V7 (Proposition 12), V8 (stride rule). `theory/checks/m1_location.py` (written for the $K = 1$ arms): [1] zero-input collapse of all encoders and conditioners (errors $\le 5 \cdot 10^{-16}$); [2] at fixed c, FiLM, LoRA and hypernetwork decoders equal base-architecture networks with explicit weights ($\le 3 \cdot 10^{-13}$); [3] the injection's z-Lipschitz constant is $0, 0.61, 1.22, 2.33, 4.26, 6.82, 7.29$ at $|c| = 0, 0.25, 0.5, 1, 2, 8, 32$ (linear, then saturating), and for two observer copies the measured incremental rate is below the sup log-norm on the visited segments ($-1.52 \le -1.23$; $-1.05 \le -0.29$, where the small-gain margin is -5.47 and certifies nothing; at the largest gain the log-norm is $+8.1$ and also certifies nothing while the copies still contract, so the log-norm condition is sufficient only); [4] the rotating oscillator; [5] *at random initialisation* (trained decoders are not tested here) the z-Jacobian of concat stays below its uniform-in-c bound (≤ 1.25 against 12 for $|c| \le 256$) while FiLM, LoRA and hypernetwork reach 81, $3.7 \cdot 10^5$, $2.3 \cdot 10^6$ and then fall (LoRA and hyper to 0 and 2.9 at $|c| = 256$): c-uniformly bounded, with a far larger constant, as stated in Proposition 8(b); [6] the linear windowed map is affine with w-independent slope to 10^{-15}; [7] the Koopman resolvent identity $p_i = b_i C(F - \lambda_i I)^{-1}$ (to 10^{-16}) and its Laplace-transform form (to 10^{-8}); [8] the a-priori form of Proposition 8(a): with $\mu = 0.46$, $|e(0)| = 0.5$, $\sup|d| = 0.3$ the error never leaves $B(r)$, $r = 1.15$. `theory/checks/check_theory.py`: `where_input_enters` (Proposition 10: measured rate $u - \alpha$ to 10^{-3}; windowed-context error equals $\frac{\ell}{2}\varepsilon(L)$ to three digits at $L = 6$, i.e. the bound is attained for a constant input; the exact context's $m(t)$ reaches 10^{11}–10^{19}).

2 How much memory the observer needs

Setting. Assumptions (A1)–(A4) of Section 1: forward invariance of a compact $\mathcal{X}$, $\|h\| \le \bar{y}$ on $\mathcal{X}$, $\|e^{At}\| \le \kappa e^{-\alpha t}$, and Lipschitz injectivity (modulus ℓ, uniform in $L \ge L_0$) of the windowed map. Here we also use that $-\alpha$ is *exactly* the spectral abscissa of A and that the eigenvalues on $\operatorname{Re}\lambda = -\alpha$ are semisimple (otherwise no finite κ exists for that α). The Butterworth real-Jordan A is normal: each block is $sI + \omega J$, so $\|e^{At}\| = e^{-\alpha t}$ and $\kappa = 1$; its real parts are $\alpha_j = \alpha \sin\frac{(2j-1)\pi}{2n_z} / \sin\frac{\pi}{2n_z}$, $j = 1, \ldots, \lceil n_z/2 \rceil$, so $\alpha = \alpha_1 \le \alpha_j \le \alpha_{\max} = \alpha / \sin\frac{\pi}{2n_z}$. The *label* latent starts at $z(0) = 0$ (the convention of Section 1), hence $z(t) = T^0(x(t), t; u)$ exactly; the deployed observer's initial transient

is a separate term (Proposition 23) and plays no role in this section. $T_L(x, w)$ is the windowed map (1) (zero history before the window), and

$$\varepsilon(L) := \frac{2\kappa\|B\|\bar{y}}{\alpha}\, e^{-\alpha L}.$$

[v3-M2-1: scope] The default decoder reads the latent window Z_k of Definition 1 ($K = 20$ entries at stride s, span $S = (K-1)s\,dt$, $\alpha S \in [5.50, 5.90]$). Lemma 14, Theorem 16(a),(b) and Corollary 17 are statements about decoders $D(z, u_t)$ of the current latent ($K = 1$, the observer of the C2 nodes); Lemma 15, Theorem 16(c) and Proposition 18(d) are about the window.

Lemma 14 (fading-memory tail)**.** *(a) [proven] Along every admissible trajectory and $t \geq L$, $\|T^0(x(t), t; u) - T_L(x(t), u_t)\| \leq \frac{1}{2}\varepsilon(L)$. (b) [proven under its hypothesis] If $u = v$ on $[t-L, t]$ and* both *backward solutions from x stay in $\mathcal{X}$ on $[0, t]$, then*

$$T^0(x,t;u) - T^0(x,t;v) = e^{AL}\big[T^0(\xi, t-L; u) - T^0(\xi, t-L; v)\big], \qquad \|T^0(x,t;u) - T^0(x,t;v)\| \leq \varepsilon(L),$$

with ξ the common backward state at $t - L$. In particular the map forgets at exactly rate α: for a unit left eigenvector p of an eigenvalue with real part $-\alpha$, $|p^*[T^0(x,t;u) - T^0(x,t;v)]| = e^{-\alpha L}\,|p^*[T^0(\xi, t-L; u) - T^0(\xi, t-L; v)]|$.

Intuition. Inside the window the two backward solutions are one curve; everything older reaches the latent through e^{AL}.

Proof. (a) $T^0 - T_L$ is the integral over $[0, t-L]$ of $e^{A(t-s)}Bh(x(s))$ along the true past, which lies in $\mathcal{X}$ by (A1); bound the integrand by $\kappa\|B\|\bar{y}e^{-\alpha(t-s)}$. (b) On $[t-L, t]$ both backward solutions solve the same ODE from x, so they coincide; split (1)-type integrals at $t - L$ and use the flow property: the window parts cancel and the rest is $e^{AL}T^0(\xi, t-L; \cdot)$. Bound each map's remainder as in (a). Finally $p^*e^{AL} = e^{\lambda L}p^*$, $|e^{\lambda L}| = e^{-\alpha L}$. □

No Lipschitz property of f in u is used; the plant enters only through $\bar{y}$. *Where (b) does not apply.* Part (a) uses only the true past and (A1), so it holds for every benchmark. Part (b) needs the *counterfactual* backward solution (from x under v) to stay in $\mathcal{X}$; for a dissipative chaotic flow this fails generically, because backward time expands at the rate λ_b of the most contracting forward direction (Lorenz: $\lambda_b \approx 14.6$, so a counterfactual past leaves any compact set within a fraction of a second). For Lorenz and Rössler, (b) and everything that rests on it (Proposition 6(b)'s $\ell\varepsilon(L)$-sufficiency of a context that determines the window; the lower bounds below) are not available; the sufficient direction (a) is. The zero-order hold changes nothing: if the held samples agree for $j = k-M, \ldots, k-1$, apply the lemma at t_k with $L = M\,dt$ (the sample u_k has not acted yet); the deployed discrete filter has the same bounds with $e^{-\alpha L} \mapsto e^{-\alpha M dt}$ and constants no larger than the continuous ones, plus e_{disc} (Lemma 42).

Lemma 15 (the latent window is an exact finite-memory map)**.** [v3-M2-2: new] *[proven] Let z obey $\dot{z} = Az + By$ on $[0, t]$ from any $z(0)$, with $y = h(x)$ noise-free along an admissible trajectory, and let $0 < T \leq t$. Then*

$$z(t) - e^{AT}z(t-T) = T_T\big(x(t), u_t^{(T)}\big) \qquad \text{exactly,}$$

with $u_t^{(T)}$ the input on $[t-T, t]$ shifted to $[-T, 0]$ and T_T the windowed map (1) at $L = T$. Consequently:

(a) the left side depends neither on $z(0)$ nor on anything before $t - T$: no truncation tail and no initial transient;

(b) *if (A4) holds at $L = T$, then $G^*\big(z(t) - e^{AT}z(t-T), u_t^{(T)}\big) = x(t)$ exactly, so a decoder of the window whose entries include lags 0 and $T = js\,dt$ ($1 \le j \le K-1$) and whose input context determines $u_t^{(T)}$ has zero noise-free error for $t \ge T$;*

(c) *for the deployed (sampled) latent, $\hat{z}_k - A_d^m \hat{z}_{k-m} = \sum_{i=0}^{m-1} A_d^i (W_0 \tilde{y}_{k-1-i} + W_1 \tilde{y}_{k-i})$ exactly (a finite impulse response of the last $m+1$ samples), it is within $(1 + \kappa e^{-\alpha m\,dt})e_{\text{disc}}$ of $T_{m\,dt}(x(t_k), u_{t_k}^{(m\,dt)})$, and a sensor deviation v enters it with gain at most $\kappa\|B\|/\alpha$.*

For a linear plant $T_T(x,w) = P_T x + Q_T w$ with $P_T = P - e^{AT}Pe^{-FT}$ (Proposition 6(a)).

Intuition. The latent is a linear filter: subtracting its own propagated past removes everything older than T, including the unknown initial condition, and leaves the windowed map exactly.

Proof. Variation of constants: $z(t) = e^{AT}z(t-T) + \int_{t-T}^{t} e^{A(t-\tau)}Bh(x(\tau))\,d\tau$. Along the true past, $x(t+\sigma) = X(x(t), 0; \sigma; u_t^{(T)})$ for $\sigma \in [-T, 0]$ (uniqueness of solutions under the hold), and the substitution $\tau = t + \sigma$ gives (1) at $L = T$. (a) is read off. (b) (A4) makes $x \mapsto T_T(x,w)$ injective on $\overline{\mathcal{P}_T}$ with left inverse $G^*(\cdot, w)$ (Lemma 5 with $E = \text{id}$). (c) Unroll the exact step m times. For the defect, let $\delta_k := \hat{z}_k - z(t_k)$ for the continuous latent from the same initial condition; by linearity δ does not depend on that condition, and $\|\delta_k\| \le e_{\text{disc}}$ (Lemma 42(b)); the feature differs from the continuous one by $\delta_k - A_d^m \delta_{k-m}$. The noise gain is a partial sum of the merged kernels of Lemma 42(c). □

What it buys, and what it does not. At the default span the removed tail is the factor $e^{-\alpha S} \in [0.0027, 0.0041]$ of the $K = 1$ tail bound, so in the noise-free bound the gain over $K = 1$ is the transient and the input history older than S, not a large truncation term. With noise, inverting a short window is ill-conditioned: for the linear check plant the white-noise gain of P_T^+ applied to the FIR feature is 5.6 at $\alpha T = 0.3$ (one stride), 1.0 at $\alpha T = 1$, 0.66 at $\alpha T = 2$ and 1.0 at the span (check V1d: not monotone in T; the check's monotone prediction FAILED). The lemma says that the window class *contains* an exact finite-memory decoder; it does not say that the trained decoder uses it. *Check* V1: identity to $4 \cdot 10^{-15}$, independent of $z(0)$ to $3 \cdot 10^{-16}$, the FIR decoder exact to 10^{-14} while the $K = 1$ decoder with the same input window errs by $1.8 \cdot 10^{-2}$.

Theorem 16 (window rule for the observer)**.** *Let $L^*(\eta)$ be the shortest window for which some decoder $D(z, u_t)$ has worst-case error at most η.*

(a) *(*sufficient; every plant under (A1)–(A4)*) [proven] $D = T_L^*$ achieves $\frac{\ell}{2}\varepsilon(L)$, so*

$$L^*(\eta) \;\le\; \frac{1}{\alpha}\ln\frac{\kappa\ell\|B\|\bar{y}}{\alpha\eta} \;=\; \frac{\ln(1/\eta) + \ln\ell + \ln(\kappa\|B\|\bar{y}/\alpha)}{\alpha}.$$

(b) *(*necessary; proven for linear plants; decoders $D(z, u_t)$, $K = 1$*) For every such decoder and every plant,* $\sup\|D - x\| \ge \frac{1}{2}\delta(L)$*, where the fibre diameter $\delta(L)$ is the largest $\|x - x'\|$ over* admissible *pairs with $T^0(x, t; u) = T^0(x', t; v)$ and $u = v$ on $[t-L, t]$ (each state reached at t under its own past). For a linear plant $\dot{x} = Fx + Gu$, $y = Cx$, normal A, (A, PG) controllable, symmetric inputs $|u| \le \bar{u}$, under*

(H1) washout*: $e^{At}Pe^{-Ft} \to 0$, so that after a transient $z = Px + q$ with $PF = AP + BC$, P injective, $q(t) = -\int_{-\infty}^{t} e^{A(t-r)}PGu(r)dr$ (this holds iff α exceeds the fastest* contraction *rate of F; automatically for neutrally stable F);*

(H2) overlapping reachable sets*: for any two admissible input histories and every d with $\|d\| \le 2\kappa R_q e^{-\alpha L}/\sigma_{\min}(P)$ there is a state x reachable at t under the first history with $x+d$ reachable under the second. (Every $x \in \mathcal{X}$ being reachable under every history is* not *required and is false under (A1) whenever the forced response is nonzero: the reachable set under one history is the translate $e^{Ft}\mathcal{X}_0+$(forced response), a strict subset of $\mathcal{X}$. (H2) holds for F skew-symmetric or $F=0$ when $\mathcal{X}_0$ contains a ball whose radius exceeds the forced-response bound plus $\|d\|/2$, as in the oscillator and integrator examples; it fails for contracting F, where the reachable set under a fixed history shrinks to a point and the fibre, hence the necessary window, is* smaller*);*

the fibre diameter satisfies

$$\frac{r_q}{\|P\|}\, e^{-\alpha_k L} \;\le\; \delta(L) \;\le\; \frac{2\kappa R_q}{\sigma_{\min}(P)}\, e^{-\alpha L},$$

where $R_q = \kappa\|PG\|\bar{u}/\alpha$, $r_q > 0$ is the radius of a ball contained in the set Q of reachable q, and α_k is the real part of the first mode group such that the slowest groups up to it span a dimension $\ge n_z - n_x + 1$. Hence $\alpha \le \alpha_k \le \alpha_{\max}$ and

$$\frac{1}{\alpha_k}\ln\frac{r_q}{2\|P\|\eta} \;\le\; L^*(\eta) \;\le\; \frac{1}{\alpha}\ln\frac{\kappa\ell\|B\|\bar{y}}{\alpha\eta}, \qquad L^*(\eta) = \Theta\Big(\frac{1}{\alpha}\ln\frac{1}{\eta}\Big) \quad \textit{(linear plants)},$$

with constants that depend on n_z only through $\alpha_k/\alpha \le 1/\sin\frac{\pi}{2n_z}$ ($\alpha_k/\alpha = 2.618$ at $n_z = 5$, $n_x = 2$; 4.049 at $n_z = 7$, $n_x = 3$).

*(c) (*decoders of the latent window*)* [v3-M2-3: new; (b) does not transfer to the default] *Let $L_K^*(\eta)$ be the shortest input window for which some decoder $D(Z, u_t)$ of the K-entry window has worst-case error at most η. [proven] $L_K^*(\eta) \le L^*(\eta)$ (the class contains the $K=1$ decoders), and if (A4) holds at $L = s\,dt$ then $L_K^*(\eta) \le s\,dt$ for every $\eta > 0$ (Lemma 15(b), noise-free). [proven, linear plants] The fibre over (Z, u_t) is $\{0\}$ whenever $L \ge s\,dt$ and $P_{s\,dt}$ is injective. Hence the lower bound of (b), and the two-sided $\Theta(\ln(1/\eta)/\alpha)$, are statements about $K=1$ decoders only. For $K \ge 2$ the memory requirement that remains in the noise-free setting is Proposition 18(d) ($L < s\,dt$); with noise, it is the information limit of Lemma 31.*

Intuition. The decoder cannot recover what the window does not contain, and what it does not contain reaches the latent through e^{AL}; a decoder may lean on the faster modes, but never faster than $\alpha_{\max}$.

Proof. (a) Lemma 5 with Lemma 14(a). (b) Two points of the fibre receive the same input (z, u_t), so D errs by $\ge \frac{1}{2}\|x - x'\|$ on one of them. Linear case: by (H1) the window parts of q coincide, so $Px+q = Px'+q'$ iff $P(x-x') = e^{AL}(m'-m)$ with $m, m' \in Q$ the history terms at $t-L$; every such $d = x' - x$ obeys the upper bound below, so by (H2) the pair can be realised by admissible states, and $\delta(L) = \sup\{\|d\| : e^{-AL}Pd \in Q - Q\}$. Upper: $\|d\| \le \|Pd\|/\sigma_{\min}(P) \le \|e^{AL}\|\, 2R_q/\sigma_{\min}(P)$. Lower: $\dim \operatorname{range} P = n_x$ and the slow span S_k has dimension $\ge n_z - n_x + 1$, so some unit d has $Pd \in S_k$; for normal A, $\|e^{-AL}Pd\| \le e^{\alpha_k L}\|P\|$, and $Q - Q \supseteq Q \supseteq B(r_q)$ ($0 \in Q$, controllability), so $d\, r_q e^{-\alpha_k L}/\|P\|$ is feasible. □

Proof of (c). The first inequality is inclusion of classes. The second is Lemma 15(b) with $T = s\,dt$. Linear case: two points of the fibre have equal entries 0 and 1 and equal input on $[t - s\,dt, t]$, so by Lemma 15 $P_{s\,dt}(x - x') = 0$, hence $x = x'$. *Check* V2: an explicit pair with equal $z(t)$ and equal

input on the last stride has $|x(t) - x'(t)| = 1.1 \cdot 10^{-3}$ (so the $K = 1$ fibre is not trivial); the second entry differs by $2.3 \cdot 10^{-5}$ and $P^{+}_{s\,dt}$ recovers the state gap to $2 \cdot 10^{-15}$.

What is proven and what is not. The two-sided $\Theta(\ln(1/\eta)/\alpha)$ is a statement about linear plants and $K = 1$ decoders (with the hidden factor α_k/α in the lower constant at $n_z = 5$). For nonlinear plants only (a) is proven; "(b) with rate between α and $\alpha_{\max}$" is a **conjecture**, and for the chaotic benchmarks even the counterfactual-past step is unavailable (Lemma 14). *The fibre is a needle.* Writing $Q = \{\int_0^\infty e^{As} PGu\, ds : |u| \le \bar{u}\}$, a symmetric convex body with support function $h_Q(w) = \bar{u} \int_0^\infty |w^\top e^{As} PG|\, ds$, the fibre diameter is computed exactly as $\delta(L) = \max_{|d|=1} 2/g_Q(e^{-AL}Pd)$ with g_Q the gauge of Q (a linear programme on a Riemann discretisation of Q). For the oscillator with $n_x = 2$ the maximiser is the single direction d with $Pd \in S_k$: $\delta(L)$ decays at rate 1.000α ($n_z = 3$) and 2.617α ($n_z = 5$; $\alpha_k = 2.618\alpha$), every other direction decays at $\approx \alpha_{\max}$, and the angular half-width of the needle shrinks from 3° to 1° over $L = 2$–6 (Table 2). An RMS error curve averages over directions, so a sweep's knee on a log scale is expected between the slopes $-\alpha_k$ and $-\alpha_{\max}$, with the slow needle visible only at the largest L. *Tested by* (all with the $K = 1$ recipe A0, not the default) C2-window-a1 (six dev systems, one bandwidth; pre-registered rule "$e(L \ge \ln(1/\eta)/\alpha)$ within 5% of $e(512)$ on $\ge 5/6$": **FAIL**, `id` 2/5, `sequence_ood` 3/5) and C2-window-a2 (battery, duffing, vdp at 2α, the only test of the $1/\alpha$ scaling; rule "knee in samples changes by $\le 2\times$ on $\ge 2/3$": **FAIL**, 1/2). The ℓ-free rule that was pre-registered is the sufficient bound with ℓ dropped; the corollary below restores ℓ and is *post hoc* consistent with the largest failure (battery), not with lorenz (Table 1). Source: `paper/generated/RESULTS.md`.

Corollary 17 (practical window rule, and why one plant needs more)**.** *[proven: (a) of the theorem, rewritten; its use as an explanation of the sweep is* post hoc*, not pre-registered; sufficient also for the default, whose class contains the $K = 1$ decoders [v3-M2-4: scope]] For a target relative error η the window that is* sufficient *is*

$$L^*(\eta) \;\approx\; \frac{\ln(1/\eta) + \ln \ell}{\alpha} \qquad \textit{(plus the design constant } \ln(\kappa \|B\| \bar{y}/\alpha)/\alpha\textit{)},$$

where ℓ is the Lipschitz constant of the inverse of the windowed map (A4). The $\ln(1/\eta)/\alpha$ part is plant-independent; the plant enters through $\ln \ell$ and, in the design constant, through $\bar{y}$. The lower bound of the theorem (linear plants) can only shorten *the necessary window (rate $\alpha_k \ge \alpha$), so a plant whose sufficient window exceeds $\ln(1/\eta)/\alpha$ does so through ℓ (or $\bar{y}$), not through the fading-memory rate.*

Integrator-type states (coulomb counting; a heuristic reading, not a proven statement). If a state enters the output only weakly, as the battery's state of charge enters the terminal voltage through the open-circuit-voltage slope $\partial V_{\mathrm{oc}}/\partial \mathrm{SOC}$, then the latent carries that state at gain $\propto$ slope and inverting it costs $\ell \propto 1/\text{slope}$: large ℓ, long L^*. The state itself is the integral of the input (coulomb counting), so a decoder must integrate the window, which is possible only over the window's length; the memory needed is $\ln \ell/\alpha$ longer than for a well-observed state. This is the sufficient side of the theorem; the digest's earlier attribution of the battery to "the Lemma 1′ worst case" is withdrawn: Lemma 14(b) is plant-independent (rate exactly α for every plant) and the lower bound at $n_z - n_x + 1 = n_z$ (scalar integrator) gives $\alpha_k = \alpha_{\max}$, the *shortest* necessary window.
Proxy for ℓ. No node measures ℓ itself; the phase-1 diagnostic `1_dec_q95` (random-direction quotient of the *stationary* base decoder, $\approx \|J\|_F/\sqrt{n_z}$, Remark 29) is used as its proxy in Table 1. It is an estimate on data of a related constant, not the constant of the bound.
Reading the table (post hoc). The pre-registered ℓ-free rule fails where ℓ is largest (battery, $1.73\times$ the $L = 512$ error at $L = 256$) and marginally on lorenz (1.07). With $\ln \ell$ restored the rule can be evaluated only on bioreactor, lorenz and vdp: on battery, duffing and quad_tank the refined

Table 1: Post-hoc reading of the C2-window-a1 sweep (lora, `sequence_ood`, NRMSE at $L \in \{1, \dots, 512\}$ samples, $\eta = 0.05$). $L_0 = \ln(1/\eta)/(\alpha dt)$ is the pre-registered ℓ-free rule and $L^* = (\ln(1/\eta) + \ln \ell)/(\alpha dt)$ the rule of Corollary 17 with $\ell =$ `l_dec_q95` of the node's own phase 1 (both in samples; the design constant is omitted; not pre-registered). "ratio" is $\max_{L \geq L_\cdot} e(L)/e(512)$ on the dyadic grid; the node's criterion is ≤ 1.05; n.t. = not testable (no grid window in $[L_\cdot, 512)$, so the ratio is $e(512)/e(512)$ and says nothing; the results layer leaves these out). ℓ and L^* are the results-layer values. Monotone: no window worse than a shorter one by $> 10\%$.

system	n_z	α	αdt	ℓ	L_0	L^*	ratio, $L \geq L_0$	ratio, $L \geq L^*$	monotone
battery	7	1.5	0.0223	806	135	435	1.73 (fails)	n.t.	yes
bioreactor	5	0.5	0.0309	44	97	219	1.00	1.00	no (min at 128; 2.4× worse at 512)
duffing	5	3.0	0.0309	295	97	281	1.00	n.t.	yes
lorenz	7	2.0	0.0223	10	135	239	1.07 (fails)	1.07 (fails)	yes
quad_tank	6	0.1	0.0050	14	599	1121	n.t.	n.t.	yes (falling at 512)
vdp	5	1.5	0.0309	23	97	199	1.00	1.00	yes

L^* leaves no grid window below 512, so the refined rule is *untested* there (it does not "hold"); it holds on bioreactor and vdp and still fails by 7% on lorenz (2/3 on `id` and 2/3 on `sequence_ood` in the results layer). For battery the refined rule only says that its failure at $L \geq L_0$ is not in contradiction with the theory. This is what a sufficient-side statement can do: the measured knee sits at or below it (duffing's knee is at 32 samples against $L^* = 281$; the bound is far from tight there), and it does not predict the knee. Bioreactor is non-monotone (longer windows are *worse*: an optimisation or over-fitting effect that no truncation bound describes) and quad_tank's fixed $dt = 0.05$ puts both rules beyond the grid, as predicted below. The refined rule is offered as the theory's explanation of why the pre-registered test failed, not as a passed test; a pre-registered test of it needs a new node. The $L = 1$ row of the same sweep (current input only, Proposition 18(a)) is +143% to +1522% worse than $L = 512$ on every dev system (`sequence_ood`, lora; paired changes from the C2-window-a1 node report, not a number of the generated results layer).

Proposition 18 (the current input is not enough)**.** *(a) [proven] Under the zero-order hold, $T^0(\cdot, t_k; u)$ does not depend on u_k. (b) [proven, linear SISO plants] For the linear plant of Theorem 16 with $n_y = n_u = 1$ under (H1)–(H2), but without assuming (A, PG) controllable, (z, u_t) determines x iff* range $P \cap \langle A \,|\, PG \rangle = \{0\}$*, for every $L \geq 0$. If (A, B) is controllable, the eigenvalues of A are distinct and none is a transmission zero of $H(s) = C(sI - F)^{-1}G$, then $\langle A|PG\rangle = \mathbb{R}^{n_z}$ and the condition fails* for every *n_z: no latent dimension makes the latent store the history. (For $n_y > 1$ the condition $p_i^* BH(\lambda_i) \neq 0$ is stronger than "not a transmission zero".) (c) [proven, local; $K = 1$] Parametrise the input before $t - L$ by $\theta \in \mathbb{R}^p$ (e.g. held levels in the interior of U), keep the window fixed, and let $\Psi(x, \theta)$ be the resulting latent at t. If Ψ is C^1 near $(\bar{x}, \bar{\theta})$ with trajectories in the interior of $\mathcal{X}$ and* rank $\partial_\theta \Psi(\bar{x}, \bar{\theta}) = n_z$*, then every x in a neighbourhood of $\bar{x}$ has a past $\theta(x)$ with the same latent and the same window; any decoder $D(z, u_t)$ errs by at least half that neighbourhood's radius. (d)* [v3-M2-5: new] *[proven, local; latent window, $L < s\,dt$] Keep the input before $t - s\,dt$ and on $[t - L, t]$ fixed and parametrise the $N = s - N_L$ held samples on $[t - s\,dt, t - L)$ by θ in the interior of U^N, with trajectories in the interior of $\mathcal{X}$. All stack entries other than $z(t)$ lie at or before $t - s\,dt$ and do not depend on θ. Let $\Psi(\theta) = (x(t;\theta), z(t;\theta))$ be C^1 near $\bar{\theta}$. If* rank $\partial_\theta z(t; \bar{\theta}) = n_z$ *and $\partial_\theta x(t; \bar{\theta})$ does not vanish on* ker $\partial_\theta z(t; \bar{\theta})$*, then there is a C^1 curve $\theta(\tau)$, $\theta(0) = \bar{\theta}$, along which the whole stack Z and the input window are constant while $x(t)$ moves with nonzero speed; every decoder $D(Z, u_t)$ errs by at least half of that displacement on one of the two ends. In particular the current input alone ($L = 0$) is not enough for the default either.*

Intuition. With a time-varying map the past input is part of what must be inverted, and the

latent alone cannot hold it.

Proof. (a) The backward solution on $(-\infty, t_k]$ uses u on $[t_j, t_{j+1})$, $j \le k-1$, only. (b) By Theorem 16(b), ambiguity exists iff $Pd = e^{AL}\Delta m$ for some $d \neq 0$ and admissible Δm; the reachable differences form a neighbourhood of 0 in $\langle A|PG\rangle$, which e^{AL} maps onto itself. For distinct eigenvalues, controllability of (A, PG) is $p_i^* PG \neq 0$ for every left eigenvector, and $p_i^* PG = -p_i^* B\, H(\lambda_i)$ (from $p_i^* P(F - \lambda_i I) = p_i^* BC$). (c) Implicit function theorem: choose n_z components of θ with invertible $\partial\Psi$-block and solve $\Psi(x, \theta) = \Psi(\bar{x}, \bar{\theta})$ for them as a C^1 function of x. (d) A held sample on $[t_i, t_{i+1})$ acts only on states after t_i, so entries at $t - js\,dt$, $j \ge 1$, do not depend on θ. By the rank condition $\theta \mapsto z(t;\theta)$ is a submersion near $\bar{\theta}$, so its level set through $\bar{\theta}$ is a C^1 manifold with tangent space $\ker \partial_\theta z(t;\bar{\theta})$; take a curve in it with tangent v, $\partial_\theta x\, v \neq 0$. The decoder receives identical inputs at both ends. □

Check V3 (linear plant, stride 10): pairs with identical stack ($|\Delta Z| \le 10^{-16}$) and input window and $|\Delta x(t)| = 9.1 \cdot 10^{-3}, 3.9 \cdot 10^{-3}, 2.4 \cdot 10^{-4}$ at $L = 0, 1, 3$ samples for input gaps ≤ 0.5: the ambiguity exists but shrinks fast as L approaches one stride; Van der Pol: a constructed pair with identical latent at t and $|\Delta x(t)| = 4.9 \cdot 10^{-4}$. The size of the ambiguity is not bounded below by the proposition. The rank condition in (c) is checkable on a simulator: for Van der Pol with the saturating actuator, twelve held levels before the window give $\operatorname{rank} \partial_\theta \Psi = 5 = n_z$ (smallest singular value $1.6 \cdot 10^{-4}$: the check shows existence of the ambiguity, not its size, which is $\sigma_{\min}\times$ the admissible range of θ). Example: the oscillator $\dot{x}_1 = \omega_0 x_2$, $\dot{x}_2 = -\omega_0 x_1 + u$, $y = x_1$, $n_z = n_x = 2$, with $v \equiv 0$, $u = a\mathbf{1}_{(-\infty, t-L)}$: x and $x' = x - P^{-1}e^{AL}A^{-1}PGa$ share latent, window and current input. On the stationary route the current input does suffice (the injection $\Phi(z, u)$ depends on $u(t)$ only; Section 1); the price there is a small-gain margin (Proposition 8). *Tested by* the $L = 1$ row of C2-window-a1 (above). No current node compares the context encoders (current, window, GRU, SFM) at a matched budget; C1-ctx only moves the SFM's parameters into the budget and does not test this proposition.

2.1 Predictions for the window sweeps (the $K = 1$ recipe of the C2 nodes)

With $dt = 0.1 \sin(\pi/2n_z)/\alpha$, $\alpha\, dt$ does not depend on the bandwidth scale s, and a window of L samples covers $(L-1)\, dt$ seconds of relevant history (Proposition 18a).

1. The knee in seconds scales as $1/(s\alpha)$; in *samples* it is the same for all s (vdp, duffing, battery in C2-window-a2). One e-fold of the rate-α bound is $1/(\alpha dt) = 32.4$ samples ($n_z = 5$) or 44.9 ($n_z = 7$); at rate α_k it is α_k/α times fewer (2.6 at $n_z = 5$, $n_x = 2$; 4.0 for lorenz, $n_z = 7$, $n_x = 3$; the `a2/a` column of `c2_predictions` is the second-slowest mode group, which equals α_k only when $n_z - n_x + 1 \le 4$). The worst-case window of Theorem 16(a) at $\eta = 10^{-3} z_{\text{scale}}$ (ℓ omitted) is $\approx 311, 313, 320, 467$ samples; measured tails are 10^2–$5 \cdot 10^2$ times below the bound, i.e. the bound overstates the window by $\ln(485)/(\alpha\, dt) \approx 200$ samples. At one bandwidth (C2-window-a1) the knees within 5% of $e(512)$ are 32 (vdp, `id` and `sequence_ood`; duffing and bioreactor on `sequence_ood`), 64 (bioreactor `id`), 128 (quad_tank `id`) and not reached on the grid for battery and lorenz (both tiers), duffing (`id`) and quad_tank (`sequence_ood`); see Table 1. At 2α (C2-window-a2) the knee in samples moved by $\le 2\times$ on only 1/2 evaluable systems: the $1/\alpha$ scaling is *not* supported by the current data.

2. quad_tank has a fixed $dt = 0.05$: $\alpha dt = 0.0025, 0.005, 0.01$ for $s = \frac{1}{2}, 1, 2$; the knee in samples scales as $1/s$ and the grid ($L \le 512$) spans only $1.3, 2.6, 5.1$ e-folds, so the curve should still fall at $L = 512$ for $s \le 1$ (it does at $s = 1$): this plant cannot test the scaling.

3. On a dyadic grid "knee×α varies by less than 2×" holds only if the knee sits at the same grid point for every s (neighbours differ by exactly 2); use "$\le 2\times$" or a knee interpolated in $\log L$.

2.2 Context encoders: forgetting by construction

Lemma 19 (SFM context)**.** *[proven] Let $\xi_{k+1} = A_d^\xi \xi_k + B_d u_k/u_{\text{ref}}$, $c_k = \psi([\xi_k, u_k/u_{\text{ref}}])$, with A^ξ block-normal, $s_b = -\text{softplus}(\cdot) < 0$, $\alpha_\xi = \min_b |s_b|$, and ψ ℓ_ψ-Lipschitz. Then for $l \ge 1$*

$$\textit{(a)}\ \Big\|\frac{\partial c_k}{\partial u_{k-l}}\Big\| \le \frac{\ell_\psi}{u_{\text{ref}}}\|B_d\| e^{-\alpha_\xi (l-1)dt}, \qquad \textit{(b)}\ \ell_E \le \frac{\ell_\psi}{u_{\text{ref}}}\Big(1 + \frac{\|B_d\|}{1 - e^{-\alpha_\xi dt}}\Big),$$

*and the per-lag bound is attained (ψ isometric on the slowest block, B_d in that block). (c) (*the memory probe*) If two input records agree on the last L samples and differ arbitrarily before, with $|u| \le \bar{u}$,*

$$\|c_k - c_k'\| \le \frac{\ell_\psi \|B_d\|\, 2\bar{u}}{u_{\text{ref}}} \frac{e^{-\alpha_\xi L\, dt}}{1 - e^{-\alpha_\xi dt}}.$$

Proof. $\xi_k = \sum_{l\ge 1} (A_d^\xi)^{l-1} B_d u_{k-l}/u_{\text{ref}}$, $\|(A_d^\xi)^{l-1}\| = e^{-\alpha_\xi (l-1)dt}$; chain rule. For ℓ_E add the direct path u_k and sum the geometric series. (c) Only lags $l \ge L+1$ differ; sum (a) over them. □

The rate exists for every parameter value but is not bounded below uniformly (softplus→ 0); a floor $s_b \le -\alpha_{\min}$ makes it uniform (the C2-floor nodes set $\alpha_\xi \ge c\,\alpha$, $c \in \{0.25, 0.5, 1\}$).

What the probe measures. The C1-where and C2-floor memory probes keep the last L inputs, swap everything older, and fit a rate to the decoded change $|\Delta\hat{x}| \le \ell_c \|c - c'\|$. This is the quantity of (c): a test of the *encoder's* memory, not of Lemma 14 (the latent z is unchanged by the probe). Since (c) is an upper envelope, a fitted rate below α_ξ over a finite L range does not contradict it unless the curve is in its exponential regime over several decades; the probe tests the rate only there.

At initialisation the SFM's fill-in poles are $\alpha/1.5^b$, $b = 1, \dots, 5$, so its slowest mode is $\alpha/1.5^5 = 0.132\alpha$ and the next 0.198α: at initialisation the SFM forgets $\approx 7.6\times$ more slowly than the latent. *Measured (un-floored, C1-where, lora, median over seeds; `paper/generated/RESULTS.md`):* the learned SFM's slowest pole is 0.14–0.18 α (battery 0.18, bioreactor 0.14, duffing 0.14, lorenz 0.18, quad_tank 0.17, vdp 0.15), i.e. between the two slowest *initial* poles: training moves the slowest pole only slightly [v3-M2-6: default values] (recipe A0; in the stored default models the slowest learned pole is 0.21α on battery and 0.30–0.33α on cartpole, two seeds each, check V7) (by at most $\approx 40\%$, never past the second initial pole), so the "5–7× extra memory" of the un-floored SFM is largely inherited from initialisation rather than learned. *The floor intervention (C2-floor-025/05/1) fails its pre-registered rule.* At 0.5α: no significant loss on 4/6 dev systems (rule: $\ge 5/6$; battery +23–36% and bioreactor lose), TOST non-inferiority at ±10% on 0/6, and the strict probe check "fitted rate ≥ floor" on 0/6 (ratios 0.73–0.96: battery 0.78, bioreactor 0.77, duffing 0.73, quad_tank 0.96, vdp 0.82; lorenz −0.02, flat). At 0.25α: no significant loss on 6/6 but TOST 3/6 and probe 1/6; at 1α: 4/6, 0/6, 0/6 (battery $\approx 2\times$ worse). So the data are *consistent with* the memory being unused on four systems but do not establish it (three seeds; see the TOST counts), and battery, the large-ℓ plant of Corollary 17, is the one system where a floor at 0.5α clearly costs accuracy, as the corollary says it should (its sufficient window $\ln(1/\eta)/\alpha + \ln \ell/\alpha$ is attenuated by $e^{-0.5\alpha L^*}$ under the floor); the bioreactor loss has no such explanation. The probe ratios below one do not contradict Lemma 19(c), which is an upper envelope, but they mean the pre-registered check of its *rate* fails on every system, and the flat lorenz curve ($d(L)$ within ×2–3 over $L \le 5.7$ s) is unexplained: a probe outside its exponential regime, to be reported as such. For the window prediction the SFM has no

truncation loss on the latent's time scale (it should match the window encoder's plateau beyond the knee, not its knee), but an input glitch persists 7.6× longer in c and ℓ_E is correspondingly larger ($\|B_d\|/(1-e^{-\alpha_\xi dt}) \approx 7.6/(\alpha dt) \approx 246\|B_d\|$ for $n_z = 5$).

Remark 20 (SFM is an exact context for linear plants)**.** If $\mathrm{spec}(A^\xi) \supseteq \mathrm{spec}(A)$ with nonzero input weights (true at initialisation), then $q(t_k) = -\sum_{i<k} A_d^{k-1-i} A^{-1}(A_d - I)PGu_i$ is a fixed linear function of ξ_k (each 2×2 block commutes with J): no truncation, whereas a window of L samples truncates at $e^{-\alpha(L-1)dt}$. Once the poles are trained this holds only approximately.

Remark 21 (a GRU has no such guarantee)**.** (1) An update-gate bias b gives $h_k = \sigma(b)h_{k-1} + \ldots$, forgetting at rate $-\ln\sigma(b) \to 0$ as $b \to \infty$. (2) No fading memory in the sense of Boyd and Chua [4]: with reset gate $\equiv 1$ and update gate $\equiv 0$, $h_{k+1} = \tanh(3h_k + 2u_k)$ is bistable; one pulse flips the state and the context difference stays 1.99 for all later k, although the two inputs agree after the pulse. Contractivity of recurrent networks needs weight constraints [18]; the SFM has it by construction. *Tested by* no current node (an encoder ablation SFM vs GRU at a matched budget is requested; C1-ctx does not contain a GRU variant).

Table 2: Numerical checks of this section (CPU).

result	check	outcome
Lemma 14	integrator; oscillator $n_z = 5$; Van der Pol	rate $1.5000 = \alpha$; $1.0000 = \alpha$ (both); bound loose 135–485×
Thm. 16(b)	$\delta(L)$ by gauge/LP, oscillator $n_x = 2$	rate $1.000, 2.617$ at $n_z = 3, 5$ ($\alpha_k = 1, 2.618$); needle at predicted angle
	bounds $r_q e^{-\alpha_k L}/\|P\| \le \delta \le 2R_q e^{-\alpha L}/\sigma_{\min}$	hold at every L (slack 21×, 1.5–106×)
	$\sigma_{\min}(e^{-AL}P)$ (arithmetic)	same rates $1.000, 2.618$
Cor. 17	Table 1 from the C2-window-a1 report	ℓ-free rule fails 1.73× (battery); with $\ln\ell$: holds on 2/3 testable systems (lorenz 1.07), untestable on battery, duffing, quad_tank
Prop. 18(c)	Van der Pol, 12 past levels	$\mathrm{rank}\,\partial_\theta\Psi = 5$
sampled form	ZOH step identity; discrete constant	exact to 10^{-15}; $c_d = 0.985 \le 1/\alpha$
Lemma 19(a,b)	tight SFM; random SFM	ratio 1.0000000000; ≤ 0.13
Lemma 19(c)	history swap, floor 0 and 0.5; initial poles	observed/bound ≤ 0.01; $\alpha_\xi = \alpha/1.5^5$ at init, $\ge 0.5\alpha$ with floor
GRU remark	latch in the code's GRU encoder	difference 1.990 at $k = 100, 1000, 4999$
Lemma 15 (v3)	V1, linear plant, $K = 20$	identity $4\cdot 10^{-15}$; FIR decoder exact; noise gain not monotone in T
Thm. 16(c) (v3)	V2, explicit $K = 1$ fibre pair	$\|\Delta x\| = 1.1\cdot 10^{-3}$, separated by the stack
Prop. 18(d) (v3)	V3, linear and Van der Pol	same stack and window, $\|\Delta x\| > 0$

3 Error bound for the deployed observer

Setting. Plant $\dot{x} = f(x,u)$, $y = h(x)$, input held on $[t_k, t_{k+1})$, $t_k = k\,dt$. The decoder-side observer, exactly as deployed, is

$$\hat{z}_{k+1} = A_d\hat{z}_k + W_0\tilde{y}_k + W_1\tilde{y}_{k+1}, \qquad \hat{x}_k = D(\hat{Z}_k;\hat{c}_k), \qquad \hat{c}_k = E(\tilde{u}_0,\ldots,\tilde{u}_k), \tag{2}$$

[v3-M3-1: stacked latent] with $\hat{Z}_k$ the latent window of Definition 1 (K entries at stride s, zero before the start) and $D(Z;c) = T^+(z_0;[c,g(Z)])$; for $K = 1$, $D(Z;c) = \hat{T}^+(\hat{z}_k;\hat{c}_k)$ as before. with A_d, W_0, W_1 the exact step of Lemma 42(a), the context held between samples, and $\hat{z}_0$ a fixed function of the

prior $\hat{x}_0$. Its response to a sensor sequence (v_k) is the *latent noise filter*, the sampled form of $(sI - A)^{-1}B$:

$$H_0 = W_1, \qquad H_m = A_d^{m-1}W_0 + A_d^m W_1 \ (m \geq 1), \qquad \gamma_y := \sum_{m\geq 0} \|H_m\| \leq \frac{\kappa\|B\|}{\alpha}, \qquad \|H\|_2^2 := \sum_{m\geq 0} \|H_m\|_F^2, \tag{3}$$

where the inequality holds at every dt (Lemma 42(c)).

Reference condition. Every bound is stated relative to the *nominal* deployed observer $\hat{x}_k^{\mathrm{nom}}$: the same episode, the nominal sensor model (the one used in training, `data.sensor`), the true input record and the nominal prior. Deviations are $v_k := \tilde{y}_k - \tilde{y}_k^{\mathrm{nom}}$ (sensor faults, extra noise), $\hat{c}_k - c_k^{\mathrm{nom}}$ (input corruption) and $\hat{z}_0 - \hat{z}_0^{\mathrm{nom}}$ (initialisation). This is the condition the decoder was trained in: phase 2 regresses x_k on the deployed latent of the nominal-noise record, so its validation error measures the nominal error directly, and no noise-free quantity is needed.

Assumptions. (A1)–(A4) and the regularity clause ($f(\cdot, u) \in C^1$, $h \in C^2$, hence $M_2 := \sup\|\ddot{y}\|$, $L_y := \sup\|\dot{y}\|$ on each hold interval) of Assumption 1, and

(L) $D \in C^1$ in (Z, c) (for $K = 1$: $\hat{T}^+ \in C^1$), and the context encoder satisfies $\|E(\tilde{u}_{0:k}) - E(u_{0:k})\| \leq \ell_E \sup_{j\leq k}\|\tilde{u}_j - u_j\|$ (for SFM, $\ell_E \leq (\ell_\psi/u_{\mathrm{ref}})(1 + \|B_d\|/(1 - e^{-\alpha_\xi dt}))$, Lemma 19).

Write ℓ_z, ℓ_c for the sup of $\|\partial_Z D\|$ (Euclidean norm of the stack; $\|\partial_z \hat{T}^+\|$ for $K = 1$), $\|\partial_c D\|$ over the segments a given episode traverses (defined in Proposition 23), and $\ell_z^{\mathrm{g}}, \ell_c^{\mathrm{g}}$ for the global constants (finite for tanh/SiLU networks on a bounded context set, for every conditioner; for FiLM, LoRA and hyper the available bound is a product of layer norms that is orders of magnitude larger, Proposition 8(b); with the stack, ℓ_z^{g} is the constant in the Euclidean norm of Z). The latent deviations are linear, input-free and decoder-independent, so they are fixed before any Lipschitz constant is used.

Definition 22 (nominal error, held-out quantile)**.** For an episode $\omega \sim \mathcal{D}$ (initial state and input path from the training episode distribution; for the default, the mixture $D_0 \cup D_1$ of standard and event-structured histories [v3-M3-2: mixed data]) let $e_{\mathrm{nom}}(\omega) := \sup_{k\in S}\|\hat{x}_k^{\mathrm{nom}} - x(t_k)\|$, S the steady window, and let ε^β be its $(1-\beta)$-quantile. For n i.i.d. held-out episodes, $\hat{\varepsilon} := \max_i e_{\mathrm{nom}}(\omega_i)$ satisfies

$$\mathbb{P}\big(\hat{\varepsilon} \geq \varepsilon^\beta\big) \geq 1 - (1-\beta)^n \qquad (n = 64,\ \beta = 0.05 :\ 0.96), \tag{4}$$

because all n draws fall below the quantile with probability at most $(1-\beta)^n$.

ε^β contains everything the nominal observer gets wrong: approximation, window truncation (by Lemma 14(a) and Theorem 16(a) at most $\frac{\ell}{2}\varepsilon(L)$ for a sufficient context), nominal noise and the discretisation (Lemma 42(b) only relates the deployed latent to the continuous map, $\|\bar{z}_k - z(t_k)\| \leq e_{\mathrm{disc}} = O(dt^2)$). It is a quantile with distribution-free confidence, not a supremum, and only for $\mathcal{D}$. It is defined on the *true* plant's episodes (phase 2 regresses true-plant state labels on the deployed latent): no model of f enters it, which is the point of Proposition 27.

Proposition 23 (decoder-side observer: ISS relative to the nominal observer)**.** *[v3-M3-3: stacked latent: (ii)–(iv) restated with the window; (i) unchanged] [(i), (ii), (iv) proven and exact; (iii) proven as a decomposition, see the paragraph after the proof] Assume (A1)–(A4), (L). Let* $\Delta z_k := \hat{z}_k - \hat{z}_k^{\mathrm{nom}}$ *(*$:= 0$ *for* $k < 0$*),* $\Delta Z_k := \hat{Z}_k - \hat{Z}_k^{\mathrm{nom}}$ *the stack of* Δz_{k-js}*,* $j < K$*, and* $\Delta c_k := \hat{c}_k - c_k$*.*

(i) *(latent deviation, exact)* $\Delta z_k = A_d^k \Delta z_0 + \sum_{i\le k} G_i^{(k)} v_i$ *with* $G_i^{(k)} = H_{k-i}$ *(*$i \ge 1$*),* $G_0^{(k)} = A_d^{k-1} W_0$*; hence* $\|\Delta z_k\| \le \kappa e^{-\alpha t_k}\|\Delta z_0\| + \frac{\kappa\|B\|}{\alpha}\sup_{i\le k}\|v_i\|$ *for every* k*, and* $\limsup_k \|\Delta z_k\| \le \gamma_y \limsup_k \|v_k\|$.

(ii) *(path identity, exact)*

$$\hat{x}_k - x(t_k) = \big(\hat{x}_k^{\mathrm{nom}} - x(t_k)\big) + \bar{J}_{Z,k}\Delta Z_k + \bar{J}_{c,k}\Delta c_k, \tag{5}$$

$\bar{J}_{Z,k} = \int_0^1 \partial_Z D(\hat{Z}_k^{\mathrm{nom}} + s\Delta Z_k; \hat{c}_k)ds$, $\bar{J}_{c,k} = \int_0^1 \partial_c D(\hat{Z}_k^{\mathrm{nom}}; c_k + s\Delta c_k)ds$.

(iii) *(ISS bound) If* $\|v_k\| \le \delta_y$ *and* $\|\tilde{u}_k - u_k\| \le \delta_u$*, then for every* k*, with* n_k *the number of stack entries with nonnegative index and* $r_K^{(k)} := (\sum_{j<K,\, k-js\ge 0} e^{-2\alpha(t_k - js\,dt)})^{1/2}$,

$$\|\hat{x}_k - x(t_k)\| \le \|\hat{x}_k^{\mathrm{nom}} - x(t_k)\| + \ell_z\Big(\kappa\, r_K^{(k)}\|\Delta z_0\| + \sqrt{n_k}\,\frac{\kappa\|B\|}{\alpha}\delta_y\Big) + \ell_c\,\ell_E\,\delta_u, \tag{6}$$

with ℓ_z, ℓ_c *the sup of the Jacobian norms on the traversed segments* $[\hat{Z}_k^{\mathrm{nom}}, \hat{Z}_k] \times \{\hat{c}_k\}$ *and* $\{\hat{Z}_k^{\mathrm{nom}}\} \times [c_k, \hat{c}_k]$*; on the steady window the first term is* $\le \varepsilon^\beta$ *for a* $(1-\beta)$ *fraction of episodes, and in the limsup* $\kappa\|B\|/\alpha$ *may be replaced by the exact* γ_y*.* $r_K^{(k)} \le e^{-\alpha(t_k - S)}/\sqrt{1 - e^{-2\alpha s\,dt}}$ *for* $t_k \ge S$*; for* $K = 1$*,* $r_1^{(k)} = e^{-\alpha t_k}$ *and* $n_k = 1$*, the bound of v2.*

(iv) *(stochastic deviations) For zero-mean i.i.d.* v_k *with covariance* $\sigma_v^2 I$*,* $\limsup_k(\mathbb{E}\|\hat{x}_k - x(t_k)\|^2)^{1/2} \le \limsup_k(\mathbb{E}\|\hat{x}_k^{\mathrm{nom}} - x(t_k)\|^2)^{1/2} + \ell_z^{\mathrm{g}}\sigma_v\sqrt{K}\|H\|_2 + \ell_c\ell_E\delta_u$*, since* $\lim_k \mathbb{E}\|\Delta Z_k\|^2 = K\sigma_v^2\|H\|_2^2$ *exactly (each entry has the same limit second moment; the Euclidean norm of the stack ignores the correlation between entries);* $\|H\|_2^2 = dt\,\mathrm{tr}W_c + o(dt)$*,* $AW_c + W_cA^\top + BB^\top = 0$*. The global constant is needed because unbounded noise leaves every neighbourhood of the data with positive probability.*

Intuition. The learned map never sees raw measurements: sensor deviations and initialisation reach it through a known, input-free stable filter, a wrong input record moves only the context, and each deviation is multiplied by the decoder's slope along the path it actually travels; the latent window only repeats the filter output at K lags.

Proof. (i) $\hat{z}$ and $\hat{z}^{\mathrm{nom}}$ obey the same linear recursion with forcing difference $W_0 v_k + W_1 v_{k+1}$; unroll it, and use $\|A_d^k\| \le \kappa e^{-\alpha t_k}$ and $\sum_i \|G_i^{(k)}\| \le \kappa\|B\|/\alpha$ (Lemma 42(c)). (ii) Split $D(\hat{Z};\hat{c}) - D(\hat{Z}^{\mathrm{nom}}; c)$ at $D(\hat{Z}^{\mathrm{nom}};\hat{c})$ and apply the fundamental theorem of calculus on each segment. (iii) Norms in (5); (i) applied to each entry; Minkowski in $\mathbb{R}^{n_k}$: $\|\Delta Z_k\| \le \|(\kappa e^{-\alpha t_{k-js}}\|\Delta z_0\|)_j\| + \|(\kappa\|B\|\delta_y/\alpha)_j\|$; (L); Definition 22. The bound on r_K is the geometric series in j. (iv) $L^2(\mathbb{P})$ norms of the pointwise inequality (Minkowski) with $\|\bar{J}_Z\| \le \ell_z^{\mathrm{g}}$; $\mathbb{E}\|\Delta Z_k\|^2 = \sum_j \mathbb{E}\|\Delta z_{k-js}\|^2$ and $\mathbb{E}\|\sum_i G_i^{(k)} v_i\|^2 = \sigma_v^2 \sum_i \|G_i^{(k)}\|_F^2 \to \sigma_v^2\|H\|_2^2$, a Riemann sum ($H_m = e^{At_m}B\,dt + O(dt^2)$) of $dt\int_0^\infty \|e^{As}B\|_F^2 ds$. □

Check V4 (linear plant, $K = 20$, stride 10): stack RMS 0.03218 (Monte Carlo) against $\sqrt{K}\sigma_v\|H\|_2 = 0.03206$; bound (iii) holds at every k (max ratio 0.98); path identity on the stack to $3 \cdot 10^{-16}$.

When the nominal sensor is noise-free, (6) is the classical decomposition: fit error plus decoder slope times the ISS gain $\kappa\|B\|/\alpha$ of the latent filter [cf. 19, Lemma 17 and Theorem 19 of arXiv v2; Lemma 6 and Theorem 4 of v3], here for the sampled observer and with an input-corruption term. The latent error equation of Shaaban et al. [23, Proposition 1] assumes no injection; with injection $\Phi \ne 0$ the latent deviation is no longer input-free and the margin of Proposition 8 is

needed (sampled form: Lemma 42(d)). The bound has no amplitude-dependent condition, but its constants ℓ_z, ℓ_c may depend on the context.

(iii) is a decomposition with measured constants, not a certificate. [v3-M3-4: numbers are for $K = 1$] With the stack the noise term picks up $\sqrt{n_k} \le \sqrt{K} \approx 4.5$, so the a priori values below (computed for $K = 1$) are, if anything, more vacuous for the default. With ℓ_z, ℓ_c the suprema over the traversed segments, (iii) is (ii) with norms taken, and the constants are only known after the fact (lower bounds of the suprema when estimated from samples). Evaluated *a priori*, with the phase-1 proxy `l_dec_q95` for the global constant ℓ_z^{g} and the nominal sensor level $\delta_y = 0.03\, y_{\mathrm{std}}$, the second term alone is $\ell_z \kappa \|B\| \delta_y / \alpha = 66$ (battery; $\kappa\|B\|/\alpha = 2.68$, $\ell = 816$), 15 (quad_tank; 37.4, 13.5), 6.7 (duffing), 5.9 (bioreactor), 1.1 (vdp), 0.40 (lorenz), against measured nominal NRMSE 0.016–0.052 (C1-where, lora, `id`): 15× to 1400× too large (`check_apriori_vacuity`). The bound is therefore vacuous as a guarantee on every benchmark and is *not* presented as one. What it offers is the exact attribution (ii) and the exact stochastic form (iv), both of which are measurable (Definition 28).

Remark 24 (reading the bound)**.** Three quantities set the error: the nominal error ε^β (capacity, data, window), the decoder slope ℓ_z times a design-only latent deviation ($\propto 1/\alpha$), and the input sensitivity $\ell_c \ell_E$, which is zero for the autonomous observer. The architecture enters only through these constants: the bound says which one a conditioning mechanism must keep small, not which mechanism wins.

What would test it (eval-only; implemented by no current node). Under `input_noise/bias/delay` the excess error is $\bar{J}_c \Delta c$ exactly: the autonomous observer is unaffected, and variants would order by their realised input gain $\|\bar{J}_c \Delta c\| / \|\Delta c\|$ if that gain were measured on saved checkpoints (Definition 28; C3-faults has the conditions, no node records the gains). Under `poor_init` the excess is $\bar{J}_z A_d^k \Delta z_0$, gone after a few $1/\alpha$ (C3-faults records the convergence time, which is the observable consequence). Upper bounds do not rank errors, so none of this is implied by the bound.

Check (`m3_checks.py::check_noise_iss`, `check_input_corruption`, `check_identity`; linear plant with exact maps $z = Px + \nu$, decoder $P^+(z - c)$, $\ell_z = \ell_c = 12.48$, $\ell_E = 0.048$, $n_z = 5$, $\alpha = 1.5$, $dt = 0.0206$ as for vdp). Worst-case bounded deviation: latent $0.0372 \le \gamma_y \delta_y = 0.0537 \le \kappa\|B\|\delta_y/\alpha = 0.0745$. White deviation, RMS latent 0.00427 (Monte Carlo) vs $\sigma_v \|H\|_2 = 0.00430$; the sup form at $\delta_y = 4\sigma_v$ is 30× looser, so only (iv) is informative for Gaussian conditions. Input bias 0.1 / delay 5 / bounded noise: state error $0.022 \le 0.061$, $0.197 \le 1.21$, $0.024 \le 0.18$. Identity (5) for a tanh-MLP decoder holds to 4×10^{-10} (finite-difference Jacobians); its realised gain differs from the on-data $\|\partial_z \hat{T}^+\|_2$ by a factor 0.01–1.38 (median 0.32), so on-data slopes neither bound nor estimate it. Coverage of (4) at $n = 64$, $\beta = 0.05$: 0.961 vs 0.963.

Corollary 25 (sensor faults: bounded and forgotten, uniformly in the state)**.** *[v3-M3-5: stated for $K = 1$; the stacked form is added at the end] For a decoder-side observer and any sensor deviation v, pathwise $\|\hat{x}_k - x(t_k)\| \le \|\hat{x}_k^{\mathrm{nom}} - x(t_k)\| + \ell_z^{\mathrm{g}} \|\sum_i G_i^{(k)} v_i\|$. In particular: an outlier of size M at k_0 adds at most $\ell_z^{\mathrm{g}} \kappa \|B\| M \left(w_1 + e^{\alpha dt} w_0\right) e^{-\alpha(t_k - t_{k_0})} \le \ell_z^{\mathrm{g}} \kappa \|B\| M\, dt\, e^{\alpha dt} e^{-\alpha(t_k - t_{k_0})}$ (w_0, w_1 of Lemma 42(c), $w_0 + w_1 \le dt$), with time integral at most $\ell_z^{\mathrm{g}} \kappa \|B\| M\, dt/\alpha$, for every state, input and M; an outlier process (each sample with probability p, size M, random sign) adds at most $\ell_z^{\mathrm{g}} \gamma_y p M$ in mean and $\ell_z^{\mathrm{g}} \sqrt{p} M \|H\|_2$ in RMS; sample-and-hold dropout adds at most $\ell_z^{\mathrm{g}} \gamma_y L_y\, dt\, p/(1-p)$ in mean; a drift b adds at most $\ell_z^{\mathrm{g}} \gamma_y \sup \|b\|$, and for slow drift $\Delta z_k \approx -A^{-1} B b_k$, since the discrete DC gain is $(I - A_d)^{-1}(W_0 + W_1) = -A^{-1}B$ exactly; coloured deviations obey $\mathbb{E}\|\Delta z\|^2 = \sum_{m,m'} \mathrm{tr}(H_m R_v(m' - m) H_{m'}^\top)$.* With the latent window *the same statements hold with ℓ_z^{g} the stack constant and the following changes: the mean, RMS, dropout and drift terms are multiplied by at most $\sqrt{K}$; an outlier at k_0 obeys $\|\Delta Z_k\| \le a\, e^{-\alpha(t_k - t_{k_0} - J_k s\, dt)} / \sqrt{1 - e^{-2\alpha s\, dt}}$ with $a = \kappa\|B\|M(w_1 + e^{\alpha dt} w_0)$ and*

$J_k = \min\{K-1, \lfloor (k-k_0)/s \rfloor\}$, *so it is not attenuated during the first span and decays at rate* α *only after it.*

Intuition. There is no feedback from the estimate to the latent: a fault is a forcing of a fixed stable filter, so the observer's own error can never amplify it.

Proof. Proposition 23(ii) with $\Delta c = 0$, $\Delta z_0 = 0$, $\|\bar{J}_Z\| \le \ell_z^{\mathrm{g}}$. Stack: Minkowski over the K entries for the mean, RMS, dropout and drift terms; for the outlier, $\|\Delta z_i\| \le a\, e^{-\alpha(t_i - t_{k_0})}$ for $i \ge k_0$ and 0 before, so $\|\Delta Z_k\|^2 \le a^2 e^{-2\alpha(t_k - t_{k_0})} \sum_{j \le J_k} e^{2\alpha j s\, dt}$. Outlier: $\Delta z_{k_0} = W_1 M e$ and $\Delta z_k = (A_d^{k-k_0-1} W_0 + A_d^{k-k_0} W_1) M e$, bounded with the merged kernels of Lemma 42(c); the integral is $dt \sum_m \|H_m\| M \le dt\, \kappa \|B\| M/\alpha$. Process: $\mathbb{E}\|v_i\| \le pM$ in (i), and $R_v(m) = pM^2 \delta_m$. Dropout: a held sample differs from the current one by at most $L_y\, dt\, N_k$, N_k geometric with $\mathbb{E}N_k \le p/(1-p)$. DC gain: $W_0 + W_1 = A^{-1}(A_d - I)B$ and A, A_d commute; the quasi-static error is bounded by summation by parts, $\|\Delta z_k + A^{-1} B b_k\| \le \sum_{m \ge 1} \| \sum_{j \ge m} H_j \| \sup_j \|b_j - b_{j-1}\|$. □

Check V4c: outlier envelope never violated (max ratio 0.84); one span after the outlier the stack still holds 0.92× the one-step deviation, against $e^{-\alpha S} = 0.003$ for a $K = 1$ latent.

What this proves about filters, and what it does not. The corollary is a *qualitative* ISS guarantee: the excess is bounded and decays at rate α for every state, input and fault size, given ℓ_z^{g}, with no feedback from the estimate to the latent. The EKF's stochastic stability is local (small initial error and noise, 21) and an outlier moves its linearisation point. A qualitative guarantee is not a comparative advantage, and the data say so: on *exact* models no C3-faults condition (sensor, input, plant-drift and outage) reaches the learned-wins regime of $\ge 4/6$ systems (learned-wins regime: none of 15 conditions; the learned observer beats the best filter on 0/6 under every sensor-side condition and on at most 2/6 against the best filter of either kind under input bias, input delay or 10% plant drift), and it wins on 0/6 under colored Student-t noise (C3-noise-ct), as Remark 26 predicts for linear plants and as the EKF's 2.7–14.1× (median 5.9×) smaller exact-model `sequence_ood` error on the dev systems (C1-where) makes expected. Worse, the best *learned* observer is itself above climatology (a failed estimator) under exact-model `outliers` (battery, duffing, quad_tank) and under 30% post-training plant drift (bioreactor, quad_tank): the corollary bounds the excess by $\ell_z^{\mathrm{g}}\times$(filter response), and with ℓ_z^{g} in the hundreds that bound is consistent with failure. Where learned observers win is model incompleteness (C3-mismatch10/20/30/50, C3-partial, C3-realistic), and the theoretical lever there is not robustness to sensor faults but the model term of Proposition 27.

Exact filter-level ratios (design-invariant). For the Butterworth design with $dt = 0.1/\|A\|$, the *latent* deviations of the stress conditions, relative to the nominal noise ($0.03\, y_{\mathrm{std}}$), do not depend on the system: RMS ×3.4 for `colored` ($\varrho = 0.95$), ×57 for `outliers` ($p = 0.02$, $M = 12\, y_{\mathrm{std}}$), and ×30–32 at the quasi-static peak for `sensor_drift` ($0.3\, y_{\mathrm{std}}$); `dropout` costs $O(L_y dt)$ per held sample. These are exact properties of the known filter; the *state*-level excess multiplies them by the variant's realised $\|\bar{J}_z\|$, which no current node records (Definition 28). *Check* `m3_checks.py::check_fault_ratios` computes these ratios for vdp, duffing, bioreactor and lorenz (all agree to two digits; drift ×32.1, 32.1, 32.1, 30.5). `check_outliers`, `check_dropout_drift`: outlier peak $0.519 \le 0.536$, envelope never violated, integral $0.266 \le 0.369$; process mean $0.174 \le 0.258$, RMS 0.241 vs 0.243; dropout ($p = 0.1$) mean $0.00105 \le 0.00737$; DC identity to 10^{-16}; random-walk drift peak $0.137 \le 0.322$, quasi-static deviation $0.0052 \le 0.049$.

Remark 26 (Kalman filter vs. learned KKL on linear plants)**.** For $\dot{x} = Fx + Gu$, $y = Cx$ the exact maps are $z = Px + \nu$ with $PF = AP + BC$, $\dot{\nu} = A\nu - PGu$ and decoder $P^+(z - \nu)$, so the KKL estimate is an LTI filter of (y, u) with sensor-to-error response $\Gamma_m = P^+ H_m$; for

$n_z = n_x$ it is the Luenberger observer with gain $P^{-1}B$. The steady KF has the same structure with gain K. (1) Gaussian noise with correct (Q, R): the KF is the MMSE estimator, so no KKL design beats it. (2) White non-Gaussian noise (Gaussian plus outliers): every LTI filter has steady MSE $\mathrm{MSE}_w + (\sigma_v^2 + pM^2)\|\Gamma\|_{H_2}^2$, the KF with $R = \sigma_v^2 + pM^2$ is the best linear filter [1], and the KKL observer against a KF tuned to the nominal R is decided by $\|\Gamma\|_{H_2}$ alone. *Check* (`check_kf_comparison`; oscillator, $\sigma_v = 0.03$, $p = 0.02$, $M = 12$): Gaussian MSE KF 1.98×10^{-3} < KKL 6.75×10^{-3}; with outliers KF(nominal R) 0.514, KF(R_{eff}) 0.0115, KKL 2.40, each within 2% of the formula. The KKL loses to both here ($\|P^+\| = 12.5$): outlier robustness is not a general property of KKL observers, and on the benchmarks any advantage over the EKF must be attributed to its mis-tuning, local stability and linearisation, or, as the data show, to the filter's model being wrong (next).

3.1 Model incompleteness: the term that decides C3

Proposition 27 (persistent model error: filter bias floor vs. learned fit error; linear plants)**.** *[proven for linear plants] Let the true plant be $x_{k+1} = Fx_k + Gu_k + d_k + w_k$, $y_k = Cx_k + v_k$, where d_k is the part of the dynamics the modeller's model lacks (a wrong F or a missing term: $d_k = \Delta F\, x_k + \Delta G\, u_k$), w_k, v_k white with covariances Q, R. Any fixed-gain predict–correct observer built on the model (F, G, C), in particular the steady-state Kalman filter and every Luenberger observer, $\hat{x}_{k+1} = \hat{x}_{k+1}^- + K(y_{k+1} - C\hat{x}_{k+1}^-)$, $\hat{x}_{k+1}^- = F\hat{x}_k + Gu_k$, has error $e_k = x_k - \hat{x}_k$ obeying*

$$e_{k+1} = (I - KC)F\, e_k + (I - KC)(d_k + w_k) - Kv_{k+1}.$$

*(i) (*bias*) For a constant $d_k \equiv \bar{d}$ (e.g. a constant input and $\Delta F\bar{x}$ at the steady state), the error converges to the non-decaying bias $\bar{e} = \big(I - (I - KC)F\big)^{-1}(I - KC)\,\bar{d}$; in continuous time, for $\dot{e} = (F - KC)e + \Delta F\, x$, $\bar{e} = -(F - KC)^{-1}\Delta F\,\bar{x}$, the DC value of $(sI - (F - KC))^{-1}\Delta F\, x$.*

*(ii) (*bias–variance floor*) Let $Q \succ 0$, $R \succ 0$. For a stationary d_k independent of (w, v) (e.g. d evaluated along the noise-free trajectory, as in the check; with $d_k = \Delta F\, x_k$ along the noisy trajectory the part $\Delta F\, x_k^{\text{sto}}$ is an extra process noise correlated with w, and Σ must then be taken from the joint Lyapunov equation of (x^{sto}, e)) the steady mean-square error splits exactly as $\mathrm{MSE}(K) = b^2(K) + \operatorname{tr}\Sigma(K)$, with b^2 the mean square of the deterministic response to d through $\Phi_K = (I - KC)F$ and $\Sigma = \Phi_K\Sigma\Phi_K^\top + (I - KC)Q(I - KC)^\top + KRK^\top$. Inflating the process covariance to $Q + \lambda I$ (the oracle filters' `q_inflate`) moves $K(\lambda)$ along a one-parameter curve that trades b^2 against $\operatorname{tr}\Sigma$ (in the check: $\|K\|$ and $\operatorname{tr}\Sigma$ increase, b^2 decreases, monotonically); the best achievable error $\mathrm{MSE}^\star := \inf_K \mathrm{MSE}(K) \geq \inf_K \operatorname{tr}\Sigma(K) > 0$ depends on $(F, G, C, \Delta F, \Delta G, Q, R)$ and the input, and on no data. Only n_y directions of d can be cancelled by K, so b^2 keeps a positive limit as $\lambda \to \infty$ whenever d has a component K cannot cancel (e.g. $C\bar{d} = 0$, as for the missing stiffness term of the check).*

*(iii) (*learned decoder-side observer*) Its error obeys Proposition 23 with ε^β measured on the true plant: $\|\hat{x}_k - x_k\| \leq \varepsilon^\beta + \ell_z\|\Delta z_k\| + \ell_c\|\Delta c_k\|$ on a $(1 - \beta)$ fraction of episodes, and no term depends on the model. In the linear case, a decoder fitted on true-plant data is $P_{\text{true}}^+(z - \nu_{\text{true}})$ with $P_{\text{true}}F_{\text{true}} = AP_{\text{true}} + BC$ (continuous-time Sylvester equation of the true plant; for the sampled plant its discrete counterpart): its error is driven by w, v only (no bias), with a mean-square error $\mathrm{MSE}_{\text{KKL}}$ that is a property of the true plant and the noise; at a fixed true plant it does not depend on the modeller's model at all. The model enters the learned observer only through the phase-1 PDE regulariser of T_0 (and through `kkl_mb`, which is a model-based filter and belongs to the other class).*

Prediction. *The learned observer beats the best (oracle-tuned) filter iff* $\mathrm{MSE}^\star > \varepsilon^2$*, where* ε *is its fit error on the true plant. For a missing term* $s\,d$ *of fixed shape and scale* s *(at a fixed true plant),* $\mathrm{MSE}^\star(s) = \inf_K [s^2 b_1^2(K) + \operatorname{tr}\Sigma(K)]$ *(*b_1^2 *the bias term at* $s = 1$*;* d *independent of* (w, v) *as in (ii)) is nondecreasing in* s *and* ε *does not depend on* s*, so there is at most one crossing: filters win below a threshold in the model error (possibly* 0*) and learned observers above it. The threshold depends on the plant and the noise.*

Intuition. A model-based filter corrects its wrong model only through the n_y measured directions and only at the cost of trusting the noisy measurement more; a decoder trained on the true plant has no model to be wrong about, it pays its fit error instead.

Proof. Subtract the predict–correct recursion from the true one: $x_{k+1} - \hat{x}^-_{k+1} = Fe_k + d_k + w_k$ and $e_{k+1} = (I - KC)(x_{k+1} - \hat{x}^-_{k+1}) - Kv_{k+1}$. (i) Set $e_{k+1} = e_k$ with $w = v = 0$; $(I - (I - KC)F)$ is invertible when the filter is stable. Continuous time: set $\dot{e} = 0$. (ii) Linearity: the response to d (deterministic) and to (w, v) (zero mean, independent of d) are orthogonal in L^2, so the mean squares add; Σ is the stationary covariance of the noise response (discrete Lyapunov equation). $\operatorname{tr}\Sigma(K)$ is the error of the linear filter K under the nominal model and the true noises, so it is bounded below by the nominal-model MMSE, which is positive for $Q \succ 0$, $R \succ 0$; hence $\mathrm{MSE}^\star > 0$ for every d. As $\lambda \to \infty$ the gain can cancel d only on the n_y directions that C measures, so b^2 has a positive limit when $(I - KC)\bar{d}$ cannot be made zero. (iii) Proposition 23 with Definition 22; in the linear case $z - P_{\mathrm{true}}x - \nu_{\mathrm{true}}$ obeys a stable linear recursion forced by $W_0 v_k + W_1 v_{k+1} - P_{\mathrm{true}} w_k$ only. □

Check (`m3_checks.py::check_model_mismatch`; damped oscillator, missing stiffness term $\Delta F = \epsilon\,[0\ 0; -1\ 0]$, $q = 0.05$, $\sigma_v = 0.03$, PRBS input). (i) the steady bias agrees with the formula to 10^{-8} (discrete, $\lambda = 0$ and 10^{-3}) and 10^{-6} (continuous Luenberger). (ii) at $\epsilon = 0.3$ (relative model error $|\Delta F x|/|Fx| = 0.29$), over $\lambda \in [0, 10^{-1}]$: $\|K\|$ rises $0.25 \to 1.19$, b^2 falls $5.24 \to 4.80 \cdot 10^{-2}$, $\operatorname{tr}\Sigma$ rises $2.0 \to 3.0 \cdot 10^{-3}$, MSE is U-shaped with $\mathrm{MSE}^\star = 5.09 \cdot 10^{-2}$; Monte Carlo agrees with $b^2 + \operatorname{tr}\Sigma$ within 1%. (iii) the true-plant KKL has $\mathrm{MSE} = 6.6 \cdot 10^{-3}$ and mean error 10^{-3} at every ϵ. Threshold: filters win at relative model error 0 and 0.05 ($\mathrm{MSE}^\star = 2.0, 4.2 \cdot 10^{-3}$), learned wins from 0.10 on ($10.0 \cdot 10^{-3}$ and above), one crossing.

What it predicts, and where it is tested (verdicts from `paper/generated/RESULTS.md`, the mechanical application of each node's pre-registered rule; model errors are the generated macros). With model error the C3 nodes measure, the proposition predicts a single crossing in the relative model error: filters win below it, learned observers above, per plant. On the exact-model side no condition reaches a learned-wins regime (C3-faults: none of 15 conditions; C3-noise-ct 0/6). On the incomplete-model side the learned observer beats the best *working* filter (a filter above climatology counts as failed) at every mismatch level, 10% (5/6 `id`, 4/6 `sequence_ood`), 20% (5/6, 5/6), 30% (5/6, 5/6), 50% (6/6, 5/6), under a missing term (C3-partial 5/6, 5/6) and in the realistic node (5/5, 5/5); the pre-registered crossover hypothesis "between 10% and 30%" therefore **fails** in the direction of the learned observer (first winning level 10%; the crossing lies below the smallest tested mismatch for the $\geq 4/6$ level rule, although duffing and battery are non-wins at 10%). Across the 36 model-wrong cells (`sequence_ood`, vs. the best working filter) the learned observer wins in 30/36. *Within* each system the outcome is ordered by model error: the six non-wins are duffing under every parameter mismatch (model error 0.02 at 30%), while duffing under the missing term (0.13) is a win; battery at 10% mismatch, while battery wins from 20% (model error 0.064); and quad_tank under the missing term (0.15), while quad_tank wins under parameter mismatch (0.69 at 30%); the systems with the largest errors (bioreactor 2.15, lorenz 0.71, vdp 0.66) are won at every level. *Across* systems the ordering does *not* hold: the smallest model error with a win (0.064,

mismatch20/battery) is below the largest without one (0.152, partial/quad_tank). This is what the proposition allows (the threshold depends on the plant and the noise; the toy crosses at 0.05–0.10): the data are consistent with one crossing per plant, at plant-dependent thresholds that overlap in the band 0.06–0.15, not with a single cross-system threshold. This reading is post hoc, not a pre-registered test; TEST-C3 (frozen recipe, held-out plants) is **inconclusive** at the time of writing (6/18 jobs). The proposition is proven for linear plants; on the nonlinear benchmarks "learned wins iff the floor exceeds the fit error" is the reading of an **empirical** result. The "no model term" half of (iii) has one caveat: the phase-1 PDE term does use the (wrong) model; whether that helps or hurts is *not* established (C3-partial-modelfree: the partial model beats a model-free phase 1 with a CI excluding zero on 0/6 systems, so no claim that the partial model helps is made). Two further comparisons are needed before C3 can be read as a statement about KKL rather than about "data beats a wrong model": a Koopman/EDMDc Kalman observer and a grey-box (partial model plus learned residual) EKF trained on the same data (`baselines_data.py`; not yet in the reported nodes).

3.2 Measurement and certification

Definition 28 (measured decomposition and predicted bound)**.** For a trained model, a stress condition and the steady window, in the σ-normalised norm of `metrics.py`, compute exactly: Δz_k by running the known filter on $v = \tilde{y} - \tilde{y}^{\mathrm{nom}}$ (and $A_d^k \Delta z_0$ for the prior); Δc_k by running E on $\tilde{u}$ and u; and $\bar{J}_{z,k}, \bar{J}_{c,k}$ in (5) by Gauss quadrature along the traversed segments. Report the three terms of (5) (an exact attribution of the excess error) and the *realised effective gains* $g_z = \mathrm{RMS}\|\bar{J}_z \Delta z\|/\mathrm{RMS}\|\Delta z\|$, $g_c = \mathrm{RMS}\|\bar{J}_c \Delta c\|/\mathrm{RMS}\|\Delta c\|$. The *predicted bound* is $\widehat{\mathcal{B}} = \mathrm{NRMSE}_{\mathrm{nom}} + \hat{g}_z \mathrm{RMS}\|\Delta z\| + \hat{g}_c \mathrm{RMS}\|\Delta c\|$ with $\hat{g}_z, \hat{g}_c$ measured on *other* conditions (or the a priori $\kappa\|B\|\delta_y/\alpha$, $\sigma_v\|H\|_2$, $\ell_E \delta_u$ in place of the RMS terms).

With the gains of the same condition, $\widehat{\mathcal{B}} \geq \mathrm{NRMSE}$ is the triangle inequality and tests nothing. With transferred gains it is a prediction, and the test is one-sided: a point above the diagonal falsifies the transferred gain (the decoder is steeper along this condition's paths); a point below the diagonal confirms nothing. *Status.* This procedure is evaluation-only on saved checkpoints (the filter, the encoder and the decoder are all available) and has not been run by any node in the current tree; until it is, the transferred-gain prediction is untested, and this section claims only (i), (ii), (iv) and the exact filter-level ratios above.

Remark 29 (what can be certified)**.** The result is a decomposition into measurable terms, not a certificate. *Exact* (no learning): $\kappa, \alpha, \gamma_y, \|H\|_2$, the latent deviation of any recorded fault, the path Jacobians on recorded data, and an upper bound on the SFM gain ℓ_E (product of layer norms, Lemma 19). *Lower bounds only*: any supremum of Jacobian norms estimated from samples. Exact Lipschitz constants are NP-hard to compute even for two-layer ReLU networks [27]; certified upper bounds (layer-norm products, LipSDP, 7) are valid but loose. On-data Jacobians are the wrong set for sensor faults: with $n_z > n_x$ the nominal latents lie near an n_x-dimensional manifold (times the context fibre) and $\hat{z} = \hat{z}^{\mathrm{nom}} + \Delta z$ leaves it, where the decoder's slope is inductive bias; hence the traversed-segment Jacobians of Definition 28. The random-direction quotient l_{dec} of `train.phase1_diagnostics` estimates about $\|J\|_F/\sqrt{n_z}$ on the base decoder's data (check: $q_{0.95} = 0.81\|J\|_2$ for a linear decoder) and is not the constant of the bound. *Statistical*: the nominal error only through (4).

4 An information limit for observers with finite memory

Summary. [v3-LI-1: rewritten with the decided wording] A KKL observer forgets its measurements at the rate α of its latent filter, so up to a restart term it uses only a finite window of data. Lemma 31 bounds what *any* observer with a memory of W samples can achieve: a window-limited Bayesian Cramér–Rao (van Trees) bound on the per-state error, equivalently a bit count. It is a *necessary condition*: no observer whose memory is W reaches an error below it, whatever its architecture, training data or size; it is exact and attained for linear-Gaussian plants; it is not a sufficiency result otherwise. The bound is a theorem with the Fisher information J_π of the state law at the window start. The *numbers* reported at finite windows replace J_π by the inverse climatology, a Gaussian stand-in; they are therefore an approximation, exact only for Gaussian laws, and in general larger than the rigorous bound (Remark 32). In particular, the "window starvation" of some plants (a finite-window number well above the full-past one) is a property of that stand-in and is *not proven*: the rigorous window floor A_Q is close to the full-past floor on every plant where the stand-in differs. Remark 33 turns the latent eigenvalues into a window; with the z-window of the default, the effective window is at least $6/\alpha$ (to be precise, the span $S \in [5.50, 5.90]/\alpha$ plus a logarithmic term, Remark 33), so the examples below use $W = 6/\alpha$ instead of the $3/\alpha$ of the pre-registered reading. Lemma 34 adds a local-expansion statement for linear time-varying plants; the dependence of the learned observer's error on the local Lyapunov expansion remains an empirical observation (Remark 35). Nothing here says that an observer reaching the bound exists (outside the linear-Gaussian case) or that HyperKKL reaches it.

4.1 Setting

The sampled plant, exactly as the data generator produces it,

$$x_{k+1} = \Phi(x_k, u_k) + w_k, \qquad y_k = h(x_k) + v_k, \tag{7}$$

Φ the RK4 step under the zero-order hold, $F_k := \partial_x \Phi(x_k, u_k)$, $H_k := \partial_x h(x_k)$. σ_j is the normalisation of state j (the training standard deviation), and for a set S of evaluation times the *pooled* normalised RMS of an estimator $\hat{x}$ is $\mathrm{RMS}_j(\hat{x}) := \big(\mathrm{mean}_{k\in S}\mathbb{E}[(\hat{x}_{k,j} - x_{k,j})^2]\big)^{1/2}/\sigma_j$ (never smaller than the mean of per-episode RMS values, by Jensen).

Assumption 2 (information lemma)**.** (I1) *Process noise.* w_k i.i.d. $\mathcal{N}(0, Q)$, $Q \succ 0$; $\Phi(\cdot, u) \in C^1$ for every u.

(I2) *Sensor noise.* v_k i.i.d. with a positive C^1 density p_v of finite Fisher information $J_v := \mathbb{E}[\nabla \log p_v(v) \nabla \log p_v(v)^\top]$; $h \in C^1$. Gaussian: $J_v = R^{-1}$. Unit-variance Student-t_ν with scale s: $J_v = J_t s^{-2} I$, $J_t = \frac{\nu+1}{\nu+3} \cdot \frac{\nu}{\nu-2}$ ($= 2$ at $\nu = 3$). AR(1)-coloured noise $v_k = \rho v_{k-1} + \sqrt{1-\rho^2}\, s\, e_k$ is covered after whitening (Remark 32(d)).

(I3) *Exogenous inputs.* The input record is random, independent of (x_0, w, v), and observed by the observer as specified by its class (Definition 30).

(I4) *Moments.* $\mathbb{E}\|x_k\|^2 < \infty$ for every k, and the estimators considered have finite second moments.

No smoothness of the *prior* is assumed: (I1) supplies it (Remark 32(a)).

Definition 30 (observer classes; window information)**.** Fix k and a window of W samples. $\mathcal{O}_W$: estimators $\hat{x}_k = \psi(y_{k-W:k}, u_{k-W:k})$ (memory W in both measurements and inputs). $\mathcal{O}^u_W$: $\hat{x}_k = \psi(y_{k-W:k}, u_{0:k})$ (memory W in the measurements, the whole input record known). $\mathcal{O}_\infty$: $\psi(y_{0:k}, u_{0:k})$.

Let π_W be the conditional law of x_{k-W} given the inputs the class observes, $J_\pi := \mathbb{E}[\nabla \log \pi_W \nabla \log \pi_W^\top]$ its (location) Fisher information averaged over those inputs, and define the *window information*

$$J_{k-W} = J_\pi + \mathbb{E}[H_{k-W}^\top J_v H_{k-W}], \qquad J_{i+1} = D_i^{22} - (D_i^{12})^\top (J_i + D_i^{11})^{-1} D_i^{12} \quad (k-W \le i < k),$$
$$D_i^{11} = \mathbb{E}[F_i^\top Q^{-1} F_i], \qquad D_i^{12} = -\mathbb{E}[F_i^\top] Q^{-1}, \qquad D_i^{22} = Q^{-1} + \mathbb{E}[H_{i+1}^\top J_v H_{i+1}],$$

expectations over the joint law of states and inputs [25]. Write $P_k(W) := J_k^{-1}$,

$$\beta_j(W) := \big(\mathrm{mean}_{k\in S}[P_k(W)]_{jj}\big)^{1/2}/\sigma_j, \qquad A(W) := \mathrm{mean}_{j\notin \text{sensors}}\, \beta_j(W)$$

(the unmeasured window bound). $A_\Sigma(W)$, $A_Q(W)$ denote the same quantity with J_π replaced by Σ_x^{-1} (inverse climatology: what `budgets.py` and `BUDGET_FIX` compute) and by Q^{-1}.

4.2 The information-limit lemma

Lemma 31 (information limit of a finite memory)**.** *Under Assumption 2, for every $k \ge W+1$:*

*(a) (*window van Trees bound*) every $\hat{x}_k \in \mathcal{O}_W$ satisfies $\mathbb{E}[(\hat{x}_k - x_k)(\hat{x}_k - x_k)^\top] \succeq P_k(W)$; hence $\mathrm{RMS}_j(\hat{x}) \ge \beta_j(W)$ for every state j, and the mean unmeasured RMS is at least $A(W)$. The same holds for $\mathcal{O}_W^u$ with π_W^u (law of x_{k-W} given $u_{0:k}$) in place of π_W, and for $\mathcal{O}_\infty$ with the recursion started at $k=1$.*

*(b) (*necessary condition*) consequently, if $\varepsilon < \beta_j(W)$, no observer with memory W reaches $\mathrm{RMS}_j \le \varepsilon$; if a competitor reaches mean unmeasured error $\varepsilon_c < A(W)$, no observer in $\mathcal{O}_W$ can match it. Both statements are independent of the observer's architecture, training data and size.*

*(c) (*bits form*) let $X = x_{k,j}$. If $R_X(\varepsilon^2) > I(x_k; y_{k-W:k}, u_{k-W:k})$, where R_X is the rate–distortion function of X for squared error, then no $\hat{x}_k \in \mathcal{O}_W$ has $\mathbb{E}[(\hat{x}_{k,j} - X)^2] \le \varepsilon^2$.*

*(d) (*linear-Gaussian: exact and attained*) [v3-LI-2: Gaussian inputs required for the bits part] if $\Phi(x,u) = Fx + Gu$, $h(x) = Cx$, v Gaussian and π_W Gaussian with covariance Σ_W not depending on the observed inputs (e.g. a stationary plant driven by white inputs), then $J_\pi = \Sigma_W^{-1}$, $P_k(W)$ is the covariance of the Kalman filter started at $k-W$ with prior Σ_W, that filter attains (a) with equality, and (c) is equivalent to (b): with $I_j(W) = \frac{1}{2}\log_2\big((\Sigma_x)_{jj}/[P_k(W)]_{jj}\big)$ and $R_j(\varepsilon) = \frac{1}{2}\log_2\big((\Sigma_x)_{jj}/\varepsilon^2\big)$, $R_j(\varepsilon) > I_j(W) \iff \varepsilon < \sqrt{[P_k(W)]_{jj}}$, the bits statement requiring in addition that the inputs be Gaussian, so that $x_{k,j}$ is Gaussian (with white binary inputs it is not: check V11b). In this case the bound is also sufficient: every $\varepsilon \ge \beta_j(W)$ is achieved by a window observer.*

Intuition. A memory of W samples is a noisy channel from the current state to the observer, opened at $k-W$ with only the prior knowledge of the state there; the Bayesian information of that channel caps the precision of every function of it.

Proof. (a) Condition on the inputs V the class observes and let $\theta = (x_{k-W}, \dots, x_k) \in \mathbb{R}^{(W+1)n_x}$. Its conditional density is $\pi_W(x_{k-W} \mid V) \prod_{i=k-W}^{k-1} \mathcal{N}(x_{i+1}; \Phi(x_i, u_i), Q)$ and the data density is $\prod_{i=k-W}^{k} p_v(y_i - h(x_i))$. The transitions inside the window use only $u_{k-W:k-1}$, which every class observes; inputs the class does not observe enter only through π_W, the law of x_{k-W} given V (this is what separates $\mathcal{O}_W$ from $\mathcal{O}_W^u$; Remark 32(c)). By (I3) conditioning on V leaves the noise laws unchanged. By Remark 32(a) the prior density of θ is positive, absolutely continuous, and has

finite Fisher information; the data density is C^1 in θ with finite Fisher information by (I2). The multivariate van Trees inequality [26, 8] gives, for any estimator $\hat{\theta}(Y, V)$ with finite second moments, $\mathbb{E}[(\hat{\theta}-\theta)(\hat{\theta}-\theta)^\top|V] \succeq \mathcal{J}(V)^{-1}$, where $\mathcal{J}(V)$ is the expected outer product of the score of the joint density of (θ, Y) given V. The score is a sum of factor scores: the transition factor contributes $F_i^\top Q^{-1} w_i$ to the x_i-block and $-Q^{-1}w_i$ to the x_{i+1}-block, the measurement factor $-H_i^\top \nabla \log p_v(v_i)$ to the x_i-block, and factor scores are uncorrelated because each noise is independent of everything before it and has zero-mean score. Hence $\mathcal{J}$ is block tridiagonal with blocks D^{11}, D^{12}, D^{22} (conditional on V) and first block $J_\pi + \mathbb{E}[H^\top J_v H]$; only C^1 regularity of Φ, h is used (score form, no Hessians). Any estimator of x_k extends to one of θ; the (k,k) block of $\mathcal{J}^{-1}$ is the inverse of the Schur complement computed by the recursion of Definition 30 [25, Prop. 1 and Thm. 1]. Finally average over V: $M \mapsto [M^{-1}]_{kk}$ is matrix convex on positive definite matrices, so $\mathbb{E}_V[[\mathcal{J}(V)^{-1}]_{kk}] \succeq [(\mathbb{E}_V \mathcal{J}(V))^{-1}]_{kk}$, which is the recursion with expectations over states and inputs jointly. Per state: take diagonals, average over $k \in S$, divide by σ_j^2; the unmeasured statement is the termwise mean. $\mathcal{O}_W^u$, $\mathcal{O}_\infty$: same argument with the stated prior.
(b) Contrapositive of (a). A trained observer is a fixed map once its training data are independent of the evaluation episode, which is the protocol.
(c) If $\mathbb{E}[(\hat{x}_{k,j}-X)^2] \le \varepsilon^2$, the joint law of $(X, \hat{x}_{k,j})$ is a test channel meeting the distortion constraint, so $I(X;\hat{x}_{k,j}) \ge R_X(\varepsilon^2)$ by the definition of R_X [6, Ch. 10]; and $I(X;\hat{x}_{k,j}) \le I(x_k; y_{k-W:k}, u_{k-W:k})$ by the data-processing inequality, $\hat{x}_{k,j}$ being a function of the window. Contradiction.
(d) With linear dynamics, Gaussian noises and a Gaussian π_W, the expectations are trivial and the recursion is the information form of the Riccati recursion (matrix-inversion lemma), so $P_k(W)$ is the Kalman covariance; the posterior of x_k given the window is Gaussian with covariance $P_k(W)$ and its mean attains the bound. For a Gaussian scalar X of variance $(\Sigma_x)_{jj}$ (Gaussian inputs), $R_X(D) = \frac{1}{2}\log_2((\Sigma_x)_{jj}/D)$, and (X, window) is jointly Gaussian with $\mathrm{Var}(X|\text{window}) = [P_k(W)]_{jj}$, so $I(X;\text{window}) = I_j(W)$; the equivalence is algebra, and the window Kalman filter achieves $\varepsilon = \beta_j(W)$. □

Remark 32 (the prior at the window start: what is proven, what is computed)**.** (a) *Regularity is automatic, and $J_\pi \preceq Q^{-1}$.* $x_{k-W} = \Phi(x_{k-W-1}, u_{k-W-1}) + w_{k-W-1}$ with w independent of the first term and of the inputs, so π_W is a Gaussian convolution: positive, smooth, and by Blachman's identity $\nabla \log \pi_W(s) = \mathbb{E}[\nabla \log p_w(w) \,|\, x_{k-W} = s]$, whence $J_\pi = \mathbb{E}\big[\mathbb{E}[\nabla \log p_w|\cdot]\mathbb{E}[\nabla \log p_w|\cdot]^\top\big] \preceq \mathbb{E}[\nabla \log p_w \nabla \log p_w^\top] = Q^{-1}$ [24, 28]. The recursion is monotone in its initial information, so $A_Q(W) \le A(W)$: A_Q is a *rigorous, computable* floor for every observer of the class. (b) *The computed number is a Gaussian-prior surrogate.* For any density with covariance Σ, $J_\pi \succeq \Sigma^{-1}$ (integrate by parts: $\mathbb{E}[(x-\mu)\nabla \log \pi^\top] = -I$, then Cauchy–Schwarz). With $\Sigma = \Sigma_x$ (climatology), $A(W) \le A_\Sigma(W)$, with equality when π_W is Gaussian with covariance Σ_x (Lemma 31(d)). So the reported A_Σ is exact for linear-Gaussian plants and in general an *upper* approximation of the rigorous bound: $A_Q(W) \le A(W) \le A_\Sigma(W)$. How far apart the ends are is a property of the plant (check [S4]): vdp 0.0038 = 0.0038; quad_tank 0.0178 vs 0.0297; duffing 0.0046 vs 0.0113; battery 0.0033 vs 0.0084 ($W = 3/\alpha$), the rigorous ends being close to the full-past bounds (0.0198, 0.0053, 0.0036). [v3-LI-3: 6/alpha values] At the corrected window $W = 6/\alpha$ (check V9): quad_tank $A_Q = 0.0194$, $A_\Sigma = 0.0222$; duffing 0.0053, 0.0054; battery 0.0035, 0.0048; cartpole 0.0079, 0.0229; highway_traffic 0.0144, 0.0368. Since $\mathcal{O}_W \subset \mathcal{O}_\infty$, the rigorous floor of $\mathcal{O}_W$ is at least $\max\{A_Q(W), A_\infty\}$ with A_∞ the rigorous full-past floor. The surrogate can be beaten: for a scalar plant whose state law at the window start is a two-component mixture, the exact posterior mean has mean-square error 0.0080 while the climatology-prior number is 0.0385; the bound with the true J_π is 0.0080 (check V12). Hence: the *rigorous* window floors and the full-past floors computed with the rigorous start (prior information Q^{-1} at $k = 1$) are theorems; the stored full-past floors (climatology at $k = 0$) agree

with them to three digits on eleven plants and are 5.4% higher on cartpole (check V9b) [v3-LI-7: qualified]; the *window deficit* $\rho(W) = A_\Sigma(W)/A_\Sigma(\infty)$ of the window-starved plants is a statement about the Gaussian-prior surrogate, i.e. about observers that know no more about x_{k-W} than its climatology. (c) *The class matters.* An observer that knows the input before the window (class $\mathcal{O}^u_W$) can predict x_{k-W} open-loop with the model: its prior is the open-loop prediction law, much tighter than climatology on stable plants. In the linear-Gaussian check [L2] $A^u(W)$ is attained and lies below $A(W)$ (unmeasured x_2, $W = 5$: 0.039 vs 0.161); on the benchmarks the surrogate $A^u_\Sigma(3/\alpha)$ is 0.0199 (quad_tank), 0.0057 (duffing), 0.0036 (battery), essentially the full-past floor. A long *input* memory therefore changes the bound, which is why Remark 33 counts the context's memory. (d) *Coloured and heavy-tailed sensor noise.* For $v_k = \rho v_{k-1} + \sqrt{1-\rho^2}\, s\, e_k$, conditioning on y_{k-W} and using $\tilde{y}_{i+1} = y_{i+1} - \rho y_i = h(x_{i+1}) - \rho h(x_i) + \sqrt{1-\rho^2}\, s\, e_{i+1}$ keeps the factor graph a chain with pairwise factors, so the proof of (a) goes through with $D^{11} \mathrel{+}= \rho^2 \mathbb{E}[H_i^\top J_e H_i]$, $D^{12} \mathrel{-}= \rho \mathbb{E}[H_i^\top J_e H_{i+1}]$, $D^{22} = Q^{-1} + \mathbb{E}[H_{i+1}^\top J_e H_{i+1}]$, $J_e = J_t/((1-\rho^2)s^2)$ (the recursion of `BUDGET_FIX.md`). At the window start `BUDGET_FIX` uses J_t/s^2 for the marginal of v_{k-W}, an upper bound on its true Fisher information (the marginal is a sum of independent terms; Stam's inequality), which can only lower the bound: the noise side remains rigorous; the state-prior caveat (b) is unchanged. (e) *Where Monte Carlo enters.* For nonlinear plants the expectations $\mathbb{E}[F^\top Q^{-1} F]$, $\mathbb{E}[F]$, $\mathbb{E}[H^\top J_v H]$ are averages over the evaluation episodes (64 in `budgets.json`); the bound is a theorem about the exact expectations and the reported numbers carry their sampling error (a few per cent on slow states, M4 check [7]). Linearisation enters nowhere in (a): F and H are exact Jacobians inside expectations. The *local* recursion of 36(i) (an EKF covariance along one trajectory) is not a bound for nonlinear plants.

4.3 Fading memory and the window: the latent eigenvalues

The deployed default HyperKKL (method_v2.tex) is $\hat{x}_k = D(Z_k, c_k)$ with $Z_k = (\hat{z}_k, \hat{z}_{k-s}, \dots, \hat{z}_{k-(K-1)s})$, $\hat{z}_{k+1} = A_d \hat{z}_k + W_0 y_k + W_1 y_{k+1}$, $\|A_d^m\| \le \kappa e^{-\alpha m\, dt}$, and $c_k = \psi([\xi_k, u_k])$, $\xi_{k+1} = A_d^\xi \xi_k + B_d u_k$, $\|(A_d^\xi)^m\| \le e^{-\alpha_\xi m\, dt}$ (the z-window features $g(Z_k)$ are part of D). Let ℓ_Z, ℓ_c be global Lipschitz constants of D in Z (Euclidean norm of the stack) and in c, ℓ_ψ that of ψ, and let span $:= (K-1)s$.

Remark 33 (fading memory is a window)**.** [v3-LI-4: was Corollary; kept as a remark by decision; statement and proof unchanged, (iii) added] [proven] Under Assumption 2, for every $W \ge$ span and every state j,

$$\mathrm{RMS}_j(\hat{x}) \;\ge\; \beta_j(W) - \frac{\Delta(W)}{\sigma_j}, \qquad \Delta(W) := \ell_Z\, \kappa\, r_K(W)\, \bar{Z}_W + \ell_c \ell_\psi\, e^{-\alpha_\xi W dt}\, \bar{\Xi}_W,$$

$r_K(W) := \big(\sum_{j=0}^{K-1} e^{-2\alpha(W-js)dt}\big)^{1/2} \le e^{-\alpha(W-\mathrm{span})dt}/\sqrt{1-e^{-2\alpha s\, dt}}$, $\bar{Z}_W := (\mathrm{mean}_{k\in S}\mathbb{E}\|\hat{z}_{k-W}\|^2)^{1/2}$, $\bar{\Xi}_W := (\mathrm{mean}_{k\in S}\mathbb{E}\|\xi_{k-W}\|^2)^{1/2}$. Hence $\mathrm{RMS}_j(\hat{x}) \ge \sup_{W\ge \mathrm{span}}[\beta_j(W) - \Delta(W)/\sigma_j]$, and $\Delta(W) \le \eta$ as soon as

$$W\, dt \;\ge\; \max\left\{\mathrm{span}\, dt + \frac{1}{\alpha} \ln \frac{2\ell_Z \kappa \bar{Z}}{\eta\sqrt{1-e^{-2\alpha s\, dt}}},\; \frac{1}{\alpha_\xi} \ln \frac{2\ell_c \ell_\psi \bar{\Xi}}{\eta}\right\} =: W_{\mathrm{eff}}(\eta)\, dt.$$

If the observer is compared within $\mathcal{O}^u_W$ (the input record known), the context term drops: $\mathrm{RMS}_j \ge \beta^u_j(W) - \ell_Z \kappa r_K(W) \bar{Z}_W/\sigma_j$. (iii) If the decoder depends on the latent only through the FIR feature $\hat{z}_k - A_d^m \hat{z}_{k-m}$ with $m\, dt \le W$ (Lemma 15), the latent restart term is exactly 0 (check V1c); the default decoder is not of this form.

Intuition. Restart the latent and the context at $k - W$: the restarted observer is a window observer, and it differs from the real one only through e^{AW} and $e^{A^\xi W}$ acting on the state at the restart.

Proof. Define $\hat{x}_k^W := D(Z_k^W, c_k^W)$ where $\hat{z}^W$ and ξ^W obey the same recursions from $\hat{z}_{k-W}^W = 0$, $\xi_{k-W}^W = 0$. They use $y_{k-W:k}$ and $u_{k-W:k}$ only, and D is fixed, so $\hat{x}_k^W \in \mathcal{O}_W$ and Lemma 31(a) gives $\mathrm{RMS}_j(\hat{x}^W) \geq \beta_j(W)$. The recursions are linear, so $\hat{z}_i - \hat{z}_i^W = A_d^{i-k+W}\hat{z}_{k-W}$ and $\xi_k - \xi_k^W = (A_d^\xi)^W \xi_{k-W}$; with $W \geq$ span every entry of the stack lies after the restart, so $\|Z_k - Z_k^W\| \leq \kappa r_K(W)\|\hat{z}_{k-W}\|$ and $\|c_k - c_k^W\| \leq \ell_\psi e^{-\alpha_\xi W dt}\|\xi_{k-W}\|$. Lipschitz continuity of D and Minkowski's inequality in $L^2(\mathbb{P} \times \mathrm{Unif}(S))$: $\mathrm{RMS}_j(\hat{x}) \geq \mathrm{RMS}_j(\hat{x}^W) - \mathrm{RMS}_j(\hat{x} - \hat{x}^W) \geq \beta_j(W) - \Delta(W)/\sigma_j$. The threshold: bound each term of Δ by $\eta/2$ (with $\bar{Z} \geq \bar{Z}_W$, $\bar{\Xi} \geq \bar{\Xi}_W$ uniform second moments). The $\mathcal{O}_W^u$ form keeps c_k unchanged. □

Reading for the default. [v3-LI-5: decided wording; precise span] The measurement memory is the z-window span S ($\alpha S \in [5.50, 5.90]$ on the twelve plants, check V8) plus $\ln(\cdot)/\alpha$; it is at least $6/\alpha$ as soon as the logarithm exceeds 0.50, i.e. $2\ell_Z \kappa \bar{Z}/(\eta\sqrt{1 - e^{-2\alpha s\, dt}}) \geq 1.65$, which holds for every $\eta \leq \ell_Z \kappa \bar{Z}$ because $\sqrt{\cdot} \leq 1$; the input memory is $\ln(\cdot)/\alpha_\xi$ with $\alpha_\xi \approx 0.14$–$0.18\,\alpha$ for the learned SFM (Lemma 19), i.e. 5.6–7× longer per logarithmic unit. Two consequences. (i) The window $W = 3/\alpha$ used for A in `PREREG.md`, `RESULTS.md` and `BUDGET_FIX.md` was a reading for the pre-A3 decoder ($Z_k = \hat{z}_k$, $K = 1$); for the final method $W_{\mathrm{eff}} \geq 6/\alpha$ in the measurements, and longer still if the input memory is counted. Because the floor decreases with W, a bound computed at $6/\alpha$ is the *largest* floor the remark allows for the default, not a floor it is guaranteed to obey: the inequality holds at W_{eff}, which may be longer. Which observer class to use for the SFM input memory ($\mathcal{O}_W$ or $\mathcal{O}_W^u$) is open (STATUS.md). (ii) The a priori Δ uses global Lipschitz constants, which for trained networks are norm products and make it vacuous at moderate W (as for 23(iii)); in check [L4] the a priori bound is informative only at $W \approx 400$ samples, set by the context term. The tight version replaces Δ by the measured restart term $\mathrm{RMS}_j(\hat{x} - \hat{x}^W)$ (a *restart probe*: rerun the stored observer with latent and context restarted at $k - W$), with which the inequality is exact; this probe has not been run on the stored models.

4.4 Local expansion

Lemma 34 (information along expanding directions saturates; LTV linear-Gaussian)**.** *Let $x_{i+1} = F_i x_i + G_i u_i + w_i$, $w_i \sim \mathcal{N}(0, Q_i)$, $Q_i \succeq 0$, $y_i = H_i x_i + v_i$, $v_i \sim \mathcal{N}(0, R_i)$, $R_i \succ 0$, F_i invertible, prior $x_{k-W} \sim \mathcal{N}(m, P_0)$, $P_0 \succ 0$, all independent; $\Phi_{k,i} := (F_{k-1} \cdots F_i)^{-1}$ ($\Phi_{k,k} = I$), so that $\Phi_{k,i} v$ is the perturbation at time i that the noise-free dynamics carry into v at time k. Let P_k be the Kalman covariance of x_k given $(y_{k-W:k}, u)$ and $\mathcal{I}_k := \Phi_{k,k-W}^\top P_0^{-1} \Phi_{k,k-W} + \sum_{i=k-W}^{k} \Phi_{k,i}^\top H_i^\top R_i^{-1} H_i \Phi_{k,i}$.*

(a) $P_k \succeq \mathcal{I}_k^{-1}$ (process noise only removes information).

(b) For every unit vector v and every estimator $\hat{x}_k$ of x_k from $(y_{k-W:k}, u)$: $\mathbb{E}[(v^\top(\hat{x}_k - x_k))^2] \geq v^\top P_k v \geq 1/(v^\top \mathcal{I}_k v)$.

(c) If v is expanded by the dynamics over the window at finite-time rate at least $\lambda > 0$, i.e. $\|\Phi_{k,i} v\| \leq C e^{-\lambda(k-i)dt}$ for $k - W \leq i \leq k$, then with $\bar{r} := \max_i \|R_i^{-1/2} H_i\|$, $\bar{p} := \|P_0^{-1}\|$,

$$\mathbb{E}[(v^\top(\hat{x}_k - x_k))^2] \;\geq\; \Big[C^2 \bar{p}\, e^{-2\lambda W dt} + \frac{C^2 \bar{r}^2}{1 - e^{-2\lambda dt}}\Big]^{-1} \;\xrightarrow{W \to \infty}\; \frac{1 - e^{-2\lambda dt}}{C^2 \bar{r}^2} \approx \frac{2\lambda\, dt}{C^2 \bar{r}^2}.$$

The floor along v increases with the expansion rate λ and, unlike Lemma 31, does not vanish as the window grows: information along an expanding direction saturates.

Proof. (a) Both covariances start from the same update at $k - W$. The prediction map $P \mapsto F_i P F_i^\top + Q_i$ is monotone in P and in Q_i, and the update $P \mapsto (P^{-1} + H^\top R^{-1} H)^{-1}$ is monotone;

by induction $P_k(Q) \succeq P_k(0)$ [1]. With $Q = 0$, $x_i = \Phi_{k,i} x_k + (\text{known})$, so the window is a static linear-Gaussian model of x_k with design rows $H_i \Phi_{k,i}$ and prior covariance $\Phi_{k,k-W}^{-1} P_0 \Phi_{k,k-W}^{-\top}$; its posterior information is $\mathcal{I}_k$. (b) The Kalman filter is the MMSE estimator, so every estimator has error covariance $\succeq P_k$; then $v^\top \mathcal{I}_k^{-1} v \geq (v^\top v)^2/(v^\top \mathcal{I}_k v)$ (Cauchy–Schwarz with $\mathcal{I}_k^{1/2}$). (c) $v^\top \mathcal{I}_k v \leq \bar{p} \|\Phi_{k,k-W} v\|^2 + \bar{r}^2 \sum_{m=0}^{W} \|\Phi_{k,k-m} v\|^2$; insert the hypothesis and sum the geometric series. □

If v is the leading left singular vector of the forward map $F_{k-1} \cdots F_i$, then $\|\Phi_{k,i} v\| = 1/\sigma_1(F_{k-1} \cdots F_i) = e^{-(k-i)dt\, \lambda_1^{\mathrm{FT}}}$ with λ_1^{FT} the finite-time Lyapunov exponent over $[i, k]$; C absorbs the rotation of the singular directions across lags.

Remark 35 (scope of Lemma 34, and the Lyapunov finding)**.** The lemma is exact for linear time-varying Gaussian plants (the Kalman filter is optimal), and therefore applies to the *local* recursion of a nonlinear plant (the EKF covariance along a trajectory). It is *not* a bound on the nonlinear Bayesian floor: the Bayesian recursion of Lemma 31 averages F before inverting, so expansion along individual trajectories does not appear in it, and we have no proof that would put it there. Empirically (Lorenz-96, $n_x = 20$, half the nodes measured; `experiments/chaos_deep` Stage 3 and `experiments/l96_default`), HyperKKL's error tracks the finite-time expansion of the *preceding* 0.5 time units (within-episode Spearman +0.27 to +0.37, top/bottom FTLE-quintile error ratio 1.26–1.47, unchanged by the final default; three interventions targeting it did not help). The *direction* agrees with Lemma 34 (the floor along expanding directions rises with the preceding expansion and is not removed by a longer memory). The *magnitude* does not: with $\sigma_R = 0.03\, y_{\mathrm{std}}$ the lemma's floor is $O(\sigma_R \sqrt{2\lambda dt})$, far below HyperKKL's error (≈ 0.2 unmeasured for A3; Bayesian floor 0.0018; NODE+EKF ≈ 0.07 unmeasured). The Lyapunov dependence of the *learned* observer therefore remains an empirical observation; the lemma explains why an optimal observer would show the same sign, not why HyperKKL shows it at this size.

4.5 Empirical check

All numbers are produced by `theory/checks/lemma_checks.py` (log: `lemma_checks.log`) from the stored files of `experiments/information`, `p2_rescore`, `beat_greybox` and `chaos_deep`. [v3-LI-6: rewritten at the corrected window $6/\alpha$; 3/alpha kept as the pre-registered reading]

- *Guarantee, full past.* The stored full-past floors start the recursion from the climatology at $k = 0$; the rigorous start (prior information $Q^{-1} \succeq J_\pi$ at $k = 1$) agrees to three digits on eleven plants and is 5.4% lower on cartpole (unmeasured 0.0101 vs 0.0107; per state x_2: 0.0134 vs 0.0145; check V9b, which FAILED its 2% agreement criterion on this plant). Recount of P0 over the 448 cells of `achieved.json`: no learned observer and no grey-box filter is below $0.95\times$ the full-past floor. With tier-matched budgets two cells were below, both the exact-model EKF on cartpole (`id`, x_2 0.0134, total 0.0079); against the rigorous floor they sit *at* it (0.0134, 0.0079), i.e. the optimal filter attains the bound, as Lemma 31(d) predicts near the linear regime. The pre-registered scoring found three (also quad_tank x_3 on `sequence_ood`). The newer arms are above the full-past floor in all 64 arm-cells checked (A2/A4 of `p2_rescore`, A3 of `beat_greybox`, seed means).

- *Competitor-accuracy form at the corrected window* (necessary condition, Lemma 31(b)). "HyperKKL cannot beat the best grey-box filter (unmeasured error ε) if the window floor exceeds ε", over the 22 (plant, tier) pairs of `bits_rule_c3.json`, $W = 6/\alpha$ (check V9c): with the surrogate $A_\Sigma(6/\alpha)$, "cannot" on 4 pairs (quad_tank and highway_traffic, both tiers); with the rigorous $A_Q(6/\alpha)$ on 1 (quad_tank `sequence_ood`, 0.0194 > 0.0185), and with $\max\{A_Q, A_\infty\}$

also on quad_tank `id` ($0.0198 > 0.0196$). Violations (predicted "cannot", observed win): 0 in every form. Against the $3/\alpha$ reading, duffing (both tiers) and cartpole `id` change from "cannot" to "can"; all three were losses. The rule is one-sided: "can win" is not a prediction of a win (fhn, two_link_arm, cartpole lose although they can). The ε values are those of the one-step grey-box comparator, which the corrected grey-box replaces in the consolidated re-run (CLAIMS_v3_draft.md, Disclosure 7); with a smaller ε more pairs become "cannot", so the rule must be re-scored.

- *A3 against the corrected window.* No A3 cell of `beat_greybox` (total NRMSE, 12 cells) lies below $A_\Sigma(6/\alpha)$; at $3/\alpha$ two did (duffing and quad_tank `id`), which is what a memory longer than $3/\alpha$ predicts. (NRMSE is a mean of per-episode RMS, so "below" is inconclusive and "above" conclusive.)
- *Honest caveats.* (1) P0 failed by the letter of its pre-registered 5% tolerance. (2) The pre-registered grouping thresholds (rich / window-starved) were defined at $3/\alpha$ on the surrogate, and the competitor form was introduced post hoc (`BUDGET_FIX.md`); the $6/\alpha$ rescoring here is also post hoc. (3) Window starvation is not proven (Remark 32(b)). (4) Lorenz-96: the window carries all the bits (floor 0.0018 unmeasured, window = full past); the rule says "can win" against NODE+EKF and the default HyperKKL loses (2.3× all states, 2.7× unmeasured): an extraction-limited failure the lemma does not address.

What the lemmas do not claim. Neither lemma is a sufficiency result outside the linear-Gaussian case: an observer meeting $A(W)$ need not exist for a nonlinear plant (on Lorenz and Rössler the exact-model EKF is 4.6× and 30× above the Bayesian floor), and nothing here bounds a learned observer from above or says that more data, a larger network or a different conditioner would approach the floor. The lemmas say only what an observer with a given memory *cannot* do, and Remark 33 says which memory HyperKKL has.

Numerical checks (`lemma_checks.py`, CPU; rerun for v3: 21/21 PASS, `theory/v3/lemma_checks_rerun.log`; v3 additions V1c, V8, V9, V11, V12 in `v3_checks.py`). [L1] Linear-Gaussian plant, white Gaussian input: the window Kalman filter attains $A(W)$ for $W = 5$–200 (Monte Carlo, $6 \cdot 10^4$ episodes), the information recursion equals the Kalman covariance, $A(W)$ is nonincreasing. [L2] The y-window filter that knows the input record attains $A^u(W) < A(W)$. [L3] Bits $\Leftrightarrow$ Fisher equivalence of (d). [L4] A KKL-type observer (stable latent, strided window, slow input context, least-squares decoder) satisfies Remark 33 at every W; its restarted copy obeys Lemma 31. [L5] Lemma 34(a)–(c) and the monotonicity in λ; the expanding direction saturates with W while a contracting one keeps gaining. [N1] Forced Van der Pol with Gaussian initial law: the full-past EKF and a window EKF respect the rigorous bounds. [S1]–[S4] the stored runs, the rule, and the prior sensitivity quoted above.

A How much information the measurements carry, and when any observer must fail

Setting. The sampled plant exactly as the data generator produces it (`data.integrate`):

$$x_{k+1} = \Phi(x_k, u_k) + w_k, \qquad y_k = h(x_k) + v_k, \qquad w_k \sim \mathcal{N}(0, Q), \quad v_k \sim \mathcal{N}(0, R), \tag{8}$$

Φ the RK4 step of $\dot{x} = f(x, u)$ under the zero-order hold, $Q = dt\,\mathrm{diag}(\sigma_p x_{\mathrm{std}})^2$ (Euler–Maruyama increment after the step, $\sigma_p = 0.005$), $R = \mathrm{diag}(\sigma_y y_{\mathrm{std}})^2$ ($\sigma_y = 0.03$), w, v independent and white.

$F_k := \partial_x \Phi(x_k, u_k)$, $H_k := \partial_x h(x_k)$. $D_\sigma = \mathrm{diag}(x_{\mathrm{std}})$ is the normalisation of `metrics.py`; for a matrix M write $\tilde{M} := D_\sigma^{-1} M D_\sigma^{-1}$. An *observer* is any measurable map $\hat{x}_k = \psi_k(y_{0:k}, u_{0:k})$ (the deployed decoder-side observer, the EKF and the grey-box filter are such maps; the prior $\hat{x}_0$ is a constant). Errors are $e_k = \hat{x}_k - x_k$; the paper's per-state NRMSE on the settled window S is a per-episode RMS of $e_{k,j}/\sigma_j$ averaged over episodes, and the quantity a Cramér–Rao bound controls is the *pooled* RMS $\big(\frac{1}{|S|}\sum_{k\in S}\mathbb{E}[e_{k,j}^2]\big)^{1/2}/\sigma_j$, which is never smaller than the mean of per-episode RMS values (Jensen); every comparison below uses the pooled form. Σ_x is the *climatology*: the covariance of the settled state under the evaluation inputs, the reference against which NRMSE $= 1$ means "no information".

Definition 36 (information matrices along a window; information budget)**.** Fix an episode distribution (initial condition, input record, noises) and a window length W (samples).

(i) *Local recursion* (along one realised trajectory): $P_{k-W} = \Sigma_x$ and, for $i = k-W, \dots, k-1$, $P_{i+1}^- = F_i P_i F_i^\top + Q$, $P_{i+1} = \big((P_{i+1}^-)^{-1} + H_{i+1}^\top R^{-1} H_{i+1}\big)^{-1}$ (after the update with y_{k-W} at the start). $P_k(W)$ is the Kalman covariance of the linearised plant.

(ii) *Bayesian recursion* [25]: $J_{k-W} = \Sigma_x^{-1} + \mathbb{E}[H^\top R^{-1} H]$ and

$$J_{i+1} = D_i^{22} - (D_i^{12})^\top (J_i + D_i^{11})^{-1} D_i^{12},$$
$$D_i^{11} = \mathbb{E}[F_i^\top Q^{-1} F_i], \qquad D_i^{12} = -\mathbb{E}[F_i^\top] Q^{-1}, \qquad D_i^{22} = Q^{-1} + \mathbb{E}[H_{i+1}^\top R^{-1} H_{i+1}],$$

the expectations over the episode distribution (estimated as averages over the evaluation episodes). Write $P_k^{\mathrm{B}}(W) := J_k^{-1}$. $W = \infty$ ("full past") starts the recursion at $k = 0$. [v3-M4-1: status of the prior] For $W < \infty$ the prior information Σ_x^{-1} at the window start is the Gaussian stand-in of Remark 32(b): $P_k^{\mathrm{B}}(W)$ is then A_Σ in the notation of Section 4, not a bound in general. The rigorous window floor uses J_π, or the computable $Q^{-1} \succeq J_\pi$ (A_Q).

(iii) *Budget and bounds.* $\mathrm{bits}_k(W) := \frac{1}{2}\log_2 \det \tilde{\Sigma}_x - \frac{1}{2}\log_2 \det \tilde{P}_k^{\mathrm{B}}(W)$ (information about x_k delivered by the last W samples of (y, u) beyond climatology); $\beta_j(W) := \big(\mathrm{mean}_{k\in S}[\tilde{P}_k^{\mathrm{B}}(W)]_{jj}\big)^{1/2}$ (per-state floor); $\beta_{\mathrm{unm}}(W)$ its mean over unmeasured states; $\rho(W) := \beta_{\mathrm{unm}}(W)/\beta_{\mathrm{unm}}(\infty)$ (window deficit).

Proposition 37 (Fisher information is the observability Gramian)**.** *[proven] (a) For a linear plant $x_{k+1} = F_d x_k + G_d u_k$, $y_k = C x_k + v_k$ without process noise, the Fisher information about x_k carried by $y_{k-W:k}$ (given u) is*

$$\mathcal{I}_k(W) = \sum_{i=0}^{W} \Phi_i^\top C^\top R^{-1} C\, \Phi_i, \qquad \Phi_i = \partial x_{k-i}/\partial x_k = F_d^{-i},$$

the backward observability Gramian of the window weighted by R^{-1} (for $R = \sigma^2 I$: the Gramian over σ^2), and [v3-M4-2: corrected: the prior is transported] $J_k(W) = \Phi_W^\top \Sigma_x^{-1} \Phi_W + \mathcal{I}_k(W)$ is the $Q \to 0$ limit of both recursions of Definition 36, the prior being placed on $x_{k-W} = \Phi_W x_k + (\text{known})$. It equals $\Sigma_x^{-1} + \mathcal{I}_k(W)$ only when $\Phi_W^\top \Sigma_x^{-1} \Phi_W = \Sigma_x^{-1}$ (e.g. $W = 0$, or a noise-free stationary law); v2 stated the latter form without this condition (check V10: corrected form to $7 \cdot 10^{-6}$, old form off by $5 \cdot 10^{-3}$ relative on a non-orthogonal F_d; the check's pre-set 10^{-2} threshold for "off" was not reached, so the verdict is logged as FAIL although the discrepancy is real). It does not depend on u. (b) With process noise the two recursions coincide and equal the Kalman filter's posterior covariance (information form of the Riccati recursion). (c) For the nonlinear plant (8) *the Bayesian*

recursion is the exact Fisher information of the joint density of $(x_{0:k}, y_{0:k})$ *given* u *[25, Prop. 1]:* $-\mathbb{E}[\nabla^2_{x_k,x_{k+1}} \log p(x_{k+1}|x_k, u_k)]$ *has the blocks* $D^{11}, D^{12}, D^{22} - Q^{-1}$ *exactly, because the transition noise is additive Gaussian, and* $-\mathbb{E}[\nabla^2 \log p(y|x)] = \mathbb{E}[H^\top R^{-1} H]$.

Intuition. Each measurement in the window sees the current state through the backward flow; the Gramian adds up how far the window's outputs move when the current state moves, in units of the noise.

Proof. (a) $y_{k-i} = CF_d^{-i} x_k +$ (known function of u) $+ v_{k-i}$, a Gaussian linear model in x_k with design matrix CF_d^{-i}; the Fisher information of independent Gaussian observations adds, and the prior information on x_{k-W} becomes $\Phi_W^\top \Sigma_x^{-1} \Phi_W$ in the frame of x_k. The $Q \to 0$ limit of (i): with $Q = 0$ the covariance recursion is the information recursion $J \mapsto F^{-\top} J F^{-1} + H^\top R^{-1} H$ in the frame of x_k (check [2]). (b) Linear plant: F, H are constant, the expectations in (ii) are trivial, and the block elimination $D^{22} - D^{12\top}(J + D^{11})^{-1} D^{12}$ is the matrix-inversion-lemma form of the Riccati update. (c) Differentiate $\log p(x_{k+1}|x_k) = -\frac{1}{2}\|x_{k+1} - \Phi(x_k, u_k)\|^2_{Q^{-1}} + c$ twice; the term $\mathbb{E}[(x_{k+1} - \Phi)^\top Q^{-1} \partial_x^2 \Phi]$ vanishes because the residual is the zero-mean noise, independent of x_k. □

Theorem 38 (what any observer must pay)**.** *[v3-M4-3: status restated] [(a): full past proven for the nonlinear plant* (8) *with the recursion started from the law of* x_1 *(Lemma 31, class* $\mathcal{O}_\infty$*); window form proven* with the true J_π *(Lemma 31(a)), and with the climatology* Σ_x *at the window start only for Gaussian laws: in general that number is a surrogate and can be beaten (check V12). (b): proven for* $Q = 0$ *and any prior, and for* $Q \succ 0$ *with a Gaussian prior (equality); false in general for* $Q \succ 0$ *and a non-Gaussian prior (check V13). (c): proven for a Gaussian prior. The* numbers *for nonlinear plants are Monte Carlo estimates of the expectations in Definition 36(ii).] For every observer* ψ *and every* k*:*

*(a) (*Bayesian Cramér–Rao / van Trees*)* $\mathbb{E}[e_k e_k^\top] \succeq P_k^{\mathrm{B}}(\infty)$ *for any* ψ_k*; when* ψ_k *uses the last* W *samples only,* $\mathbb{E}[e_k e_k^\top] \succeq P_k(W)$ *of Lemma 31 (prior information* J_π*), which equals* $P_k^{\mathrm{B}}(W)$ *of Definition 36 when the law of* x_{k-W} *is Gaussian with covariance* Σ_x*. Hence the pooled NRMSE of state* j *is at least* $\beta_j(W)$*, resp.* $\beta_j(\infty)$*, and the pooled total NRMSE at least* $\big(\frac{1}{n}\sum_j \beta_j^2\big)^{1/2}$*. For the linear-Gaussian plant the Kalman filter attains every one of these with equality.*

*(b) (*mutual information*)* *For a linear plant* without process noise *and any prior of covariance* Σ_x *on* x_k*,* $I(x_k; y_{0:k} \,|\, u_{0:k}) \le \frac{1}{2}\log_2 \det\big(I + \tilde{\Sigma}_x \tilde{\mathcal{I}}_k\big)$*, with equality for a Gaussian prior; with* $Q \succ 0$ *and a Gaussian prior,* $I = \mathrm{bits}_k$ *exactly. The inequality does* not *extend to* $Q \succ 0$ *with a non-Gaussian prior: for a scalar plant with a three-point prior of unit variance,* $q = r = 2$*, the information of* (y_0, y_1) *about* x_1 *exceeds the Gaussian value by* 0.021 ± 0.001 *bits (check V13; v2 stated it for any prior, and its proof used noise independent of* x_k*, which fails when* $Q \succ 0$*).*

*(c) (*rate–distortion: the guarantee in bits*)* *Let* $D := \frac{1}{n}\,\mathrm{mean}_{k\in S}\, \mathbb{E}\|D_\sigma^{-1} e_k\|^2$ *be the normalised total MSE of an observer that extracts* I *bits about* x_k *from the window (*$I \le \mathrm{bits}_k(W)$ *by (b) and the data-processing inequality, since* $\hat{x}_k$ *is a function of the window). For a Gaussian prior*

$$D \;\ge\; (\det \tilde{\Sigma}_x)^{1/n}\, 2^{-2I/n} \;\ge\; (\det \tilde{\Sigma}_x)^{1/n}\, 2^{-2\,\mathrm{bits}_k(W)/n};$$

equivalently, reaching a per-state RMS of $\sqrt{D}$ requires *at least* $I^*(D) = \frac{n}{2}\log_2\big((\det \tilde{\Sigma}_x)^{1/n}/D\big)$ *bits per window, and an observer whose window delivers fewer bits fails to reach* D*, whatever its architecture. For a non-Gaussian prior replace* $\frac{1}{2}\log_2\big((2\pi e)^n \det \tilde{\Sigma}_x\big)$ *by the differential*

entropy $h(\tilde{x}_k)$ in the Shannon lower bound $R(D) \geq h(\tilde{x}_k) - \frac{n}{2}\log_2(2\pi e D)$. The two bounds are ordered: the rate–distortion floor never exceeds the Cramér–Rao total (arithmetic–geometric mean inequality on the eigenvalues of $\tilde{P}_k^{\mathrm{B}}$), so (a) is the sharper statement per state and (c) is the same statement in bits.

Intuition. The window is a noisy channel from the current state to the observer; (a) says how precisely the state can be read off it in each direction, (c) counts the bits that reading requires. Both depend only on the plant, the sensors, the noise, the window and the input record, never on the observer.

Proof. (a) Van Trees' inequality [26, Ch. 2.4]: for a prior with the stated regularity and any estimator, $\mathbb{E}[ee^\top] \succeq J_B^{-1}$ with J_B the Bayesian information of x_k; the recursive form of J_B for the Markov model (8) is Theorem 1 of Tichavský et al. [25], whose blocks are those of Proposition 37(c); restricting the observer to the window is the same statement with the prior placed at $k - W$, whose information is J_π (Lemma 31); replacing J_π by Σ_x^{-1} is exact only for a Gaussian law (Remark 32). Attainment: for the linear-Gaussian plant the posterior is Gaussian with covariance P_k (the Kalman recursion), the conditional mean has exactly this error covariance, and $P_k = J_k^{-1}$ by Proposition 37(b). (b) For a linear plant with $Q = 0$, $y_{0:k} = \mathcal{L}x_k + \mathcal{N}$ with $\mathcal{N}$ the sensor noise, Gaussian and independent of x_k (the known input contributes a deterministic offset): among all inputs of covariance Σ_x the Gaussian maximises the mutual information of an additive Gaussian channel [6, Ch. 8–9], and for a Gaussian prior $I = \frac{1}{2}\log\det\Sigma_x - \frac{1}{2}\log\det P_k$. With $Q \succ 0$ the process noise is not independent of x_k, so this step does not apply; for a Gaussian prior the joint law is Gaussian and the equality holds directly. (c) The Shannon lower bound [6, Ch. 10]: $R(D) \geq h(\tilde{x}) - \frac{n}{2}\log_2(2\pi e D)$ for squared error nD; with the Gaussian entropy, $R(D) \geq \frac{1}{2}\log_2\left(\det\tilde{\Sigma}_x / D^n\right)$, and $R(D) \leq I$ because the observer's output achieves distortion D with I bits of information about x_k (rate–distortion converse). Solve for D. Ordering: $\frac{1}{n}\mathrm{tr}\,\tilde{P} \geq (\det\tilde{P})^{1/n} = (\det\tilde{\Sigma}_x)^{1/n} 2^{-2\,\mathrm{bits}/n}$. □

What is a guarantee here and what is not. For a linear plant with Gaussian noises (a)–(c) are theorems with the constants of the data generator, and (a) is tight. For the nonlinear benchmarks (a) in its Bayesian form remains a theorem, but its expectations are Monte Carlo averages over the evaluation episodes, so the quoted floors carry sampling error (a few per cent on the slow states, check [7]). The local recursion (i) is the EKF's own covariance and is *not* a bound for nonlinear plants; it is reported next to the Bayesian one because it is what an engineer computes and because the two differ exactly where the Bayesian bound is loose: by Jensen's inequality averaging information before inverting lowers the floor, so on the chaotic plants the Bayesian floor is far below anything achievable (lorenz 0.0014 against the exact-model EKF's 0.0064; rössler 0.0014 against 0.048 for the local recursion), while on the near-linear plants the two coincide (quad_tank 0.0198 both; Table 3). Nothing in this section bounds a learned observer *from above*.

Proposition 39 (excitation and the budget)**.** *[(a), (b) proven; (c) proven; (d) proven (data processing) with an empirical reading]*

(a) Linear plants: the input cannot buy information about the state. $J_k(W)$ *does not depend on u (Proposition 37(a,b)): the per-state floors $\beta_j(W)$ are the same for every input record. Excitation changes the climatology Σ_x, hence* $\mathrm{bits}_k(W)$ relative to climatology*, but not the absolute error floor. (This is the estimation-side counterpart of the identification-side fact that persistent excitation is needed to learn the* model*, 16, Ch. 13: with a known model the state information is a Gramian property.)*

(b) Nonlinear plants. *u enters $J_k(W)$ only through the operating points (x_i, u_i) at which F_i, H_i are evaluated; the budget moves with the evaluation inputs exactly as much as the linearisation does. On quad_tank (square-root orifice law) the full-past floor changes by less than 1% when the input time scale is varied by a factor 16 (check [4];* bits $= 21.27$ *at every level of the dose-response experiment).*

(c) Short windows starve slow modes *[v3-M4-4: proven for the linear-Gaussian recursion; the numbers are the surrogate and window starvation is not proven]. $J_k(W)$ is nondecreasing in W and converges to the steady-state information, so $\rho(W) \geq 1$ decreases to 1. For an unmeasured state that reaches the output only through a mode of rate λ_s, the window information about it keeps growing until $W \approx 1/\lambda_s$: a window short on the scale $1/\lambda_s$ is a deficit $\rho(W) \gg 1$ no matter how rich the input. quad_tank ($\lambda_s^{-1} = 84$ s, $\alpha = 0.1$): $\beta_{\text{unm}} = 0.052, 0.030, 0.020$ at $W = 1/\alpha, 3/\alpha, \infty$ ($\rho(3/\alpha) = 1.50$); battery (SOC is an integrator):* $0.033, 0.0084, 0.0036$ *($\rho = 2.3$); vdp:* $0.0051, 0.0038, 0.0038$ *($\rho = 1.0$). These are surrogate numbers A_Σ. At the corrected window $6/\alpha$: quad_tank $A_\Sigma = 0.0222$ ($\rho = 1.12$), battery* 0.0048 *(*1.32*); the rigorous $A_Q(6/\alpha)$ is* 0.0194 *and* 0.0035*, at or below the full-past floors* 0.0198*,* 0.0036*. So the deficit is a statement about observers that know only the climatology at the window start, not a theorem.*

(d) What the latent can carry. *z_k and c_k are functions of $(y_{0:k}, u_{0:k})$, so $I(x_k; z_k, c_k) \leq I(x_k; y_{0:k}, u_{0:k}) \leq$* $\text{bits}_k(\infty)$ *and every decoder-side observer obeys Theorem 38 with the full-past floor. By Lemma 14 the latent's dependence on measurements older than W is attenuated by $e^{-\alpha W}$, and the context carries u only; the* reading *(not a theorem) [v3-M4-5: corrected window] is that a decoder-side observer is, in the information sense, a window observer with the effective window of Remark 33: for the default at least $6/\alpha$ in the measurements (the span of the z-window plus a logarithmic term) and longer in the inputs (the SFM), so its floor is $\beta_j(W)$ with $W \geq 6/\alpha$, not $W \in [1/\alpha, 3/\alpha]$ (that reading was for the $K = 1$ recipe A0). On quad_tank HyperKKL's unmeasured error (*0.067 *id,* 0.093 *sequence OOD) lies above $\beta_{\text{unm}}(1/\alpha) = 0.052$, i.e.* $2.2\times$ *the $3/\alpha$ floor and* $3.4\times$ *the full-past floor, while the grey-box EKF (*0.0196*) is* at *the full-past floor (*0.0198*).*

Proof. (a), (b) are read off Definition 36 and Proposition 37. (c) Monotonicity: adding measurements adds a positive semidefinite term and the Bayesian recursion is monotone in J [25]; the convergence is that of the Riccati recursion. (d) Data processing inequality. □

Proposition 40 (the learning side: a necessary condition for any learned decoder)**.** *[proven, elementary; the sample-complexity statement is the well-specified linear case] Let the decoder be fitted by empirical risk minimisation on training triples (z', c', x') with regressor features $\varphi(z', c') \in \mathbb{R}^p$ (for a network, the features of its last layer; for the grey-box filter, the inputs of its residual network) and training covariance $C_\varphi = \mathbb{E}_{\text{train}}[\varphi\varphi^\top]$.*

*(a) (*rank / persistent excitation*) If C_φ has a null space $\mathcal{N} \neq \{0\}$, two hypotheses that differ only by a linear functional supported on $\mathcal{N}$ have the same empirical risk for every sample size, so at a query with $\varphi \notin \mathcal{N}^\perp$ the learned map is undetermined by the data: its error there is at least half the diameter of the hypotheses compatible with the training set, independent of N. The condition that excludes this uniformly in time, $\inf_t \lambda_{\min}\big(\frac{1}{W}\sum_{i=t-W}^{t} \varphi_i\varphi_i^\top\big) \geq c > 0$, is persistent excitation of the* regressors*; it is a property of the training input distribution and of the plant together, not of the evaluation data.*

*(b) (*how many informative samples*) For a well-specified linear-in-φ map with p parameters and label noise σ_ℓ^2, the excess risk of least squares is $\sigma_\ell^2 \operatorname{tr}(NC_\varphi)^{-1} \approx \sigma_\ell^2 p/N$ [10, Ch. 3]; the same*

excess therefore needs $N \propto p$. A grey-box filter that knows the mass balance and learns only a low-dimensional residual (p_g) reaches a given excess with N smaller by the factor p_g/p_f than a generic decoder (p_f), and the directions it does not *learn (the known structure) have zero excess at every N.*

Proof. (a) The empirical risk depends on the hypothesis only through its values at the training regressors, which are $\mathcal{N}^{\perp}$-measurable; two hypotheses differing on $\mathcal{N}$ agree there. The lower bound is the two-point argument of Corollary 11(b). (b) $\hat{w} - w = (\varphi^{\top}\varphi)^{-1}\varphi^{\top}\epsilon$ has covariance $\sigma_{\ell}^2(\varphi^{\top}\varphi)^{-1}$; take the trace and the law of large numbers $\varphi^{\top}\varphi \approx NC_{\varphi}$ (check [6]: ratios 0.84–1.49 at $p = 2$ and $p = 64$). □

Honest scope. (a) is necessary, not sufficient; (b) is the linear well-specified rate, and for the deep decoders of this paper only the direction of the statement is claimed (fewer effective parameters, fewer informative samples). The two sides are distinct: Theorem 38 is about the *evaluation* episodes (what any observer can extract from them) and is unchanged by any amount of training data; Proposition 40 is about the *training* episodes and is unchanged by the evaluation sensor. An observer can fail for either reason, and the two have different remedies (a longer window or a better sensor against the first; richer or structured training against the second).

Remark 41 (Kolmogorov complexity: why it cannot give the number, and what it does say)**.** The Kolmogorov complexity $K(s)$ of a string (the length of the shortest program producing it) is not computable: no algorithm outputs $K(s)$ for every s, and every computable lower bound of K is bounded above [15, Thm. 2.3.2 and its corollaries]; it is also defined only up to an additive constant that depends on the reference machine. A "guarantee" of the form "the observer fails because $K(\text{data})$ is too small" is therefore not a statement one can evaluate on quad_tank, and the per-system numbers of Table 3 are Shannon and Fisher quantities, which are computable from the model and the noise. The honest connection is through the minimum description length principle [22, 9]: a two-part code pays $L(\text{model}) + L(\text{data} \,|\, \text{model})$ bits. The grey-box filter's first term is the few parameters of its residual network (the mass balance costs nothing, it is given); a generic decoder's first term is its weight description. The second term, the bits needed to encode the residual error of the state at the observer's precision, is bounded below by $I^*(D)$ of Theorem 38(c): no coding scheme describes the state to within D with fewer bits than the window carries. MDL thus *interprets* Proposition 40(b) (fewer model bits, fewer samples to pay them) and reproduces Theorem 38(c) in its data term; it adds no quantity beyond them, and the Kolmogorov version of the same two-part argument [15, Ch. 5] adds only uncomputability. We therefore make no Kolmogorov claim.

What it predicts, and how it is tested. Pre-registered in `experiments/information/PREREG.md` (thresholds fixed from the budgets before any comparison with achieved errors), scored mechanically by `score.py` into `RESULTS.md`; the test plants' partial-knowledge runs (TEST-C3) use coloured Student-t noise on a 30%-perturbed plant, and the post hoc recomputation of their budgets under that model (`BUDGET_FIX.md`) moves $\beta_{\text{unm}}(3/\alpha)$ to 0.065 (cartpole), 0.097 (highway_traffic) and 0.020 (two_link_arm, now marginal) without changing any group of Table 3 or any verdict below; the training-distribution budgets equal the evaluation ones to within 0.3 bits on every plant, so the `id` column stands for both. [v3-M4-7: historical] (P0)–(P4) are the pre-registered predictions and their scoring at $W = 3/\alpha$ with the surrogate; they are kept as run. The reading at the corrected window is in Section 4 (Empirical check).

Table 3: Information budget of the 12 plants (evaluation tier `id`, 64 episodes, Bayesian recursion; `experiments/information/budgets.json`). bits: information about the current state beyond climatology from the full past and from a $3/\alpha$ window. β_{unm}: floor on the pooled NRMSE of the unmeasured states (full past; the local EKF-covariance value next to it; windows $3/\alpha$ and, [v3-M4-6: 6/alpha columns replace 1/alpha] the corrected $6/\alpha$, all with the climatology prior, i.e. the surrogate A_Σ); $A_Q(6/\alpha)$: the rigorous window floor (prior information Q^{-1}; check V9); $\rho_W = \beta_{\text{unm}}(W)/\beta_{\text{unm}}(\infty)$. HyperKKL and grey-box columns are historical (recipe A0 and the one-step grey-box). EKF/β: exact-model EKF pooled unmeasured RMS over the full-past floor (tightness). HyperKKL and grey-box: achieved unmeasured NRMSE (`id`; HyperKKL from the full-knowledge runs, best grey-box filter from the partial-knowledge runs; `achieved.json`). Group by the pre-registered thresholds of `PREREG.md` (rich: $\beta_{\text{unm}}(3/\alpha) < 0.02$ and $\rho < 1.25$; window-starved: $\rho \geq 1.4$; absolutely starved: $\beta_{\text{unm}}(3/\alpha) \geq 0.03$).

system	n_x	n_y	bits$_\infty$	bits$_{3/\alpha}$	$\beta_{\text{unm}}(\infty)$	local	$\beta_{\text{unm}}(3/\alpha)$	$\beta_{\text{unm}}(6/\alpha)$	$A_Q(6/\alpha)$	ρ_3/ρ_6	EKF/β	HyperKKL	grey-box	group
battery	5	2	39.3	37.5	0.0036	0.0054	0.0084	0.0048	0.0035	2.31/1.32	1.99	0.054	0.027	window
bioreactor	2	1	8.6	8.6	0.0085	0.0087	0.0085	0.0085	0.0085	1.00/1.00	0.99	0.021	0.270	rich
duffing	2	1	16.1	14.5	0.0053	0.0056	0.0113	0.0054	0.0053	2.14/1.02	1.04	0.066	0.009	window
lorenz	3	1	27.5	27.5	0.0014	0.0064	0.0014	0.0014	0.0014	1.00/1.00	4.61	0.025	0.239	rich
quad_tank	4	2	21.3	20.3	0.0198	0.0198	0.0297	0.0222	0.0194	1.50/1.12	0.99	0.067	0.020	window
vdp	2	1	16.2	16.2	0.0038	0.0065	0.0038	0.0038	0.0038	1.00/1.00	1.67	0.021	0.025	rich
cartpole	4	2	29.4	27.0	0.0107	0.0107	0.0415	0.0229	0.0079	3.89/2.15	0.95	0.175	0.048	window
fhn	2	1	14.9	14.8	0.0077	0.0083	0.0078	0.0077	0.0077	1.01/1.00	1.07	0.036	0.014	rich
highway_traffic	5	3	21.5	18.7	0.0186	0.0187	0.0512	0.0368	0.0144	2.75/1.98	1.00	0.106	0.043	window
network	10	3	48.5	48.3	0.0076	0.0090	0.0079	0.0076	0.0076	1.04/1.00	1.17	0.046	–	rich
rossler	3	1	27.0	27.0	0.0014	0.0481	0.0014	0.0014	0.0014	1.00/1.00	30.43	0.102	0.658	rich
two_link_arm	4	2	21.0	21.0	0.0455	0.0841	0.0456	0.0500	1.00	1.85	0.172	0.315	abs.	

(P0) *Guarantee.* Every achieved pooled RMS, per state and in total, of every method in every stored run (448 method–tier cells, 12 plants, exact and partial knowledge) must lie above $0.95\,\beta_j(\infty)$. **FAIL by the letter**: 3 cells are below, all of them the exact-model EKF on slow states (quad_tank x_3 on sequence OOD, 0.0233 vs 0.0245; cartpole x_2, 0.0134 vs 0.0145, and its total), and in all three the floor lies inside the bootstrap 95% interval of the achieved pooled RMS (per-episode coefficient of variation 0.3–0.5: a slow error process yields few independent samples per episode). No learned observer and no grey-box filter is below any floor. The 5% tolerance was too strict for slow states; the post-hoc reading is "no floor exceeds the achieved error by more than its sampling uncertainty". *Tightness*: the exact-model EKF sits at the floor on the near-linear plants (ratio 0.95–1.07 on quad_tank, cartpole, bioreactor, duffing, highway_traffic, fhn) and above it where the Bayesian floor is loose (lorenz 4.6, rössler 30) or the EKF is not the optimal filter (battery 2.0, vdp 1.7, two_link_arm 1.9).

(P1) *Rank correlation across plants.* HyperKKL's unmeasured NRMSE against $\beta_{\text{unm}}(3/\alpha)$: Spearman $\rho_s = 0.65$ ($p = 0.02$, $n = 12$; 0.76 on sequence OOD) – **PASS** (rule ≥ 0.5). Exact-model EKF against $\beta_{\text{unm}}(\infty)$: 0.58 – **FAIL** (rule ≥ 0.7; the EKF is far from the Bayesian floor on the chaotic plants, where that floor is loose). Best grey-box against $\beta_{\text{unm}}(\infty)$: -0.05 – **FAIL**: on bioreactor, lorenz, rössler and two_link_arm the grey-box filter fails (0.24–0.66 NRMSE, divergence and model error), so its error is not information-limited; the floor binds only an estimator that is close to optimal.

(P2) *Where learned observers lose.* The gap (HyperKKL − best grey-box)/grey-box on the unmeasured states, partial-knowledge runs: median +464% in the window-starved group against −66% in the rich group on sequence OOD (+321% vs −63% on `id`), with 5/5 window-starved plants above the rich median – **PASS** on both tiers. The exception inside the rich group is

fhn (+533%), which the budget does not explain; two_link_arm (absolutely starved) has a small gap (+19%) because both observers are poor there.

(P3) *Dose–response on the evaluation inputs* (quad_tank, vdp, two_link_arm; input time scale $\times\{\frac{1}{4}, \frac{1}{2}, 1, 2, 4\}$ within the same amplitude box; stored models, no retraining; `dose_response.py`). Predicted: the budget is flat on quad_tank (Proposition 39(b)), every method stays above the floor at every level, the exact-model EKF's error tracks the budget where it moves and stays flat where it does not, and HyperKKL's error moves more than the EKF's (a learning-side, not an information effect). *Found* (3 seeds, 32 episodes per level): the budget is flat on all three plants (full-past floor max/min 1.000, 1.05, 1.14) and so is the exact-model EKF (1.06, 1.09, 1.16) – P3b **PASS** 3/3; every method stays above $0.95\,\beta_j(\infty)$ on vdp and two_link_arm, while on quad_tank 5 of 30 cells (exact-model EKF and grey-box, x_3/x_4) fall 5–7% below it, once more within the sampling error of a slow state at 32 episodes – P3a **PASS** 2/3 by the letter; the grey-box filter sits at 0.95–1.01 of the full-past floor at every level. P3c **FAIL**: HyperKKL's error varies by only 1.08–1.32× across the levels, less than 1.5× the EKF's variation, and on quad_tank it *falls* monotonically with slower inputs (0.076 → 0.058) while the budget does not move: a small learning-side (time-scale shift) effect, not an information effect, and smaller than predicted. HyperKKL stays at $1.9\text{–}2.6 \times \beta_{\mathrm{unm}}(3/\alpha)$ and $2.9\text{–}3.8 \times \beta_{\mathrm{unm}}(\infty)$ at every level (P3d). On two_link_arm every observer, the exact-model EKF included (0.38–0.45), is 15–20× above the floor (0.021–0.024): the floor is loose there (strong nonlinearity, Jensen) and the EKF is far from optimal; the plant is poorly observable in practice, not information-starved in the Bayesian sense.

(P4) *Prediction for the quad_tank training-enrichment experiment* (`experiments/quadtank`). The arms change the training inputs only; the evaluation episodes, hence β, are unchanged: $\beta_{\mathrm{unm}} = 0.0198$ (full past; x_3 0.0245, x_4 0.0151), 0.0201 ($10/\alpha$), 0.0222 ($6/\alpha$, added in v3), 0.0297 ($3/\alpha$), 0.0517 ($1/\alpha$). The stored grey-box EKF (0.0185–0.0197) is within 1% of the full-past floor: it is information-optimal on quad_tank, and no observer of any kind can beat it by more than that. Prediction: no arm brings HyperKKL's unmeasured error below 0.020; if the fading-memory window is the binding constraint, not below 0.030; the gap that training enrichment can close is from the present 0.065–0.095 to the 0.03–0.05 band at most, and the rest is the window deficit, which only a longer memory removes. The enriched inputs add no bits to the evaluation budget by construction (Proposition 39(a,b)); they can only change the training coverage of Proposition 40.

Real battery (`real_battery.py`, Panasonic 18650PF, identified 2-RC model driven by the measured current, 30-min windows with the prior 20% off, settled half, as in the real-data evaluation; the real sensor noise is not identified, so two settings are reported). With the tuned EKF's own (Q, R) ($R_{V_m} = 0.1$ V, which absorbs model error) the local SOC floor is 0.70–1.09% SOC (US06, HWFET, NN; Bayesian 0.13%); with sensor-level noise ($R_{V_m} = 3$ mV) it is 0.07–0.11% (0.02%). The achieved settled SOC RMSE is HyperKKL 2.18/0.91/0.73%, grey-box UKF 0.87/0.94/1.25%, EKF 8.85/1.19/1.95% (`tab_real`): every observer is 7–100× above the sensor-level floor and the best ones sit at the level of the tuned filter's own covariance. On the real cell the error is model error (the identified ECM), not a shortage of information; the floor is a statement about the model and is reported, not used as a guarantee.

Numerical checks. v3: `v3_checks.py` V9 (budgets at $6/\alpha$), V10 (Gramian prior transport), V12 (surrogate beaten), V13 (counterexample to (b) for $Q \succ 0$). `theory/checks/m4_checks.py` (CPU). [1] Linear-Gaussian oscillator with PRBS input: the Kalman covariance equals both recursions of

Definition 36 to 10^{-12}; the Kalman filter's Monte Carlo MSE equals the floor within sampling error, and the exact KKL observer of Remark 26 lies above it. [2] For a deterministic window the recursion with $Q \to 0$ reproduces the backward observability Gramian over σ^2 to $1.5 \cdot 10^{-8}$ relative. [3] $\frac{1}{2}\log_2\det(I+\tilde{\Sigma}\tilde{\mathcal{I}})$ equals the prior-minus-posterior log-determinant; the Shannon lower bound lies below the Cramér–Rao total (AM–GM) and below the attained Kalman MSE. [4] Linear plant: J is input-independent by construction; quad_tank at input time scales $\times\frac{1}{4}, 1, 4$: full-past floor spread below 1%. [5] vdp: the empirical (finite-difference) Gramian converges to the Jacobian Gramian as the perturbation shrinks. [6] A null direction of the regressor covariance is unlearnable at every N; least-squares excess risk equals $\sigma^2 p/N$ within 0.84–1.49 for $p = 2$ and $p = 64$ ($N = 200$–3200). [7] Linearised quad_tank, Kalman filter, 64 episodes: pooled RMS within 2% of the floor on every state, per-episode RMS of x_3 scattering by a factor 3.8 around it (0.0115–0.0432), mean of per-episode RMS below the pooled value (Jensen).

B Lemma D: the sampled-data observer

Lemma 42 (Lemma D)**.** *Let $\|e^{At}\| \le \kappa e^{-\alpha t}$, $\rho_0 := e^{-\alpha dt}$, and let input and context be held on each $[t_k, t_{k+1})$. Let $w_0 := \int_0^{dt} \frac{\tau}{dt} e^{-\alpha\tau} d\tau = \frac{1-\rho_0(1+\alpha dt)}{\alpha^2 dt}$, $w_1 := \int_0^{dt}(1-\frac{\tau}{dt})e^{-\alpha\tau}d\tau = \frac{\alpha dt - 1 + \rho_0}{\alpha^2 dt}$, so $w_0 + w_1 = (1-\rho_0)/\alpha$.*

(a) Exact step. *For y affine and g constant on the step, $\dot{z} = Az + By + g$ gives exactly $z_{k+1} = A_d z_k + W_0 y_k + W_1 y_{k+1} + W_g g$, with $A_d = e^{Adt}$, $W_0 = \int_0^{dt}(1-\frac{s}{dt})e^{A(dt-s)}B\,ds$, $W_1 = \int_0^{dt}\frac{s}{dt}e^{A(dt-s)}B\,ds$, $W_g = A^{-1}(A_d - I)$ (`design.latent_step`).*

(b) Interpolation. *With (A3) and the regularity clause, the clean iterate $\bar{z}_k$ from $\bar{z}_0 = z(0)$ satisfies $\sup_k \|\bar{z}_k - z(t_k)\| \le e_{\text{disc}} := \frac{\kappa\|B\|M_2 dt^3}{12(1-\rho_0)} \le \frac{\kappa\|B\|M_2 dt^2 e^{\alpha dt}}{12\alpha}$, also for switching inputs, since switches sit on the grid.*

(c) Merged kernels. *$\|A_d^m W_0\| \le \kappa\|B\|\rho_0^m w_0$ and $\|A_d^m W_1\| \le \kappa\|B\|\rho_0^m w_1$. Hence $\sum_i \|G_i^{(k)}\| \le \kappa\|B\|/\alpha$ for every k and $\gamma_y \le \kappa\|B\|/\alpha$ at every dt, and $\gamma_y \to \int_0^\infty \|e^{As}B\|ds$ as $dt \to 0$. More generally every sampled constant is bounded by its continuous counterpart: $\rho_0^M = e^{-\alpha L}$, $c_d \le 1/\alpha$ in (e), $\gamma_y \le \kappa\|B\|/\alpha$.*

(d) Injection: step-size condition. *For the deployed exponential midpoint step (an exponential integrator, 11, with an explicit Euler predictor) $z_{k+1} = A_d z_k + W_0 y_k + W_1 y_{k+1} + W_g \Phi(z_k^h, c_k)$, $z_k^h = z_k + \frac{dt}{2}(Az_k + By_k + \Phi(z_k, c_k))$, with $\Phi(\cdot, c)$ ℓ_Φ-Lipschitz, two solutions driven by the same (y, c) satisfy $\|z_k - z_k'\| \le \kappa\rho^k\|z_0 - z_0'\|$,*

$$\rho := \rho_0 + \kappa(1-\rho_0)\frac{\ell_\Phi\beta}{\alpha}, \qquad \beta := 1 + \tfrac{dt}{2}(\|A\| + \ell_\Phi).$$

So $\kappa\ell_\Phi\beta < \alpha$ suffices; it tends to $\kappa\ell_\Phi < \alpha$ (Corollary 9) as $dt \to 0$ with $-\ln\rho/dt \to \alpha - \kappa\ell_\Phi$, and $\beta \le 1.1$ under $\|A\|dt \le 0.1$ (`design.choose_dt`). If the stationary latent $\zeta_k = \hat{T}_0(x(t_k))$ satisfies $\zeta_{k+1} =$ (clean step at ζ_k) $+ \int_0^{dt} e^{A(dt-s)}R\,ds + q_k$, with $\|R\| \le \varepsilon_{\text{res}}$ the PDE residual and q_k the interpolation/quadrature defect with $\|A_d^m q_k\| \le \kappa\rho_0^m\bar{q}$, and the step is fed samples with $\|v_k\| \le \delta_y$, then

$$\|\hat{z}_k - \zeta_k\| \le \kappa\rho^k\|\hat{z}_0 - \zeta_0\| + \frac{\kappa\big(\|B\|(1+\ell_\Phi\frac{dt}{2})\delta_y + \varepsilon_{\text{res}}\big)}{\alpha - \kappa\ell_\Phi\beta} + \frac{\kappa\,\alpha\,\bar{q}}{(1-\rho_0)(\alpha-\kappa\ell_\Phi\beta)},$$

a single factor κ, with $\bar{q} = O(dt^3)$ so the last term is $O(dt^2)$.

(e) Deployed fading memory. *If $u_j = u'_j$ for $j = k-M, \dots, k-1$, the deployed map $T_k^d(x;u) = \sum_{j\ge 0} A_d^j(W_0 y_{k-1-j} + W_1 y_{k-j})$ satisfies $\|T_k^d(x;u) - T_k^d(x;u')\| \le 2\kappa\|B\|\bar{y}\,\rho_0^M c_d$, $c_d := (w_0 + \rho_0 w_1)/(1-\rho_0) \le 1/\alpha$.*

Proof. (a) Variation of constants over one step with the affine y inserted. (b) $\bar{z}_k - z(t_k) = -\int_0^{t_k} e^{A(t_k-s)}B(y-\ell)(s)ds$ with ℓ the piecewise-linear interpolant. On $[t_j, t_{j+1}]$ the input is constant, so $y \in C^2$ there, and the Peano form gives $\|y(t_j+\sigma) - \ell(t_j+\sigma)\| \le \frac{\sigma(dt-\sigma)}{2} M_2$ in any norm. Use $\|e^{A(t_k-s)}\| \le \kappa\rho_0^{k-1-j}$ on that interval and $\int_0^{dt} \frac{\sigma(dt-\sigma)}{2} d\sigma = \frac{dt^3}{12}$, sum the geometric series, and use $1-\rho_0 \ge \alpha dt\,\rho_0$. (c) $A_d^m W_0 = \int_0^{dt}(1-\frac{s}{dt})e^{A(mdt+dt-s)}B\,ds$, so $\|A_d^m W_0\| \le \kappa\|B\|\rho_0^m \int_0^{dt}(1-\frac{s}{dt})e^{-\alpha(dt-s)}ds = \kappa\|B\|\rho_0^m w_0$, and likewise for W_1. Then $\|H_0\| + \sum_{m\ge1}\|H_m\| \le \kappa\|B\|\big(w_1 + \sum_{m\ge1}(\rho_0^{m-1}w_0 + \rho_0^m w_1)\big) = \kappa\|B\|\frac{w_0+w_1}{1-\rho_0} = \frac{\kappa\|B\|}{\alpha}$, and the finite-$k$ coefficients are a sub-sum of the same terms. The limit holds because $H_m = e^{At_m}B\,dt + O(dt^2)$ with summable tails. (d) $\Delta_k = A_d^k\Delta_0 + \sum_{j<k} A_d^{k-1-j} W_g(\Phi(z_j^h) - \Phi(z_j'^h))$ with $\|z_j^h - z_j'^h\| \le \beta\|\Delta_j\|$ and the merged kernel $\|A_d^{k-1-j}W_g\| = \|\int_0^{dt} e^{A((k-j)dt-s)}ds\| \le \kappa\rho_0^{k-1-j}(1-\rho_0)/\alpha$. With $a_k = e^{\alpha t_k}\|\Delta_k\|$ and $g = \kappa(e^{\alpha dt}-1)\ell_\Phi\beta/\alpha$, $a_k \le \kappa a_0 + \sum_{j<k} g a_j$; the right side S_k obeys $S_{k+1} \le (1+g)S_k$, so $\|\Delta_k\| \le \kappa(\rho_0(1+g))^k\|\Delta_0\| = \kappa\rho^k\|\Delta_0\|$. If $\kappa\ell_\Phi < \alpha$ and $\|A\|dt \le 0.1$, then $\ell_\Phi dt < \alpha dt \le \|A\|dt$, so $\beta \le 1.1$. For the forced bound, unroll $e_k = \hat{z}_k - \zeta_k$; every forcing term is merged with the kernel that propagates it: $\|A_d^m(W_0 v_j + W_1 v_{j+1})\| \le \kappa\rho_0^m\|B\|(w_0+w_1)\delta_y$, $\|A_d^m\int_0^{dt} e^{A(dt-s)}R\,ds\| \le \kappa\rho_0^m(1-\rho_0)\varepsilon_{\rm res}/\alpha$, the predictor mismatch $\le \beta\|e_j\| + \frac{dt}{2}\|B\|\delta_y$ enters through $A_d^m W_g$, and $\|A_d^m q_j\| \le \kappa\rho_0^m\bar{q}$. So $a_k \le \kappa a_0 + \sum_{j<k}\big(g a_j + \kappa e^{\alpha(j+1)dt}D\big)$ with $D = \frac{1-\rho_0}{\alpha}\big(\|B\|(1+\ell_\Phi\frac{dt}{2})\delta_y + \varepsilon_{\rm res}\big) + \bar{q}$, hence $S_{k+1} \le (1+g)S_k + \kappa e^{\alpha(k+1)dt}D$ and $\|e_k\| \le \kappa\rho^k\|e_0\| + \kappa D/(1-\rho)$; insert $1-\rho = (1-\rho_0)(\alpha - \kappa\ell_\Phi\beta)/\alpha$. (e) The samples y_i coincide for $i \ge k-M$; bound by (c), summing from $j = M$ (W_0) and $j = M+1$ (W_1), with $\|y_i - y'_i\| \le 2\bar{y}$. □

Continuous-time interpretation. The latent step is exact for the held input, the context and any generated decoder weights are constant on each interval by construction, and the decoder runs at t_k only. The bounds are about the implemented observer, and the continuous formulas are their $dt \to 0$ limits; the sampled observer adds only $e_{\rm disc}$ and, with injection, the factor β.

Check (`m3_checks.py`). (b) `check_interpolation`: PRBS input, four step sizes, order 2.00, bound/error 3.6 (`check_theory.lemma_d` uses the same constant, bound/error 4.4–4.5). (c) `check_design_constants`: $\gamma_y = 1.074 \le \kappa\|B\|/\alpha = 1.491$ ($\int\|e^{As}B\| = 1.075$); $\gamma_y \le \kappa\|B\|/\alpha$ at $\alpha dt \in \{0.03, 0.3, 1, 3\}$ and for a non-normal A with $\kappa = 1.46$. The κ of a non-normal A is estimated by `spectral_bounds` on a 400-point grid, so those rows are consistency checks, not certificates; the deployed Butterworth A is normal and has $\kappa = 1$ exactly. (d) `check_midpoint`: for $\Phi = \ell z$ the per-step gain is below ρ (e.g. $0.99591 \le 0.99762$ at $\ell = 1.3$); 200 random linear injections: $\max\|\text{step}\|/\rho = 0.999$; for a non-normal A ($\kappa = 1.99$) $\max_k\|\text{step}^k\|/(\kappa\rho^k) = 0.92$, so a single κ suffices; $\Phi = -400z$, more stable in continuous time, gives discrete spectral radius 26.4 at $\ell dt = 8.2$, so the step-size part is needed; the numpy step equals `LatentStep` to 1.5×10^{-7}. (e) `m2_checks.py::lemma_d_constants`.

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